Manage application consistency groups

ONTAP 9.13.1 REST API reference

NetApp
February 13, 2024

This PDF was generated from https://docs.netapp.com/us-en/ontap-restapi-9131/ontap/application_consistency-groups_endpoint_overview.html on February 13, 2024. Always check docs.netapp.com for the latest.

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Manage application consistency groups

Application consistency-groups endpoint overview

Overview

A consistency group is a group of volumes that supports capabilities such as creating a snapshot of all of its member volumes at the same point-in-time with a write-fence, thus ensuring a consistent image of the volumes at that time.

Applications with datasets scoped to a single volume can have its contents saved to a Snapshot copy, replicated, or cloned in a crash-consistent manner implicitly with corresponding native ONTAP volume-granular operations. Applications with datasets spanning a group of multiple volumes must have such operations performed on the group. Typically, by first fencing writes to all the volumes in the group, flushing any writes pending in queues, executing the intended operation, that is, take Snapshot copy of every volume in the group and when that is complete, unfence and resume writes. A consistency group is the conventional mechanism for providing such group semantics.

Consistency group APIs

The following APIs are used to perform operations related to consistency groups:

– GET /api/application/consistency-groups

– POST /api/application/consistency-groups

– GET /api/application/consistency-groups/{uuid}

– PATCH /api/application/consistency-groups/{uuid}

– DELETE /api/application/consistency-groups/{uuid}

Examples

Retrieving all consistency groups of an SVM

```
# The API:
/api/application/consistency-groups

# The call:
curl -X GET "https://<mgmt-ip>/api/application/consistency-
groups?svm.name=vs1" -H "accept: application/hal+json"

# The response:
{
"records": [
  {
    "uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
    "name": "vol1",
```

```
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/6f48d798-0a7f-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
    "name": "parent_cg",
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b22c85-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b270b1-0a82-11ec-a449-005056bbcf9f",
    "name": "child_1",
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b270b1-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b270c3-0a82-11ec-a449-005056bbcf9f",
    "name": "child_2",
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b270c3-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  }
],
"num_records": 4,
"_links": {
  "self": {
    "href": "/api/application/consistency-groups"
  }
}
}
```

Retrieving details of all consistency groups of an SVM

Retrieving details of the consistency groups for a specified SVM. These details are considered to be performant and will return within 1 second when 40 records or less are requested.

```
curl -X GET "https://<mgmt-ip>/api/application/consistency-
groups?svm.name=vs1&fields=*&max_records=40"

#### Response:
{
"records": [
  {
    "uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
    "name": "vol1",
    "svm": {
      "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
        }
      }
    },
    "space": {
      "size": 108003328,
      "available": 107704320,
      "used": 299008
    },
    "replicated": false,
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/6f48d798-0a7f-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
    "name": "parent_cg",
    "svm": {
      "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
        }
      }
```

```
    },
    "snapshot_policy": {
      "name": "default-1weekly",
      "uuid": "a30bd0fe-067d-11ec-a449-005056bbcf9f",
      "_links": {
        "self": {
          "href": "/api/storage/snapshot-policies/a30bd0fe-067d-11ec-a449-
005056bbcf9f"
        }
      }
    },
    "consistency_groups": [
      {
        "uuid": "c1b270b1-0a82-11ec-a449-005056bbcf9f",
        "name": "child_1",
        "space": {
          "size": 41943040,
          "available": 39346176,
          "used": 499712
        },
        "_links": {
          "self": {
            "href": "/api/application/consistency-groups/c1b270b1-0a82-
11ec-a449-005056bbcf9f"
          }
        }
      },
      {
        "uuid": "c1b270c3-0a82-11ec-a449-005056bbcf9f",
        "name": "child_2",
        "space": {
          "size": 41943040,
          "available": 39350272,
          "used": 495616
        },
        "_links": {
          "self": {
            "href": "/api/application/consistency-groups/c1b270c3-0a82-
11ec-a449-005056bbcf9f"
          }
        }
      }
    ],
    "space": {
      "size": 83886080,
      "available": 78696448,
```

```
      "used": 995328
    },
    "replicated": false,
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b22c85-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b270b1-0a82-11ec-a449-005056bbcf9f",
    "name": "child_1",
    "parent_consistency_group": {
      "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
      "name": "parent_cg",
      "_links": {
        "self": {
          "href": "/api/application/consistency-groups/c1b22c85-0a82-11ec-
a449-005056bbcf9f"
        }
      }
    },
    "svm": {
      "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
        }
      }
    },
    "snapshot_policy": {
      "name": "default",
      "uuid": "a30b60a4-067d-11ec-a449-005056bbcf9f",
      "_links": {
        "self": {
          "href": "/api/storage/snapshot-policies/a30b60a4-067d-11ec-a449-
005056bbcf9f"
        }
      }
    },
    "space": {
      "size": 41943040,
      "available": 39346176,
      "used": 499712
```

```
    },
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b270b1-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b270c3-0a82-11ec-a449-005056bbcf9f",
    "name": "child_2",
    "parent_consistency_group": {
      "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
      "name": "parent_cg",
      "_links": {
        "self": {
          "href": "/api/application/consistency-groups/c1b22c85-0a82-11ec-
a449-005056bbcf9f"
        }
      }
    },
    "svm": {
      "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
        }
      }
    },
    "snapshot_policy": {
      "name": "default",
      "uuid": "a30b60a4-067d-11ec-a449-005056bbcf9f",
      "_links": {
        "self": {
          "href": "/api/storage/snapshot-policies/a30b60a4-067d-11ec-a449-
005056bbcf9f"
        }
      }
    },
    "space": {
      "size": 41943040,
      "available": 39350272,
      "used": 495616
    },
    "_links": {
```

```
      "self": {
        "href": "/api/application/consistency-groups/c1b270c3-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  }
],
"num_records": 4,
"_links": {
  "self": {
    "href": "/api/application/consistency-
groups?svm.name=vs1&fields=*&max_records=40"
  }
}
}
```


Retrieving details of non-nested consistency groups

Retrieves details of the consistency groups without nested consistency groups, or only the parent consistency group for a number of consistency groups of a specified SVM.

```
curl -X GET "https://<mgmt-ip>/api/application/consistency-
groups?svm.name=vs1&parent_consistency_group.uuid=null"

#### Response:
{
"records": [
  {
    "uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
    "name": "vol1",
    "svm": {
      "name": "vs1"
    },
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/6f48d798-0a7f-11ec-
a449-005056bbcf9f"
      }
    }
  },
  {
    "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
    "name": "parent_cg",
    "svm": {
      "name": "vs1"
    },
    "_links": {
      "self": {
        "href": "/api/application/consistency-groups/c1b22c85-0a82-11ec-
a449-005056bbcf9f"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/application/consistency-
groups?svm.name=vs1&parent_consistency_group.uuid=null"
  }
}
}
```

Creating a single consistency group with a new SAN volume

Provisions an application with one consistency group, each with one new SAN volumes, with one LUN, an

igroup and no explicit Snapshot copy policy, FabricPool tiering policy, storage service, and QoS policy specification. The igroup to map a LUN to is specified at LUN-granularity.

```
curl -X POST https://<mgmt-ip>/api/application/consistency-
groups?return_records=true -d '{ "svm": { "name": "vs1" }, "luns": [ {
"name": "/vol/vol1/lun1", "space": { "size": "100mb" }, "os_type":
"linux", "lun_maps": [ { "igroup": { "name": "igroup1", "initiators": [ {
"name": "iqn.2021-07.com.netapp.englab.gdl:scspr2429998001" } ] } } ] } ]
}'

#### Response:
{
"num_records": 1,
"records": [
  {
    "uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
    "name": "vol1",
    "svm": {
    "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
    "name": "vs1",
    "_links": {
      "self": {
        "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
      }
    }
    },
    "luns": [
    {
      "lun_maps": [
        {
          "igroup": {
            "name": "igroup1",
            "initiators": [
              {
                "name": "iqn.2021-
07.com.netapp.englab.gdl:scspr2429998001"
              }
            ]
          }
        }
      ],
      "name": "/vol/vol1/lun1",
      "os_type": "linux",
      "space": {
        "size": 104857600
      }
```

```
    }
    ]
  }
],
"job": {
  "uuid": "6f4907ae-0a7f-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
    "href": "/api/cluster/jobs/6f4907ae-0a7f-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Creating an Application with two consistency groups with existing SAN volumes

Provisions an application with two consistency groups, each with two existing SAN volumes, a Snapshot copy policy at application-granularity, and a distinct consistency group granular Snapshot copy policy.

```
curl -X POST https://<mgmt-ip>/api/application/consistency-
groups?return_records=true -d '{ "svm": { "name": "vs1" }, "name":
"parent_cg", "snapshot_policy": { "name": "default-1weekly" },
"consistency_groups": [ { "name": "child_1", "snapshot_policy": { "name":
"default" }, "volumes": [ { "name": "existing_vol1",
"provisioning_options": { "action": "add" } }, { "name": "existing_vol2",
"provisioning_options": { "action": "add" } } ] }, { "name": "child_2",
"snapshot_policy": { "name": "default" }, "volumes": [ { "name":
"existing_vol3", "provisioning_options": { "action": "add" } }, { "name":
"existing_vol4", "provisioning_options": { "action": "add" } } ] } ] }'

#### Response:
{
"num_records": 1,
"records": [
  {
    "uuid": "c1b22c85-0a82-11ec-a449-005056bbcf9f",
    "name": "parent_cg",
    "svm": {
      "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
      "name": "vs1",
      "_links": {
        "self": {
          "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
        }
      }
    },
```

```
    "snapshot_policy": {
      "name": "default-1weekly"
    },
    "consistency_groups": [
      {
        "uuid": "c1b270b1-0a82-11ec-a449-005056bbcf9f",
        "name": "child_1",
        "snapshot_policy": {
          "name": "default"
        },
        "volumes": [
          {
            "name": "existing_vol1"
          },
          {
            "name": "existing_vol2"
          }
        ]
      },
      {
        "uuid": "c1b270c3-0a82-11ec-a449-005056bbcf9f",
        "name": "child_2",
        "snapshot_policy": {
          "name": "default"
        },
        "volumes": [
          {
            "name": "existing_vol3"
          },
          {
            "name": "existing_vol4"
          }
        ]
      }
    ]
  }
],
"job": {
  "uuid": "c1b272b9-0a82-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/c1b272b9-0a82-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Retrieving specific details of an existing consistency group

Retrieves the details of an existing consistency group.

```
curl -X GET https://<mgmt-ip>/api/application/consistency-groups/6f48d798-
0a7f-11ec-a449-005056bbcf9f

#### Response:
{
"uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
"name": "vol1",
"svm": {
  "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
  "name": "vs1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
    }
  }
},
"space": {
  "size": 108003328,
  "available": 107724800,
  "used": 278528
},
"replicated": false,
"_links": {
  "self": {
    "href": "/api/application/consistency-groups/6f48d798-0a7f-11ec-a449-
005056bbcf9f"
  }
}
}
```

Retrieving all details of an existing consistency group

Retrieves all details of an existing consistency group. These details are not considered to be performant and are not guaranteed to return within one second.

```
curl -X GET https://<mgmt-ip>/api/application/consistency-groups/6f48d798-
0a7f-11ec-a449-005056bbcf9f?fields=**

#### Response:
{
"uuid": "6f48d798-0a7f-11ec-a449-005056bbcf9f",
"name": "vol1",
```

```
"svm": {
  "uuid": "4853f97a-0a63-11ec-a449-005056bbcf9f",
  "name": "vs1",
  "_links": {
    "self": {
      "href": "/api/svm/svms/4853f97a-0a63-11ec-a449-005056bbcf9f"
    }
  }
},
"qos": {
  "policy": {
    "uuid": "b7189398-e572-48ab-8f69-82cd46580812",
    "name": "extreme-fixed",
    "_links": {
      "self": {
        "href": "/api/storage/qos/policies/b7189398-e572-48ab-8f69-
82cd46580812"
      }
    }
  }
},
"tiering": {
  "policy": "none"
},
"create_time": "2021-08-31T13:18:24-04:00",
"volumes": [
  {
    "uuid": "6f516c6c-0a7f-11ec-a449-005056bbcf9f",
    "qos": {
      "policy": {
        "uuid": "b7189398-e572-48ab-8f69-82cd46580812",
        "name": "extreme-fixed",
        "_links": {
          "self": {
            "href": "/api/storage/qos/policies/b7189398-e572-48ab-8f69-
82cd46580812"
          }
        }
      }
    },
    "tiering": {
      "policy": "none"
    },
    "comment": "",
    "create_time": "2021-08-31T13:18:22-04:00",
    "name": "vol1",
```

```
    "snapshot_policy": {
      "name": "default",
      "uuid": "a30b60a4-067d-11ec-a449-005056bbcf9f"
    },
    "space": {
      "size": 108003328,
      "available": 107569152,
      "used": 434176,
      "snapshot": {
        "used": 151552,
        "reserve_percent": 0,
        "autodelete_enabled": false
      }
    },
    "analytics": {
      "supported": false,
      "unsupported_reason": {
        "message": "File system analytics is not supported on volumes that
contain LUNs.",
        "code": "111411207"
      },
      "state": "off"
    },
    "activity_tracking": {
      "supported": false,
      "unsupported_reason": {
        "message": "Volume activity tracking is not supported on volumes
that contain LUNs.",
        "code": "124518405"
      },
      "state": "off"
    },
    "_links": {
      "self": {
        "href": "/api/storage/volumes/6f516c6c-0a7f-11ec-a449-
005056bbcf9f"
      }
    }
  }
],
"luns": [
  {
    "uuid": "6f51748a-0a7f-11ec-a449-005056bbcf9f",
    "location": {
      "logical_unit": "lun1",
      "node": {
```

```
        "name": "johnhil-vsim1",
        "uuid": "6eb682f2-067d-11ec-a449-005056bbcf9f",
        "_links": {
          "self": {
            "href": "/api/cluster/nodes/6eb682f2-067d-11ec-a449-
005056bbcf9f"
          }
        }
      },
      "volume": {
        "uuid": "6f516c6c-0a7f-11ec-a449-005056bbcf9f",
        "name": "vol1",
        "_links": {
          "self": {
            "href": "/api/storage/volumes/6f516c6c-0a7f-11ec-a449-
005056bbcf9f"
          }
        }
      }
    },
    "lun_maps": [
      {
        "igroup": {
          "uuid": "6f4a4b86-0a7f-11ec-a449-005056bbcf9f",
          "name": "igroup1",
          "os_type": "linux",
          "protocol": "mixed",
          "initiators": [
            {
              "name": "iqn.2021-07.com.netapp.englab.gdl:scspr2429998001"
            }
          ],
          "_links": {
            "self": {
              "href": "/api/protocols/san/igroups/6f4a4b86-0a7f-11ec-a449-
005056bbcf9f"
            }
          }
        },
        "logical_unit_number": 0
      }
    ],
    "name": "/vol/vol1/lun1",
    "auto_delete": false,
    "class": "regular",
    "create_time": "2021-08-31T13:18:24-04:00",
```

```
    "os_type": "linux",
    "serial_number": "wIqM6]RfQK3t",
    "space": {
      "size": 104857600,
      "used": 0,
      "guarantee": {
        "requested": false,
        "reserved": false
      }
    },
    "status": {
      "container_state": "online",
      "mapped": true,
      "read_only": false,
      "state": "online"
    },
    "_links": {
      "self": {
        "href": "/api/storage/luns/6f51748a-0a7f-11ec-a449-005056bbcf9f"
      }
    }
  }
],
"space": {
  "size": 108003328,
  "available": 107569152,
  "used": 434176
},
"replicated": false,
"_links": {
  "self": {
    "href": "/api/application/consistency-groups/6f48d798-0a7f-11ec-a449-
005056bbcf9f?fields=**"
  }
}
}
```


Adding LUNs to an existing volume in an existing consistency group

Adds two NVMe namespaces to an existing volume in an existing consistency group, creates a new subsystem, and binds the new namespaces to it.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f48d798-0a7f-11ec-a449-005056bbcf9f' -d '{ "luns": [ { "name":
"/vol/vol1/new_luns", "provisioning_options": { "count": 2, "action":
"create" }, "space": { "size": "100mb" }, "os_type": "linux", "lun_maps":
[ { "igroup": { "name": "igroup2", "initiators": [ { "name":
"01:02:03:04:05:06:07:01" } ] } } ] } ] }'

#### Response:
{
"job": {
  "uuid": "5306ea44-0a87-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/5306ea44-0a87-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Restoring a consistency group to the contents of an existing snapshot

Restores an existing consistency group to the contents of an existing snapshot of the consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "restore_to": {
"snapshot": { "uuid": "92c6c770-17a1-11eb-b141-005056acd498"} } }' -H
"Accept: Application/hal+json"

#### Response:
{
"job": {
  "uuid": "8907bd9e-1463-11eb-a719-005056ac70af",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8907bd9e-1463-11eb-a719-005056ac70af"
    }
  }
}
}
```

Deleting a consistency group

Deletes a consistency group, where all storage originally associated with that consistency group remains in place.

```
curl -X DELETE 'https://<mgmt-ip>/api/application/consistency-
groups/6f48d798-0a7f-11ec-a449-005056bbcf9f'

#### Response:
{
}
```

Cloning an existing consistency group

The following example clones an existing consistency group with the current contents:

```
curl -X POST 'https://<mgmt-ip>/api/application/consistency-groups' -d '{
"name": "clone01_of_cg01","svm": { "name": "vs_0"},"clone": { "volume": {
"prefix": "my_clone_pfx","suffix": "my_clone_sfx"},"split_initiated":
true,"parent_consistency_group": { "name": "cg01","uuid": "ca5e76fb-98c0-
11ec-855a-005056a7693b"},"guarantee": { "type": "none"} } }' -H "accept:
application/hal+json"

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Cloning a consistency group from an existing Snapshot copy

The following example clones an existing consistency group with contents from an existing Snapshot copy:

```
curl -X POST 'https://<mgmt-ip>/api/application/consistency-groups' -d '{
"name": "clone01_of_cg01","svm": { "name": "vs_0"},"clone": { "volume": {
"prefix": "my_clone_pfx","suffix": "my_clone_sfx"},"split_initiated":
true,"parent_snapshot":{ "name":
"snap01_of_cg01"},"parent_consistency_group": { "name": "cg01","uuid":
"ca5e76fb-98c0-11ec-855a-005056a7693b"},"guarantee": { "type": "none"} }
}' -H "accept: application/hal+json"

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Adding namespaces to an existing volume in an existing consistency group

To add two NVMe Namespaces to an existing volume in an existing consistency group, create a new subsystem and bind the new namespaces to it.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "namespaces": [ {
"name": "/vol/vol1/new_namespace", "space": { "size": "10M" }, "os_type":
"windows", "provisioning_options": { "count": 2 }, "subsystem_map": {
"subsystem": { "name": "mySubsystem", "hosts": [ { "nqn": "nqn.1992-
08.com.netapp:sn.d04594ef915b4c73b642169e72e4c0b1:subsystem.host1" }, {
"nqn": "nqn.1992-
08.com.netapp:sn.d04594ef915b4c73b642169e72e4c0b1:subsystem.host2" } ] } }
} ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Add a new volume in an existing consistency group

The following example adds two new volumes to an existing consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "volumes" : [ { "name":
"new_vol_", "provisioning_options": { "count" : "2"}, "space": { "size":
"1gb"} } ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Adding an existing volume to an existing consistency group

The following example adds an existing volume to an existing consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "volumes" : [ { "name":
"existing_vol", "provisioning_options": { "action" : "add"} } ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Promoting a single consistency group to a nested consistency group

The following example promotes a single consistency group to a nested consistency group with a new child consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "provisioning_options":
{ "action":"promote" }, "consistency_groups": [ { "name":"cg_child",
"provisioning_options": { "action":"create" } } ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Demoting a nested consistency group to a single consistency group

The following example demotes (flattens) a nested consistency group to a single consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "provisioning_options":
{ "action":"demote" } }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Adding a new child consistency group to nested consistency group

The following example adds a new child consistency group to an existing nested consistency group, creating a new volume.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "consistency_groups": [
{ "name": "cg_child5", "provisioning_options": { "action" : "create" },
"volumes": [ { "name": "child5_vol_1", "space": { "size": "100mb" } } ] }
] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Removing a child consistency group from nested consistency group

The following example removes a child consistency group from a nested consistency, changing it to a single consistency group with a new consistency group name.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "consistency_groups": [
{ "name": "cg_child5", "provisioning_options": { "action" : "remove",
"name" : "new_single_cg" } } ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Create a new parent consistency group with an existing consistency group

The following example creates a new nested consistency group with an existing consistency group as child consistency group.

```
curl -X POST 'https://<mgmt-ip>/api/application/consistency-groups' -d '{
"svm": { "name": "vs1" }, "name": "cg_parent2", "consistency_groups": [ {
"name": "cg_large", "provisioning_options":{ "action": "add" } }, {
"name": "cg_standalone2", "provisioning_options":{ "action": "add" } } ]
}'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Reassign a volume to another child consistency group.

The following example reassigns a volume from a child consistency group to another child consistency group with the same parent consistency group.

```
curl -X PATCH 'https://<mgmt-ip>/api/application/consistency-
groups/6f51748a-0a7f-11ec-a449-005056bbcf9f' -d '{ "consistency_groups": [
{ "name": "cg_child1", "volumes": [ { "name": "child2_vol_1",
"provisioning_options": { "action":"reassign"} } ] }, { "name":
"cg_child2" } ] }'

#### Response:
{
"job": {
  "uuid": "8c9cabf3-0a88-11ec-a449-005056bbcf9f",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/8c9cabf3-0a88-11ec-a449-005056bbcf9f"
    }
  }
}
}
```

Retrieve details of a collection or consistency group

GET `/application/consistency-groups`

Introduced In: 9.10

Retrieve details of a collection or a specific consistency group.

Notes

When volume granular properties, such as, the storage SLC, Fabric Pool tiering are not the same for all the existing volumes of a consistency group, the corresponding property is not reported at consistency group granularity. It is only reported if all the volumes of the consistency group have the same value for that property.

If this consistency group instance has 1 or more replication relationships, the "replicated" parameter is true. If there are no associated replication relationships, it is false. This parameter is only included in the output for Single-CG and Parent-CG, not for Child-CG. If this consistency group instance has 1 or more replication relationships, the "replication_relationships" parameter is included in the output for Single-CG and Parent-CG instances. If there are no associated replication relationships, this parameter is not included in the output. Note that this parameter is an array and as such it has as many elements as the number of replication relationships associated with this consistency group. Each element of the array describes properties of one replication relationship associated with this consistency group. The "uuid" parameter identifies a specific replication relationship and the "href" parameter is a link to the corresponding SnapMirror relationship. The "is_source" parameter is true if this consistency group is the source in that relationship, otherwise it is false.

Expensive properties

There is an added computational cost to retrieving values for these properties. They are not included by default in GET results and must be explicitly requested using the `fields` query parameter. See DOC Requesting specific fields to learn more.

- `volumes`
- `luns`
- `namespaces`

Parameters

Name	Type	In	Required	Description
space.available	integer	query	False	Filter by space.available
space.used	integer	query	False	Filter by space.used
space.size	integer	query	False	Filter by space.size
replication_relationships.is_source	boolean	query	False	Filter by replication_relationships.is_source • Introduced in: 9.13
replication_relationships.uuid	string	query	False	Filter by replication_relationships.uuid • Introduced in: 9.13
metric.available_space	integer	query	False	Filter by metric.available_space • Introduced in: 9.13
metric.throughput.read	integer	query	False	Filter by metric.throughput.read • Introduced in: 9.13
metric.throughput.other	integer	query	False	Filter by metric.throughput.other • Introduced in: 9.13

Name	Type	In	Required	Description
metric.throughput.write	integer	query	False	Filter by metric.throughput.write • Introduced in: 9.13
metric.throughput.total	integer	query	False	Filter by metric.throughput.total • Introduced in: 9.13
metric.duration	string	query	False	Filter by metric.duration • Introduced in: 9.13
metric.latency.read	integer	query	False	Filter by metric.latency.read • Introduced in: 9.13
metric.latency.other	integer	query	False	Filter by metric.latency.other • Introduced in: 9.13
metric.latency.write	integer	query	False	Filter by metric.latency.write • Introduced in: 9.13
metric.latency.total	integer	query	False	Filter by metric.latency.total • Introduced in: 9.13
metric.used_space	integer	query	False	Filter by metric.used_space • Introduced in: 9.13

Name	Type	In	Required	Description
metric.size	integer	query	False	Filter by metric.size • Introduced in: 9.13
metric.timestamp	string	query	False	Filter by metric.timestamp • Introduced in: 9.13
metric.status	string	query	False	Filter by metric.status • Introduced in: 9.13
metric.iops.read	integer	query	False	Filter by metric.iops.read • Introduced in: 9.13
metric.iops.other	integer	query	False	Filter by metric.iops.other • Introduced in: 9.13
metric.iops.write	integer	query	False	Filter by metric.iops.write • Introduced in: 9.13
metric.iops.total	integer	query	False	Filter by metric.iops.total • Introduced in: 9.13
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name

Name	Type	In	Required	Description
consistency_groups.parent_consistency_group.uuid	string	query	False	Filter by consistency_groups.parent_consistency_group.uuid
consistency_groups.parent_consistency_group.name	string	query	False	Filter by consistency_groups.parent_consistency_group.name
consistency_groups.namespaces.os_type	string	query	False	Filter by consistency_groups.namespaces.os_type • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.os_type	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.os_type • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.name	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.name • Introduced in: 9.12 • maxLength: 96 • minLength: 1
consistency_groups.namespaces.subsystem_map.subsystem.comment	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.comment • Introduced in: 9.12 • maxLength: 255 • minLength: 0

Name	Type	In	Required	Description
consistency_groups.namespaces.subsystem_map.subsystem.hosts.nqn	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.hosts.nqn • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.uuid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.uuid • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.nsid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.nsid • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.anagrpid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.anagrpid • Introduced in: 9.12
consistency_groups.namespaces.space.used	integer	query	False	Filter by consistency_groups.namespaces.space.used • Introduced in: 9.12
consistency_groups.namespaces.space.guarantee.requested	boolean	query	False	Filter by consistency_groups.namespaces.space.guarantee.requested • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.space.guarantee.reserved	boolean	query	False	Filter by consistency_groups.namespaces.space.guarantee.reserved • Introduced in: 9.12
consistency_groups.namespaces.space.size	integer	query	False	Filter by consistency_groups.namespaces.space.size • Introduced in: 9.12 • Max value: 140737488355328 • Min value: 4096
consistency_groups.namespaces.space.block_size	integer	query	False	Filter by consistency_groups.namespaces.space.block_size • Introduced in: 9.12
consistency_groups.namespaces.enabled	boolean	query	False	Filter by consistency_groups.namespaces.enabled • Introduced in: 9.12
consistency_groups.namespaces.status.mapped	boolean	query	False	Filter by consistency_groups.namespaces.status.mapped • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.status.state	string	query	False	Filter by consistency_groups.namespaces.status.state • Introduced in: 9.12
consistency_groups.namespaces.status.read_only	boolean	query	False	Filter by consistency_groups.namespaces.status.read_only • Introduced in: 9.12
consistency_groups.namespaces.status.container_state	string	query	False	Filter by consistency_groups.namespaces.status.container_state • Introduced in: 9.12
consistency_groups.namespaces.comment	string	query	False	Filter by consistency_groups.namespaces.comment • Introduced in: 9.12 • maxLength: 254 • minLength: 0
consistency_groups.namespaces.create_time	string	query	False	Filter by consistency_groups.namespaces.create_time • Introduced in: 9.12
consistency_groups.namespaces.uuid	string	query	False	Filter by consistency_groups.namespaces.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.name	string	query	False	Filter by consistency_groups.namespaces.name • Introduced in: 9.12
consistency_groups.namespaces.auto_delete	boolean	query	False	Filter by consistency_groups.namespaces.auto_delete • Introduced in: 9.12
consistency_groups.luns.space.size	integer	query	False	Filter by consistency_groups.luns.space.size • Max value: 140737488355328 • Min value: 4096
consistency_groups.luns.space.guarantee.requested	boolean	query	False	Filter by consistency_groups.luns.space.guarantee.requested • Introduced in: 9.11
consistency_groups.luns.space.guarantee.reserved	boolean	query	False	Filter by consistency_groups.luns.space.guarantee.reserved • Introduced in: 9.11
consistency_groups.luns.space.used	integer	query	False	Filter by consistency_groups.luns.space.used
consistency_groups.luns.lun_maps.logical_unit_number	integer	query	False	Filter by consistency_groups.luns.lun_maps.logical_unit_number

Name	Type	In	Required	Description
consistency_groups.luns.lun_maps.igroup.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.name • maxLength: 96 • minLength: 1
consistency_groups.luns.lun_maps.igroup.uuid	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.uuid
consistency_groups.luns.lun_maps.igroup.comment	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.comment • Introduced in: 9.11 • maxLength: 254 • minLength: 0
consistency_groups.luns.lun_maps.igroup.os_type	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.os_type
consistency_groups.luns.lun_maps.igroup.protocol	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.protocol
consistency_groups.luns.lun_maps.igroup.igroups.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.igroups.name • maxLength: 96 • minLength: 1
consistency_groups.luns.lun_maps.igroup.igroups.uuid	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.igroups.uuid

Name	Type	In	Required	Description
consistency_groups.luns.lun_maps.igroup.initiators.comment	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.initiators.comment • maxLength: 254 • minLength: 0
consistency_groups.luns.lun_maps.igroup.initiators.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.initiators.name
consistency_groups.luns.os_type	string	query	False	Filter by consistency_groups.luns.os_type
consistency_groups.luns.name	string	query	False	Filter by consistency_groups.luns.name
consistency_groups.luns.uuid	string	query	False	Filter by consistency_groups.luns.uuid
consistency_groups.luns.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.luns.qos.policy.max_throughput_iops
consistency_groups.luns.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.luns.qos.policy.min_throughput_mbps
consistency_groups.luns.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.luns.qos.policy.min_throughput_iops
consistency_groups.luns.qos.policy.name	string	query	False	Filter by consistency_groups.luns.qos.policy.name

Name	Type	In	Required	Description
consistency_groups.luns.qos.policy.uuid	string	query	False	Filter by consistency_groups.luns.qos.policy.uuid
consistency_groups.luns.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.luns.qos.policy.max_throughput_mbps
consistency_groups.luns.comment	string	query	False	Filter by consistency_groups.luns.comment • maxLength: 254 • minLength: 0
consistency_groups.luns.create_time	string	query	False	Filter by consistency_groups.luns.create_time
consistency_groups.luns.enabled	boolean	query	False	Filter by consistency_groups.luns.enabled
consistency_groups.luns.serial_number	string	query	False	Filter by consistency_groups.luns.serial_number • maxLength: 12 • minLength: 12
consistency_groups.name	string	query	False	Filter by consistency_groups.name
consistency_groups.tiering.policy	string	query	False	Filter by consistency_groups.tiering.policy
consistency_groups.uuid	string	query	False	Filter by consistency_groups.uuid
consistency_groups.snapshot_policy.uuid	string	query	False	Filter by consistency_groups.snapshot_policy.uuid

Name	Type	In	Required	Description
consistency_groups.snapshot_policy.name	string	query	False	Filter by consistency_groups.snapshot_policy.name
consistency_groups.application.component_type	string	query	False	Filter by consistency_groups.application.component_type • Introduced in: 9.12
consistency_groups.application.type	string	query	False	Filter by consistency_groups.application.type • Introduced in: 9.12
consistency_groups.qos.policy.uuid	string	query	False	Filter by consistency_groups.qos.policy.uuid
consistency_groups.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.qos.policy.max_throughput_mbps
consistency_groups.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.qos.policy.min_throughput_mbps
consistency_groups.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.qos.policy.min_throughput_iops
consistency_groups.qos.policy.name	string	query	False	Filter by consistency_groups.qos.policy.name
consistency_groups.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.qos.policy.max_throughput_iops

Name	Type	In	Required	Description
consistency_groups.svm.uuid	string	query	False	Filter by consistency_groups.svm.uuid
consistency_groups.svm.name	string	query	False	Filter by consistency_groups.svm.name
consistency_groups.space.available	integer	query	False	Filter by consistency_groups.space.available
consistency_groups.space.used	integer	query	False	Filter by consistency_groups.space.used
consistency_groups.space.size	integer	query	False	Filter by consistency_groups.space.size
consistency_groups.volumes.name	string	query	False	Filter by consistency_groups.volumes.name • maxLength: 203 • minLength: 1
consistency_groups.volumes.tiering.policy	string	query	False	Filter by consistency_groups.volumes.tiering.policy
consistency_groups.volumes.uuid	string	query	False	Filter by consistency_groups.volumes.uuid
consistency_groups.volumes.snapshot_policy.uuid	string	query	False	Filter by consistency_groups.volumes.snapshot_policy.uuid
consistency_groups.volumes.snapshot_policy.name	string	query	False	Filter by consistency_groups.volumes.snapshot_policy.name

Name	Type	In	Required	Description
consistency_groups.volumes.qos.policy.uuid	string	query	False	Filter by consistency_groups.volumes.qos.policy.uuid
consistency_groups.volumes.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.volumes.qos.policy.max_throughput_mbps
consistency_groups.volumes.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.volumes.qos.policy.min_throughput_mbps
consistency_groups.volumes.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.volumes.qos.policy.min_throughput_iops
consistency_groups.volumes.qos.policy.name	string	query	False	Filter by consistency_groups.volumes.qos.policy.name
consistency_groups.volumes.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.volumes.qos.policy.max_throughput_iops
consistency_groups.volumes.comment	string	query	False	Filter by consistency_groups.volumes.comment • maxLength: 1023 • minLength: 0

Name	Type	In	Required	Description
consistency_groups.volumes.nas.security_style	string	query	False	Filter by consistency_groups.volumes.nas.security_style • Introduced in: 9.12
consistency_groups.volumes.nas.path	string	query	False	Filter by consistency_groups.volumes.nas.path • Introduced in: 9.12
consistency_groups.volumes.nas.gid	integer	query	False	Filter by consistency_groups.volumes.nas.gid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.uuid	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.uuid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.rw_rule	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.rw_rule • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.clients.match	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.clients.match • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.export_policy.rules.allow_suid	boolean	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.allow_suid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.chown_mode	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.chown_mode • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.allow_device_creation	boolean	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.allow_device_creation • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.protocols	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.protocols • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.ntfs_unix_security	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.ntfs_unix_security • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.export_policy.rules.superuser	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.superuser • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.anonymous_user	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.anonymous_user • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.index	integer	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.index • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.ro_rule	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.ro_rule • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.name	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.name • Introduced in: 9.12
consistency_groups.volumes.nas.junction_parent.name	string	query	False	Filter by consistency_groups.volumes.nas.junction_parent.name • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.junction_parent.uuid	string	query	False	Filter by consistency_groups.volumes.nas.junction_parent.uuid • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.namespace_caching	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.namespace_caching • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.encryption	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.encryption • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.no_strict_security	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.no_strict_security • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.vscan_profile	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.vscan_profile • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.allow_unencrypted_access	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.allow_unencrypted_access • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.file_umask	integer	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.file_umask • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.offline_files	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.offline_files • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.home_directory	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.home_directory • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.comment	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.comment • Introduced in: 9.12 • maxLength: 256 • minLength: 1
consistency_groups.volumes.nas.cifs.shares.change_notify	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.change_notify • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.oplocks	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.oplocks • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.unix_symlink	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.unix_symlink • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.dir_umask	integer	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.dir_umask • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.continuously_available	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.continuously_available • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.access_based_enumeration	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.access_based_enumeration • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.acls.permission	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.permission • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.acls.user_or_group	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.user_or_group • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.acls.type	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.type • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.show_snapshot	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.show_snapshot • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.name	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.name • Introduced in: 9.12 • maxLength: 80 • minLength: 1
consistency_groups.volumes.nas.unix_permissions	integer	query	False	Filter by consistency_groups.volumes.nas.unix_permissions • Introduced in: 9.12
consistency_groups.volumes.nas.uid	integer	query	False	Filter by consistency_groups.volumes.nas.uid • Introduced in: 9.12
consistency_groups.volumes.analytics.scan_progress	integer	query	False	Filter by consistency_groups.volumes.analytics.scan_progress • Introduced in: 9.13

Name	Type	In	Required	Description
consistency_groups.volumes.analytics.unsupported_reason.code	string	query	False	Filter by consistency_groups.volumes.analytics.unsupported_reason.code • Introduced in: 9.13
consistency_groups.volumes.analytics.unsupported_reason.message	string	query	False	Filter by consistency_groups.volumes.analytics.unsupported_reason.message • Introduced in: 9.13
consistency_groups.volumes.analytics.supported	boolean	query	False	Filter by consistency_groups.volumes.analytics.supported • Introduced in: 9.13
consistency_groups.volumes.analytics.state	string	query	False	Filter by consistency_groups.volumes.analytics.state • Introduced in: 9.13
consistency_groups.volumes.space.available	integer	query	False	Filter by consistency_groups.volumes.space.available
consistency_groups.volumes.space.used	integer	query	False	Filter by consistency_groups.volumes.space.used
consistency_groups.volumes.space.size	integer	query	False	Filter by consistency_groups.volumes.space.size

Name	Type	In	Required	Description
consistency_groups.volumes.activity_tracking.unsupported_reason.code	string	query	False	Filter by consistency_groups.volumes.activity_tracking.unsupported_reason.code • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.unsupported_reason.message	string	query	False	Filter by consistency_groups.volumes.activity_tracking.unsupported_reason.message • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.state	string	query	False	Filter by consistency_groups.volumes.activity_tracking.state • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.supported	boolean	query	False	Filter by consistency_groups.volumes.activity_tracking.supported • Introduced in: 9.13
consistency_groups.volumes.language	string	query	False	Filter by consistency_groups.volumes.language
snapshot_policy.uuid	string	query	False	Filter by snapshot_policy.uuid
snapshot_policy.name	string	query	False	Filter by snapshot_policy.name
qos.policy.uuid	string	query	False	Filter by qos.policy.uuid

Name	Type	In	Required	Description
qos.policy.max_throughput_mbps	integer	query	False	Filter by qos.policy.max_throughput_mbps
qos.policy.min_throughput_mbps	integer	query	False	Filter by qos.policy.min_throughput_mbps
qos.policy.min_throughput_iops	integer	query	False	Filter by qos.policy.min_throughput_iops
qos.policy.name	string	query	False	Filter by qos.policy.name
qos.policy.max_throughput_iops	integer	query	False	Filter by qos.policy.max_throughput_iops
replication_source	boolean	query	False	Filter by replication_source
replicated	boolean	query	False	Filter by replicated
luns.space.size	integer	query	False	Filter by luns.space.size • Max value: 140737488355328 • Min value: 4096
luns.space.guarantee.requested	boolean	query	False	Filter by luns.space.guarantee.requested • Introduced in: 9.11
luns.space.guarantee.reserved	boolean	query	False	Filter by luns.space.guarantee.reserved • Introduced in: 9.11

Name	Type	In	Required	Description
luns.space.used	integer	query	False	Filter by luns.space.used
luns.lun_maps.logical_unit_number	integer	query	False	Filter by luns.lun_maps.logical_unit_number
luns.lun_maps.igroup.name	string	query	False	Filter by luns.lun_maps.igroup.name • maxLength: 96 • minLength: 1
luns.lun_maps.igroup.uuid	string	query	False	Filter by luns.lun_maps.igroup.uuid
luns.lun_maps.igroup.comment	string	query	False	Filter by luns.lun_maps.igroup.comment • Introduced in: 9.11 • maxLength: 254 • minLength: 0
luns.lun_maps.igroup.os_type	string	query	False	Filter by luns.lun_maps.igroup.os_type
luns.lun_maps.igroup.protocol	string	query	False	Filter by luns.lun_maps.igroup.protocol
luns.lun_maps.igroup.igroups.name	string	query	False	Filter by luns.lun_maps.igroup.igroups.name • maxLength: 96 • minLength: 1
luns.lun_maps.igroup.igroups.uuid	string	query	False	Filter by luns.lun_maps.igroup.igroups.uuid

Name	Type	In	Required	Description
luns.lun_maps.igroup.initiators.comment	string	query	False	Filter by luns.lun_maps.igroup.initiators.comment • maxLength: 254 • minLength: 0
luns.lun_maps.igroup.initiators.name	string	query	False	Filter by luns.lun_maps.igroup.initiators.name
luns.os_type	string	query	False	Filter by luns.os_type
luns.name	string	query	False	Filter by luns.name
luns.uuid	string	query	False	Filter by luns.uuid
luns.qos.policy.max_throughput_iops	integer	query	False	Filter by luns.qos.policy.max_throughput_iops
luns.qos.policy.min_throughput_mbps	integer	query	False	Filter by luns.qos.policy.min_throughput_mbps
luns.qos.policy.min_throughput_iops	integer	query	False	Filter by luns.qos.policy.min_throughput_iops
luns.qos.policy.name	string	query	False	Filter by luns.qos.policy.name
luns.qos.policy.uuid	string	query	False	Filter by luns.qos.policy.uuid
luns.qos.policy.max_throughput_mbps	integer	query	False	Filter by luns.qos.policy.max_throughput_mbps
luns.comment	string	query	False	Filter by luns.comment • maxLength: 254 • minLength: 0

Name	Type	In	Required	Description
luns.create_time	string	query	False	Filter by luns.create_time
luns.enabled	boolean	query	False	Filter by luns.enabled
luns.serial_number	string	query	False	Filter by luns.serial_number • maxLength: 12 • minLength: 12
statistics.size	integer	query	False	Filter by statistics.size • Introduced in: 9.13
statistics.iops_raw.read	integer	query	False	Filter by statistics.iops_raw.read • Introduced in: 9.13
statistics.iops_raw.other	integer	query	False	Filter by statistics.iops_raw.other • Introduced in: 9.13
statistics.iops_raw.write	integer	query	False	Filter by statistics.iops_raw.write • Introduced in: 9.13
statistics.iops_raw.total	integer	query	False	Filter by statistics.iops_raw.total • Introduced in: 9.13

Name	Type	In	Required	Description
statistics.used_space	integer	query	False	Filter by statistics.used_space • Introduced in: 9.13
statistics.latency_raw.read	integer	query	False	Filter by statistics.latency_raw.read • Introduced in: 9.13
statistics.latency_raw.other	integer	query	False	Filter by statistics.latency_raw.other • Introduced in: 9.13
statistics.latency_raw.write	integer	query	False	Filter by statistics.latency_raw.write • Introduced in: 9.13
statistics.latency_raw.total	integer	query	False	Filter by statistics.latency_raw.total • Introduced in: 9.13
statistics.throughput_raw.read	integer	query	False	Filter by statistics.throughput_raw.read • Introduced in: 9.13
statistics.throughput_raw.other	integer	query	False	Filter by statistics.throughput_raw.other • Introduced in: 9.13

Name	Type	In	Required	Description
statistics.throughput_raw.write	integer	query	False	Filter by statistics.throughput_raw.write • Introduced in: 9.13
statistics.throughput_raw.total	integer	query	False	Filter by statistics.throughput_raw.total • Introduced in: 9.13
statistics.available_space	integer	query	False	Filter by statistics.available_space • Introduced in: 9.13
statistics.timestamp	string	query	False	Filter by statistics.timestamp • Introduced in: 9.13
statistics.status	string	query	False	Filter by statistics.status • Introduced in: 9.13
parent_consistency_group.uuid	string	query	False	Filter by parent_consistency_group.uuid
parent_consistency_group.name	string	query	False	Filter by parent_consistency_group.name
volumes.name	string	query	False	Filter by volumes.name • maxLength: 203 • minLength: 1

Name	Type	In	Required	Description
volumes.tiering.policy	string	query	False	Filter by volumes.tiering.policy
volumes.uuid	string	query	False	Filter by volumes.uuid
volumes.snapshot_policy.uuid	string	query	False	Filter by volumes.snapshot_policy.uuid
volumes.snapshot_policy.name	string	query	False	Filter by volumes.snapshot_policy.name
volumes.qos.policy.uuid	string	query	False	Filter by volumes.qos.policy.uuid
volumes.qos.policy.max_throughput_mbps	integer	query	False	Filter by volumes.qos.policy.max_throughput_mbps
volumes.qos.policy.min_throughput_mbps	integer	query	False	Filter by volumes.qos.policy.min_throughput_mbps
volumes.qos.policy.min_throughput_iops	integer	query	False	Filter by volumes.qos.policy.min_throughput_iops
volumes.qos.policy.name	string	query	False	Filter by volumes.qos.policy.name
volumes.qos.policy.max_throughput_iops	integer	query	False	Filter by volumes.qos.policy.max_throughput_iops

Name	Type	In	Required	Description
volumes.comment	string	query	False	Filter by volumes.comment • maxLength: 1023 • minLength: 0
volumes.nas.security_style	string	query	False	Filter by volumes.nas.security_style • Introduced in: 9.12
volumes.nas.path	string	query	False	Filter by volumes.nas.path • Introduced in: 9.12
volumes.nas.gid	integer	query	False	Filter by volumes.nas.gid • Introduced in: 9.12
volumes.nas.export_policy.uuid	string	query	False	Filter by volumes.nas.export_policy.uuid • Introduced in: 9.12
volumes.nas.export_policy.rules.rw_rule	string	query	False	Filter by volumes.nas.export_policy.rules.rw_rule • Introduced in: 9.12
volumes.nas.export_policy.rules.clients.match	string	query	False	Filter by volumes.nas.export_policy.rules.clients.match • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.export_policy.rules.allow_suid	boolean	query	False	Filter by volumes.nas.export_policy.rules.allow_suid • Introduced in: 9.12
volumes.nas.export_policy.rules.chown_mode	string	query	False	Filter by volumes.nas.export_policy.rules.chown_mode • Introduced in: 9.12
volumes.nas.export_policy.rules.allow_device_creation	boolean	query	False	Filter by volumes.nas.export_policy.rules.allow_device_creation • Introduced in: 9.12
volumes.nas.export_policy.rules.protocols	string	query	False	Filter by volumes.nas.export_policy.rules.protocols • Introduced in: 9.12
volumes.nas.export_policy.rules.ntfs_unix_security	string	query	False	Filter by volumes.nas.export_policy.rules.ntfs_unix_security • Introduced in: 9.12
volumes.nas.export_policy.rules.superuser	string	query	False	Filter by volumes.nas.export_policy.rules.superuser • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.export_policy.rules.anonymous_user	string	query	False	Filter by volumes.nas.export_policy.rules.anonymous_user • Introduced in: 9.12
volumes.nas.export_policy.rules.index	integer	query	False	Filter by volumes.nas.export_policy.rules.index • Introduced in: 9.12
volumes.nas.export_policy.rules.ro_rule	string	query	False	Filter by volumes.nas.export_policy.rules.ro_rule • Introduced in: 9.12
volumes.nas.export_policy.name	string	query	False	Filter by volumes.nas.export_policy.name • Introduced in: 9.12
volumes.nas.junction_parent.name	string	query	False	Filter by volumes.nas.junction_parent.name • Introduced in: 9.12
volumes.nas.junction_parent.uuid	string	query	False	Filter by volumes.nas.junction_parent.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.namespace_caching	boolean	query	False	Filter by volumes.nas.cifs.shares.namespace_caching • Introduced in: 9.12
volumes.nas.cifs.shares.encryption	boolean	query	False	Filter by volumes.nas.cifs.shares.encryption • Introduced in: 9.12
volumes.nas.cifs.shares.no_strict_security	boolean	query	False	Filter by volumes.nas.cifs.shares.no_strict_security • Introduced in: 9.12
volumes.nas.cifs.shares.vscan_profile	string	query	False	Filter by volumes.nas.cifs.shares.vscan_profile • Introduced in: 9.12
volumes.nas.cifs.shares.allow_unencrypted_access	boolean	query	False	Filter by volumes.nas.cifs.shares.allow_unencrypted_access • Introduced in: 9.12
volumes.nas.cifs.shares.file_umask	integer	query	False	Filter by volumes.nas.cifs.shares.file_umask • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.offline_files	string	query	False	Filter by volumes.nas.cifs.shares.offline_files • Introduced in: 9.12
volumes.nas.cifs.shares.home_directory	boolean	query	False	Filter by volumes.nas.cifs.shares.home_directory • Introduced in: 9.12
volumes.nas.cifs.shares.comment	string	query	False	Filter by volumes.nas.cifs.shares.comment • Introduced in: 9.12 • maxLength: 256 • minLength: 1
volumes.nas.cifs.shares.change_notify	boolean	query	False	Filter by volumes.nas.cifs.shares.change_notify • Introduced in: 9.12
volumes.nas.cifs.shares.oplocks	boolean	query	False	Filter by volumes.nas.cifs.shares.oplocks • Introduced in: 9.12
volumes.nas.cifs.shares.unix_symlink	string	query	False	Filter by volumes.nas.cifs.shares.unix_symlink • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.dir_umask	integer	query	False	Filter by volumes.nas.cifs.shares.dir_umask • Introduced in: 9.12
volumes.nas.cifs.shares.continuously_available	boolean	query	False	Filter by volumes.nas.cifs.shares.continuously_available • Introduced in: 9.12
volumes.nas.cifs.shares.access_based_enumeration	boolean	query	False	Filter by volumes.nas.cifs.shares.access_based_enumeration • Introduced in: 9.12
volumes.nas.cifs.shares.acls.permission	string	query	False	Filter by volumes.nas.cifs.shares.acls.permission • Introduced in: 9.12
volumes.nas.cifs.shares.acls.user_or_group	string	query	False	Filter by volumes.nas.cifs.shares.acls.user_or_group • Introduced in: 9.12
volumes.nas.cifs.shares.acls.type	string	query	False	Filter by volumes.nas.cifs.shares.acls.type • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.show_snapshot	boolean	query	False	Filter by volumes.nas.cifs.shares.show_snapshot • Introduced in: 9.12
volumes.nas.cifs.shares.name	string	query	False	Filter by volumes.nas.cifs.shares.name • Introduced in: 9.12 • maxLength: 80 • minLength: 1
volumes.nas.unix_permissions	integer	query	False	Filter by volumes.nas.unix_permissions • Introduced in: 9.12
volumes.nas.uid	integer	query	False	Filter by volumes.nas.uid • Introduced in: 9.12
volumes.analytics.scan_progress	integer	query	False	Filter by volumes.analytics.scan_progress • Introduced in: 9.13
volumes.analytics.unsupported_reason.code	string	query	False	Filter by volumes.analytics.unsupported_reason.code • Introduced in: 9.13

Name	Type	In	Required	Description
volumes.analytics.unsupported_reason.message	string	query	False	Filter by volumes.analytics.unsupported_reason.message • Introduced in: 9.13
volumes.analytics.supported	boolean	query	False	Filter by volumes.analytics.supported • Introduced in: 9.13
volumes.analytics.state	string	query	False	Filter by volumes.analytics.state • Introduced in: 9.13
volumes.space.available	integer	query	False	Filter by volumes.space.available
volumes.space.used	integer	query	False	Filter by volumes.space.used
volumes.space.size	integer	query	False	Filter by volumes.space.size
volumes.activity_tracking.unsupported_reason.code	string	query	False	Filter by volumes.activity_tracking.unsupported_reason.code • Introduced in: 9.13
volumes.activity_tracking.unsupported_reason.message	string	query	False	Filter by volumes.activity_tracking.unsupported_reason.message • Introduced in: 9.13

Name	Type	In	Required	Description
volumes.activity_tracking.state	string	query	False	Filter by volumes.activity_tracking.state • Introduced in: 9.13
volumes.activity_tracking.supported	boolean	query	False	Filter by volumes.activity_tracking.supported • Introduced in: 9.13
volumes.language	string	query	False	Filter by volumes.language
clone.split_initiated	boolean	query	False	Filter by clone.split_initiated • Introduced in: 9.12
clone.parent_snapshot.name	string	query	False	Filter by clone.parent_snapshot.name • Introduced in: 9.12
clone.guarantee.type	string	query	False	Filter by clone.guarantee.type • Introduced in: 9.12
clone.parent_consistency_group.uuid	string	query	False	Filter by clone.parent_consistency_group.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
clone.parent_consistency_group.name	string	query	False	Filter by clone.parent_consistency_group.name • Introduced in: 9.12
clone.volume.prefix	string	query	False	Filter by clone.volume.prefix • Introduced in: 9.12
clone.volume.suffix	string	query	False	Filter by clone.volume.suffix • Introduced in: 9.12
uuid	string	query	False	Filter by uuid
application.type	string	query	False	Filter by application.type • Introduced in: 9.12
application.component_type	string	query	False	Filter by application.component_type • Introduced in: 9.12
tiering.policy	string	query	False	Filter by tiering.policy
name	string	query	False	Filter by name
namespaces.os_type	string	query	False	Filter by namespaces.os_type • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.subsystem_map.subsystem.os_type	string	query	False	Filter by namespaces.subsystem_map.subsystem.os_type • Introduced in: 9.12
namespaces.subsystem_map.subsystem.name	string	query	False	Filter by namespaces.subsystem_map.subsystem.name • Introduced in: 9.12 • maxLength: 96 • minLength: 1
namespaces.subsystem_map.subsystem.comment	string	query	False	Filter by namespaces.subsystem_map.subsystem.comment • Introduced in: 9.12 • maxLength: 255 • minLength: 0
namespaces.subsystem_map.subsystem.hosts.nqn	string	query	False	Filter by namespaces.subsystem_map.subsystem.hosts.nqn • Introduced in: 9.12
namespaces.subsystem_map.subsystem.uuid	string	query	False	Filter by namespaces.subsystem_map.subsystem.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.subsystem_map.nsid	string	query	False	Filter by namespaces.subsystem_map.nsid • Introduced in: 9.12
namespaces.subsystem_map.anagrpid	string	query	False	Filter by namespaces.subsystem_map.anagrpid • Introduced in: 9.12
namespaces.space.used	integer	query	False	Filter by namespaces.space.used • Introduced in: 9.12
namespaces.space.guarantee.requested	boolean	query	False	Filter by namespaces.space.guarantee.requested • Introduced in: 9.12
namespaces.space.guarantee.reserved	boolean	query	False	Filter by namespaces.space.guarantee.reserved • Introduced in: 9.12
namespaces.space.size	integer	query	False	Filter by namespaces.space.size • Introduced in: 9.12 • Max value: 140737488355328 • Min value: 4096

Name	Type	In	Required	Description
namespaces.space.block_size	integer	query	False	Filter by namespaces.space.block_size • Introduced in: 9.12
namespaces.enabled	boolean	query	False	Filter by namespaces.enabled • Introduced in: 9.12
namespaces.status.mapped	boolean	query	False	Filter by namespaces.status.mapped • Introduced in: 9.12
namespaces.status.state	string	query	False	Filter by namespaces.status.state • Introduced in: 9.12
namespaces.status.read_only	boolean	query	False	Filter by namespaces.status.read_only • Introduced in: 9.12
namespaces.status.container_state	string	query	False	Filter by namespaces.status.container_state • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.comment	string	query	False	Filter by namespaces.comment • Introduced in: 9.12 • maxLength: 254 • minLength: 0
namespaces.create_time	string	query	False	Filter by namespaces.create_time • Introduced in: 9.12
namespaces.uuid	string	query	False	Filter by namespaces.uuid • Introduced in: 9.12
namespaces.name	string	query	False	Filter by namespaces.name • Introduced in: 9.12
namespaces.auto_delete	boolean	query	False	Filter by namespaces.auto_delete • Introduced in: 9.12
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Max value: 120 • Min value: 0 • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	collection_links	
num_records	integer	Number of records.
records	array[records]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "application": {
      "component_type": "data",
      "type": "oracle"
    },
    "clone": {
      "guarantee": {
        "type": "volume"
      },
      "parent_consistency_group": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "my_consistency_group",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "volume": {
        "prefix": "string"
      }
    },
    "consistency_groups": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "application": {
```

```
        "component_type": "data",
        "type": "oracle"
      },
      "luns": {
        "clone": {
          "source": {
            "name": "/vol/volume1/lun1",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          }
        },
        "comment": "string",
        "create_time": "2018-06-04 19:00:00 +0000",
        "lun_maps": {
          "igroup": {
            "comment": "string",
            "igroups": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "name": "igroup1",
              "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
            },
            "initiators": {
              "comment": "my comment",
              "name": "iqn.1998-01.com.corp.iscsi:name1"
            },
            "name": "igroup1",
            "os_type": "aix",
            "protocol": "fcp",
            "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
          }
        },
        "name": "/vol/volume1/lun1",
        "os_type": "aix",
        "provisioning_options": {
          "action": "create"
        },
        "qos": {
          "policy": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
```

```
            "max_throughput_iops": 10000,
            "max_throughput_mbps": 500,
            "min_throughput_iops": 2000,
            "min_throughput_mbps": 500,
            "name": "performance",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          }
        },
        "serial_number": "string",
        "space": {
          "size": 1073741824,
          "used": 0
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "namespaces": {
        "comment": "string",
        "create_time": "2018-06-04 19:00:00 +0000",
        "name": "/vol/volume1/qtree1/namespace1",
        "os_type": "aix",
        "provisioning_options": {
          "action": "create"
        },
        "space": {
          "block_size": 512,
          "size": 1073741824,
          "used": 0
        },
        "status": {
          "container_state": "online",
          "state": "online"
        },
        "subsystem_map": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "anagrpid": "00103050h",
          "nsid": "00000001h",
          "subsystem": {
            "comment": "string",
            "hosts": {
              "nqn": "nqn.1992-01.example.com:string"
            },
            "name": "subsystem1",
```

```
            "os_type": "aix",
            "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
          }
        },
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "parent_consistency_group": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "my_consistency_group",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "space": {
```

```
        "available": 5737418,
        "size": 1073741824,
        "used": 5737418
      },
      "svm": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "svm1",
        "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
      },
      "tiering": {
        "control": "allowed",
        "object_stores": {
        },
        "policy": "all"
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
      "volumes": {
        "activity_tracking": {
          "state": "off",
          "unsupported_reason": {
            "code": "124518405",
            "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
          }
        },
        "analytics": {
          "scan_progress": 17,
          "state": "unknown",
          "unsupported_reason": {
            "code": "111411207",
            "message": "File system analytics cannot be enabled on
volumes that contain LUNs."
          }
        },
        "comment": "string",
        "language": "ar",
        "name": "vol_cs_dept",
        "nas": {
          "cifs": {
            "shares": {
              "_links": {
                "self": {
```

```
                "href": "/api/resourcelink"
              }
            },
            "acls": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "permission": "no_access",
              "type": "windows",
              "user_or_group": "ENGDOMAIN\\ad_user"
            },
            "comment": "HR Department Share",
            "dir_umask": 18,
            "file_umask": 18,
            "name": "HR_SHARE",
            "offline_files": "none",
            "unix_symlink": "local",
            "vscan_profile": "no_scan"
          }
        },
        "export_policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "rules": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "chown_mode": "restricted",
            "clients": {
              "match": "0.0.0.0/0"
            },
            "index": 0,
            "ntfs_unix_security": "fail",
            "protocols": {
            },
            "ro_rule": {
            },
            "rw_rule": {
            },
```

```
            "superuser": {
            }
          },
          "uuid": "string"
        },
        "junction_parent": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "vs1_root",
          "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
        },
        "path": "/user/my_volume",
        "security_style": "mixed",
        "unix_permissions": 493
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
```

```
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        },
        "space": {
          "available": 0,
          "used": 0
        },
        "tiering": {
          "control": "allowed",
          "object_stores": {
          },
          "policy": "all"
        },
        "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
      }
    },
    "luns": {
      "clone": {
        "source": {
          "name": "/vol/volume1/lun1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "lun_maps": {
        "igroup": {
          "comment": "string",
          "igroups": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "igroup1",
            "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
          },
          "initiators": {
            "comment": "my comment",
            "name": "iqn.1998-01.com.corp.iscsi:name1"
          },
          "name": "igroup1",
          "os_type": "aix",
          "protocol": "fcp",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        }
      },
```

```
      "name": "/vol/volume1/lun1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "serial_number": "string",
      "space": {
        "size": 1073741824,
        "used": 0
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "metric": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "available_space": 4096,
      "duration": "PT15S",
      "iops": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "latency": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "size": 4096,
```

```
      "status": "ok",
      "throughput": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "timestamp": "2017-01-25 11:20:13 +0000",
      "used_space": 4096
    },
    "namespaces": {
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "name": "/vol/volume1/qtree1/namespace1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "space": {
        "block_size": 512,
        "size": 1073741824,
        "used": 0
      },
      "status": {
        "container_state": "online",
        "state": "online"
      },
      "subsystem_map": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "anagrpid": "00103050h",
        "nsid": "00000001h",
        "subsystem": {
          "comment": "string",
          "hosts": {
            "nqn": "nqn.1992-01.example.com:string"
          },
          "name": "subsystem1",
          "os_type": "aix",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
```

```
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "replication_relationships": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
```

```
    },
    "space": {
      "available": 5737418,
      "size": 1073741824,
      "used": 5737418
    },
    "statistics": {
      "available_space": 4096,
      "iops_raw": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "latency_raw": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "size": 4096,
      "status": "ok",
      "throughput_raw": {
        "read": 200,
        "total": 1000,
        "write": 100
      },
      "timestamp": "2017-01-25 11:20:13 +0000",
      "used_space": 4096
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "volumes": {
      "activity_tracking": {
```

```
        "state": "off",
        "unsupported_reason": {
          "code": "124518405",
          "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
        }
      },
      "analytics": {
        "scan_progress": 17,
        "state": "unknown",
        "unsupported_reason": {
          "code": "111411207",
          "message": "File system analytics cannot be enabled on
volumes that contain LUNs."
        }
      },
      "comment": "string",
      "language": "ar",
      "name": "vol_cs_dept",
      "nas": {
        "cifs": {
          "shares": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "acls": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "permission": "no_access",
              "type": "windows",
              "user_or_group": "ENGDOMAIN\\ad_user"
            },
            "comment": "HR Department Share",
            "dir_umask": 18,
            "file_umask": 18,
            "name": "HR_SHARE",
            "offline_files": "none",
            "unix_symlink": "local",
            "vscan_profile": "no_scan"
          }
        },
```

```
        "export_policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "rules": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "chown_mode": "restricted",
            "clients": {
              "match": "0.0.0.0/0"
            },
            "index": 0,
            "ntfs_unix_security": "fail",
            "protocols": {
            },
            "ro_rule": {
            },
            "rw_rule": {
            },
            "superuser": {
            }
          },
          "uuid": "string"
        },
        "junction_parent": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "vs1_root",
          "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
        },
        "path": "/user/my_volume",
        "security_style": "mixed",
        "unix_permissions": 493
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
```

```
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "space": {
        "available": 0,
        "used": 0
      },
      "tiering": {
        "control": "allowed",
        "object_stores": {
        },
        "policy": "all"
      },
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

collection_links

Name	Type	Description
next	href	
self	href	

self_link

Name	Type	Description
self	href	

application

Name	Type	Description
component_type	string	Nested consistency group tag.
type	string	Top level consistency group tag.

guarantee

Name	Type	Description
type	string	The type of space guarantee of this volume in the aggregate.

parent_consistency_group

Consistency group that is to be cloned.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

parent_snapshot

Name	Type	Description
name	string	Name of an existing Snapshot copy of a parent consistency group.

volume

Volume name suffix/prefix for the cloned volumes.

Name	Type	Description
prefix	string	Volume name prefix for cloned volumes.
suffix	string	Volume name suffix for cloned volumes.

clone

Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
guarantee	guarantee	
parent_consistency_group	parent_consistency_group	Consistency group that is to be cloned.
parent_snapshot	parent_snapshot	
split_initiated	boolean	Splits volumes after cloning. Default is false.
volume	volume	Volume name suffix/prefix for the cloned volumes.

source

The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN.

Valid in POST to create a new LUN as a clone of the source.

Valid in PATCH to overwrite an existing LUN's data as a clone of another.

Name	Type	Description
name	string	The fully qualified path name of the clone source LUN composed of a "/vol" prefix, the volume name, the (optional) qtree name, and base name of the LUN. Valid in POST and PATCH.
uuid	string	The unique identifier of the clone source LUN. Valid in POST and PATCH.

clone

This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`.

When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`.

Persistent reservations for the patched LUN are also preserved.

Name	Type	Description
source	source	The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN. Valid in POST to create a new LUN as a clone of the source. Valid in PATCH to overwrite an existing LUN's data as a clone of another.

igroups

Name	Type	Description
_links	self_link	
name	string	The name of the initiator group.

Name	Type	Description
uuid	string	The unique identifier of the initiator group.

initiators

The initiators that are members of the initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator.
name	string	Name of initiator that is a member of the initiator group.

igroup

The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator. Valid in POST and PATCH.
igroups	array[igroups]	Separate igroup definitions to include in this igroup.
initiators	array[initiators]	The initiators that are members of the group.
name	string	The name of the initiator group. Required in POST; optional in PATCH.
os_type	string	The host operating system of the initiator group. All initiators in the group should be hosts of the same operating system. Required in POST; optional in PATCH.

Name	Type	Description
protocol	string	The protocols supported by the initiator group. This restricts the type of initiators that can be added to the initiator group. Optional in POST; if not supplied, this defaults to *mixed*. The protocol of an initiator group cannot be changed after creation of the group.
uuid	string	The unique identifier of the initiator group.

lun_maps

A LUN map is an association between a LUN and an initiator group.

When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.

Name	Type	Description
igroup	igroup	The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.
logical_unit_number	integer	The logical unit number assigned to the LUN when mapped to the specified initiator group. The number is used to identify the LUN to initiators in the initiator group when communicating through the Fibre Channel Protocol or iSCSI. Optional in POST; if no value is provided, ONTAP assigns the lowest available value. • Introduced in: 9.6 • readCreate: 1 • x-nullable: true

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

policy

The QoS policy

Name	Type	Description
_links	self_link	
max_throughput_iops	integer	Specifies the maximum throughput in IOPS, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
max_throughput_mbps	integer	Specifies the maximum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_iops	integer	Specifies the minimum throughput in IOPS, 0 means none. Setting "min_throughput" is supported on AFF platforms only, unless FabricPool tiering policies are set. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_mbps	integer	Specifies the minimum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
name	string	The QoS policy group name. This is mutually exclusive with UUID and other QoS attributes during POST and PATCH.

Name	Type	Description
uuid	string	The QoS policy group UUID. This is mutually exclusive with name and other QoS attributes during POST and PATCH.

qos

Name	Type	Description
policy	policy	The QoS policy

guarantee

Properties that request and report the space guarantee for the LUN.

Name	Type	Description
requested	boolean	The requested space reservation policy for the LUN. If *true*, a space reservation is requested for the LUN; if *false*, the LUN is thin provisioned. Guaranteeing a space reservation request for a LUN requires that the volume in which the LUN resides is also space reserved and that the fractional reserve for the volume is 100%. Valid in POST and PATCH.
reserved	boolean	Reports if the LUN is space guaranteed. If *true*, a space guarantee is requested and the containing volume and aggregate support the request. If *false*, a space guarantee is not requested or a space guarantee is requested and either the containing volume or aggregate do not support the request.

space

The storage space related properties of the LUN.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the LUN.
size	integer	The total provisioned size of the LUN. The LUN size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The actual minimum and maxiumum sizes vary depending on the ONTAP version, ONTAP platform, and the available space in the containing volume and aggregate. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the LUN. This value is the total space consumed in the volume by the LUN, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways SAN filesystems and applications utilize blocks within a LUN, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the LUN blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

luns

A LUN is the logical representation of storage in a storage area network (SAN).

In ONTAP, a LUN is located within a volume. Optionally, it can be located within a qtree in a volume.

A LUN can be created to a specified size using thin or thick provisioning. A LUN can then be renamed, resized, cloned, and moved to a different volume. LUNs support the assignment of a quality of service (QoS) policy for performance management or a QoS policy can be assigned to the volume containing the LUN. See the LUN object model to learn more about each of the properties supported by the LUN REST API.

A LUN must be mapped to an initiator group to grant access to the initiator group's initiators (client hosts). Initiators can then access the LUN and perform I/O over a Fibre Channel (FC) fabric using the Fibre Channel Protocol or a TCP/IP network using iSCSI.

Name	Type	Description
clone	clone	This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`. When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`. Persistent reservations for the patched LUN are also preserved.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the LUN was created.

Name	Type	Description
enabled	boolean	The enabled state of the LUN. LUNs can be disabled to prevent access to the LUN. Certain error conditions also cause the LUN to become disabled. If the LUN is disabled, you can consult the `state` property to determine if the LUN is administratively disabled (*offline*) or has become disabled as a result of an error. A LUN in an error condition can be brought online by setting the `enabled` property to *true* or brought administratively offline by setting the `enabled` property to *false*. Upon creation, a LUN is enabled by default. Valid in PATCH.
lun_maps	array[lun_maps]	An array of LUN maps. A LUN map is an association between a LUN and an initiator group. When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.
name	string	The fully qualified path name of the LUN composed of the "/vol" prefix, the volume name, the qtree name (optional), and the base name of the LUN. Valid in POST and PATCH.
os_type	string	The operating system type of the LUN. Required in POST when creating a LUN that is not a clone of another. Disallowed in POST when creating a LUN clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
serial_number	string	The LUN serial number. The serial number is generated by ONTAP when the LUN is created. • maxLength: 12 • minLength: 12 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
space	space	The storage space related properties of the LUN.
uuid	string	The unique identifier of the LUN. The UUID is generated by ONTAP when the LUN is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

guarantee

Properties that request and report the space guarantee for the NVMe namespace.

Name	Type	Description
requested	boolean	The requested space reservation policy for the NVMe namespace. If *true*, a space reservation is requested for the namespace; if *false*, the namespace is thin provisioned. Guaranteeing a space reservation request for a namespace requires that the volume in which the namespace resides also be space reserved and that the fractional reserve for the volume be 100%. The space reservation policy for an NVMe namespace is determined by ONTAP. • Introduced in: 9.6 • x-nullable: true
reserved	boolean	Reports if the NVMe namespace is space guaranteed. This property is *true* if a space guarantee is requested and the containing volume and aggregate support the request. This property is *false* if a space guarantee is not requested or if a space guarantee is requested and either the containing volume and aggregate do not support the request.

space

The storage space related properties of the NVMe namespace.

Name	Type	Description
block_size	integer	The size of blocks in the namespace, in bytes. Valid in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone. Valid in POST.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the NVMe namespace.
size	integer	The total provisioned size of the NVMe namespace. Valid in POST and PATCH. The NVMe namespace size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The maximum size is variable with respect to large NVMe namespace support in ONTAP. If large namespaces are supported, the maximum size is 128 TB (140737488355328 bytes) and if not supported, the maximum size is just under 16 TB (17557557870592 bytes). The minimum size supported is always 4096 bytes. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the NVMe namespace. This value is the total space consumed in the volume by the NVMe namespace, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways NVMe filesystems and applications utilize blocks within a namespace, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the namespace blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

status

Status information about the NVMe namespace.

Name	Type	Description
container_state	string	The state of the volume and aggregate that contain the NVMe namespace. Namespaces are only available when their containers are available.

Name	Type	Description
mapped	boolean	Reports if the NVMe namespace is mapped to an NVMe subsystem. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
read_only	boolean	Reports if the NVMe namespace allows only read access.
state	string	The state of the NVMe namespace. Normal states for a namespace are *online* and *offline*. Other states indicate errors.

consistency_group_nvme_host

The NVMe host provisioned to access NVMe namespaces mapped to a subsystem.

Name	Type	Description
nqn	string	The NVMe qualified name (NQN) used to identify the NVMe storage target. Not allowed in POST when the `records` property is used.

consistency_group_nvme_subsystem

An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

Name	Type	Description
comment	string	A configurable comment for the NVMe subsystem. Optional in POST and PATCH.
hosts	array[consistency_group_nvme_host]	The NVMe hosts configured for access to the NVMe subsystem. Optional in POST.

Name	Type	Description
name	string	The name of the NVMe subsystem. Once created, an NVMe subsystem cannot be renamed. Required in POST.
os_type	string	The host operating system of the NVMe subsystem's hosts. Required in POST.
uuid	string	The unique identifier of the NVMe subsystem.

subsystem_map

The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems.

There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.

Name	Type	Description
_links	self_link	
anagrpid	string	The Asymmetric Namespace Access Group ID (ANAGRPID) of the NVMe namespace. The format for an ANAGRPID is 8 hexadecimal digits (zero-filled) followed by a lower case "h". There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.

Name	Type	Description
nsid	string	The NVMe namespace identifier. This is an identifier used by an NVMe controller to provide access to the NVMe namespace. The format for an NVMe namespace identifier is 8 hexadecimal digits (zero-filled) followed by a lower case "h".
subsystem	consistency_group_nvme_subsystem	An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

namespaces

An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the storage virtual machine using the NVMe over Fabrics protocol.

In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume.

An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API.

An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem’s hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol.

Name	Type	Description
auto_delete	boolean	This property marks the NVMe namespace for auto deletion when the volume containing the namespace runs out of space. This is most commonly set on namespace clones. When set to *true*, the NVMe namespace becomes eligible for automatic deletion when the volume runs out of space. Auto deletion only occurs when the volume containing the namespace is also configured for auto deletion and free space in the volume decreases below a particular threshold. This property is optional in POST and PATCH. The default value for a new NVMe namespace is *false*. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the NVMe namespace was created.
enabled	boolean	The enabled state of the NVMe namespace. Certain error conditions cause the namespace to become disabled. If the namespace is disabled, you can check the `state` property to determine what error disabled the namespace. An NVMe namespace is enabled automatically when it is created.

Name	Type	Description
name	string	The fully qualified path name of the NVMe namespace composed of a "/vol" prefix, the volume name, the (optional) qtree name and base name of the namespace. Valid in POST. NVMe namespaces do not support rename, or movement between volumes.
os_type	string	The operating system type of the NVMe namespace. Required in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
space	space	The storage space related properties of the NVMe namespace.
status	status	Status information about the NVMe namespace.
subsystem_map	subsystem_map	The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems. There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.
uuid	string	The unique identifier of the NVMe namespace.

parent_consistency_group

The parent consistency group.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

storage_service

Determines the placement of any storage object created during this operation.

Name	Type	Description
name	string	Storage service name. If not specified, the default value is the most performant for the platform.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
name	string	New name for consistency group. Required to resolve naming collisions.
storage_service	storage_service	Determines the placement of any storage object created during this operation.

snapshot

A consistency group's Snapshot copy

Name	Type	Description
name	string	The name of the consistency group's Snapshot copy to restore to.

Name	Type	Description
uuid	string	The UUID of the consistency group's Snapshot copy to restore to.

restore_to

Use to restore a consistency group to a previous Snapshot copy

Name	Type	Description
snapshot	snapshot	A consistency group's Snapshot copy

_links

Name	Type	Description
self	href	

snapshot_policy_reference

This is a reference to the Snapshot copy policy.

Name	Type	Description
_links	_links	
name	string	
uuid	string	

space

Space information for the consistency group.

Name	Type	Description
available	integer	The amount of space available in the consistency group, in bytes.
size	integer	The total provisioned size of the consistency group, in bytes.
used	integer	The amount of space consumed in the consistency group, in bytes.

svm

The Storage Virtual Machine (SVM) in which the consistency group is located.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

object_stores

Name	Type	Description
name	string	The name of the object store to use. Used for placement.

tiering

The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

unsupported_reason

Name	Type	Description
code	string	If volume activity tracking is not supported on the volume, this field provides an appropriate error code.

Name	Type	Description
message	string	If volume activity tracking is not supported on the volume, this field provides an error message detailing why this is the case.

activity_tracking

The volume activity tracking configuration for this volume.

Name	Type	Description
state	string	Activity tracking state of the volume. If this value is *on*, ONTAP collects top metrics information for the volume in real time. There is a slight impact to I/O performance in order to collect this information. If this value is *off*, no activity tracking information is collected or available to view. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["off", "on"] • Introduced in: 9.10 • x-nullable: true
supported	boolean	This field indicates whether or not volume activity tracking is supported on the volume. If volume activity tracking is not supported, the reason why is provided in the `activity_tracking.unsupported_reason` field.
unsupported_reason	unsupported_reason	

unsupported_reason

Name	Type	Description
code	string	If file system analytics is not supported on the volume, this field provides the error code explaining why.
message	string	If file system analytics is not supported on the volume, this field provides the error message explaining why.

analytics

The file system analytics configuration for this volume.

Name	Type	Description
scan_progress	integer	Percentage of files in the volume that the file system analytics initialization scan has processed. Only returned when the `state` is *initializing*.

Name	Type	Description
state	string	File system analytics state of the volume. If this value is *on*, ONTAP collects extra file system analytics information for all directories on the volume. There will be a slight impact to I/O performance to collect this information. If this value is *off*, file system analytics information is not collected and not available to be viewed. If this value is *initializing*, that means file system analytics was recently turned on, and the initialization scan to gather information for all existing files and directories is currently running. If this value is *initialization_paused*, this means that the initialization scan is currently paused. If this value is 'unknown', this means that there was an internal error when determining the file system analytics state for the volume. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["unknown", "initializing", "initialization_paused", "off", "on"] • Introduced in: 9.8 • x-nullable: true
supported	boolean	This field indicates whether or not file system analytics is supported on the volume. If file system analytics is not supported, the reason will be specified in the `analytics.unsupported_reason` field.
unsupported_reason	unsupported_reason	

acls

The permissions that users and groups have on a CIFS share.

Name	Type	Description
_links	_links	
permission	string	Specifies the access rights that a user or group has on the defined CIFS Share. The following values are allowed: • no_access - User does not have CIFS share access • read - User has only read access • change - User has change access • full_control - User has full_control access
type	string	Specifies the type of the user or group to add to the access control list of a CIFS share. The following values are allowed: • windows - Windows user or group • unix_user - UNIX user • unix_group - UNIX group
user_or_group	string	Specifies the user or group name to add to the access control list of a CIFS share.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.

Name	Type	Description
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

cifs

Name	Type	Description
shares	array[consistency_group_cifs_share]	

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

export_rules

Name	Type	Description
_links	_links	
allow_device_creation	boolean	Specifies whether or not device creation is allowed.
allow_suid	boolean	Specifies whether or not SetUID bits in SETATTR Op is to be honored.
anonymous_user	string	User ID To Which Anonymous Users Are Mapped.

Name	Type	Description
chown_mode	string	Specifies who is authorized to change the ownership mode of a file.
clients	array[export_clients]	Array of client matches
index	integer	Index of the rule within the export policy.
ntfs_unix_security	string	NTFS export UNIX security options.
protocols	array[string]	
ro_rule	array[string]	Authentication flavors that the read-only access rule governs
rw_rule	array[string]	Authentication flavors that the read/write access rule governs
superuser	array[string]	Authentication flavors that the superuser security type governs

export_policy

The policy associated with volumes to export them for protocol access.

Name	Type	Description
_links	self_link	
name	string	Name of the export policy.
rules	array[export_rules]	The set of rules that govern the export policy.
uuid	string	Identifier for the export policy.

junction_parent

Name	Type	Description
_links	self_link	

Name	Type	Description
name	string	The name of the parent volume that contains the junction inode of this volume. The junction parent volume must belong to the same SVM that owns this volume.
uuid	string	Unique identifier for the parent volume.

nas

The CIFS share policy and/or export policies for this volume.

Name	Type	Description
cifs	cifs	
export_policy	export_policy	The policy associated with volumes to export them for protocol access.
gid	integer	The UNIX group ID of the volume. Valid in POST or PATCH.
junction_parent	junction_parent	
path	string	The fully-qualified path in the owning SVM's namespace at which the volume is mounted. The path is case insensitive and must be unique within an SVM's namespace. Path must begin with '/' and must not end with '/'. Only one volume can be mounted at any given junction path. An empty path in POST creates an unmounted volume. An empty path in PATCH deactivates and unmounts the volume. Taking a volume offline or restricted state removes its junction path. This attribute is reported in GET only when the volume is mounted.

Name	Type	Description
security_style	string	Security style associated with the volume. Valid in POST or PATCH. mixed ‐ Mixed-style security ntfs ‐ NTFS/WIndows-style security unified ‐ Unified-style security, unified UNIX, NFS and CIFS permissions unix ‐ UNIX-style security.
uid	integer	The UNIX user ID of the volume. Valid in POST or PATCH.
unix_permissions	integer	UNIX permissions to be viewed as an octal number, consisting of 4 digits derived by adding up bits 4 (read), 2 (write), and 1 (execute). First digit selects the set user ID (4), set group ID (2), and sticky (1) attributes. Second digit selects permission for the owner of the file. Third selects permissions for other users in the same group while the fourth selects permissions for other users not in the group. Valid in POST or PATCH. For security style "mixed" or "unix", the default setting is 0755 in octal (493 in decimal) and for security style "ntfs", the default setting is 0000. In cases where only owner, group, and other permissions are given (as in 755, representing the second, third and fourth digit), the first digit is assumed to be zero.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

Name	Type	Description
storage_service	storage_service	Determines the placement of any storage object created during this operation.

qos

The QoS policy for this volume.

Name	Type	Description
policy	policy	The QoS policy

space

Name	Type	Description
available	integer	The available space, in bytes.
size	integer	Total provisioned size, in bytes.
used	integer	The virtual space used (includes volume reserves) before storage efficiency, in bytes.

tiering

The tiering placement and policy definitions for this volume.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

volumes

Name	Type	Description
activity_tracking	activity_tracking	The volume activity tracking configuration for this volume.
analytics	analytics	The file system analytics configuration for this volume.

Name	Type	Description
comment	string	A comment for the volume. Valid in POST or PATCH.
language	string	Language encoding setting for volume. If no language is specified, the volume inherits its SVM language encoding setting.
name	string	Volume name. The name of volume must start with an alphabetic character (a to z or A to Z) or an underscore (_). The name must be 197 or fewer characters in length for FlexGroups, and 203 or fewer characters in length for all other types of volumes. Volume names must be unique within an SVM. Required on POST.
nas	nas	The CIFS share policy and/or export policies for this volume.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	The QoS policy for this volume.
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	
tiering	tiering	The tiering placement and policy definitions for this volume.
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • readOnly: 1 • Introduced in: 9.8 • x-nullable: true

consistency_groups

Name	Type	Description
_links	self_link	
application	application	
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

Name	Type	Description
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

iops

The rate of I/O operations observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency

The round trip latency observed at the storage object, in microseconds.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput

The rate of throughput bytes per second observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

metric

Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.

Name	Type	Description
_links	_links	
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
duration	string	The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations:
iops	iops	The rate of I/O operations observed at the storage object.
latency	latency	The round trip latency observed at the storage object, in microseconds.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput	throughput	The rate of throughput bytes per second observed at the storage object.
timestamp	string	The timestamp of the performance and capacity data.

Name	Type	Description
used_space	integer	The total space used in the consistency group, in bytes.

replication_relationships

Name	Type	Description
_links	self_link	
is_source	boolean	Indicates whether or not this consistency group is the source for replication.
uuid	string	The unique identifier of the SnapMirror relationship.

iops_raw

The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency_raw

The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput_raw

Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

statistics

These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.

Name	Type	Description
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
iops_raw	iops_raw	The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.
latency_raw	latency_raw	The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput_raw	throughput_raw	Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.
timestamp	string	The timestamp of the performance data.

Name	Type	Description
used_space	integer	The total used space in the consistency group, in bytes.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.

Name	Type	Description
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true

Name	Type	Description
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.

Name	Type	Description
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

records

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code

Name	Type	Description
message	string	Error message
target	string	The target parameter that caused the error.

Create a consistency group

POST `/application/consistency-groups`

Introduced In: 9.10

Creates a consistency group with one or more consistency groups having:

- new SAN volumes,
- existing SAN, NVMe or NAS FlexVol volumes in a new or existing consistency group

Required properties

- `svm.uuid` or `svm.name` - Existing SVM in which to create the group.
- `volumes`, `luns` or `namespaces`

Naming Conventions

Consistency groups

- name or consistency_groups[].name, if specified
- derived from volumes[0].name, if only one volume is specified, same as volume name

Volume

- volumes[].name, if specified
- derived from volume prefix in luns[].name
- derived from cg[].name, suffixed by "_#" where "#" is a system generated unique number
- suffixed by "_#" where "#" is a system generated unique number, if provisioning_options.count is provided

LUN

- luns[].name, if specified
- derived from volumes[].name, suffixed by "_#" where "#" is a system generated unique number
- suffixed by "_#" where "#" is a system generated unique number, if provisioning_options.count is provided

NVMe Namespace

- namespaces[].name, if specified

- derived from volumes[].name, suffixed by "_#" where "#" is a system generated unique number
- suffixed by "_#" where "#" is a system generated unique number, if provisioning_options.count is provided

Related ONTAP commands

There are no ONTAP commands for managing consistency group.

Parameters

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "application": {
    "component_type": "data",
    "type": "oracle"
  },
  "clone": {
    "guarantee": {
      "type": "volume"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "volume": {
      "prefix": "string"
    }
  },
  "consistency_groups": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "application": {
      "component_type": "data",
      "type": "oracle"
    },
    "luns": {
      "clone": {
        "source": {
          "name": "/vol/volume1/lun1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
```

```
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "lun_maps": {
        "igroup": {
          "comment": "string",
          "igroups": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "igroup1",
            "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
          },
          "initiators": {
            "comment": "my comment",
            "name": "iqn.1998-01.com.corp.iscsi:name1"
          },
          "name": "igroup1",
          "os_type": "aix",
          "protocol": "fcp",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "name": "/vol/volume1/lun1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "serial_number": "string",
      "space": {
```

```
        "size": 1073741824,
        "used": 0
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "namespaces": {
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "name": "/vol/volume1/qtree1/namespace1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "space": {
        "block_size": 512,
        "size": 1073741824,
        "used": 0
      },
      "status": {
        "container_state": "online",
        "state": "online"
      },
      "subsystem_map": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "anagrpid": "00103050h",
        "nsid": "00000001h",
        "subsystem": {
          "comment": "string",
          "hosts": {
            "nqn": "nqn.1992-01.example.com:string"
          },
          "name": "subsystem1",
          "os_type": "aix",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
```

```
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 5737418,
      "size": 1073741824,
      "used": 5737418
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
```

```
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "volumes": {
      "activity_tracking": {
        "state": "off",
        "unsupported_reason": {
          "code": "124518405",
          "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
        }
      },
      "analytics": {
        "scan_progress": 17,
        "state": "unknown",
        "unsupported_reason": {
          "code": "111411207",
          "message": "File system analytics cannot be enabled on
volumes that contain LUNs."
        }
      },
      "comment": "string",
      "language": "ar",
      "name": "vol_cs_dept",
      "nas": {
        "cifs": {
          "shares": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "acls": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "permission": "no_access",
```

```
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
          },
          "name": "vs1_root",
          "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
        },
        "path": "/user/my_volume",
        "security_style": "mixed",
        "unix_permissions": 493
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "space": {
        "available": 0,
        "used": 0
      },
      "tiering": {
        "control": "allowed",
        "object_stores": {
        },
```

```
        "policy": "all"
      },
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  },
  "luns": {
    "clone": {
      "source": {
        "name": "/vol/volume1/lun1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "lun_maps": {
      "igroup": {
        "comment": "string",
        "igroups": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "igroup1",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        },
        "initiators": {
          "comment": "my comment",
          "name": "iqn.1998-01.com.corp.iscsi:name1"
        },
        "name": "igroup1",
        "os_type": "aix",
        "protocol": "fcp",
        "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "name": "/vol/volume1/lun1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
```

```
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "serial_number": "string",
    "space": {
      "size": 1073741824,
      "used": 0
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "available_space": 4096,
    "duration": "PT15S",
    "iops": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "latency": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "namespaces": {
```

```
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "name": "/vol/volume1/qtree1/namespace1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "space": {
      "block_size": 512,
      "size": 1073741824,
      "used": 0
    },
    "status": {
      "container_state": "online",
      "state": "online"
    },
    "subsystem_map": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "anagrpid": "00103050h",
      "nsid": "00000001h",
      "subsystem": {
        "comment": "string",
        "hosts": {
          "nqn": "nqn.1992-01.example.com:string"
        },
        "name": "subsystem1",
        "os_type": "aix",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "parent_consistency_group": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "my_consistency_group",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "provisioning_options": {
```

```
    "action": "create",
    "storage_service": {
      "name": "extreme"
    }
  },
  "qos": {
    "policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "max_throughput_iops": 10000,
      "max_throughput_mbps": 500,
      "min_throughput_iops": 2000,
      "min_throughput_mbps": 500,
      "name": "performance",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "replication_relationships": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "snapshot_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "space": {
    "available": 5737418,
    "size": 1073741824,
    "used": 5737418
  },
  "statistics": {
    "available_space": 4096,
    "iops_raw": {
      "read": 200,
```

```
      "total": 1000,
      "write": 100
    },
    "latency_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "tiering": {
    "control": "allowed",
    "object_stores": {
    },
    "policy": "all"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "volumes": {
    "activity_tracking": {
      "state": "off",
      "unsupported_reason": {
        "code": "124518405",
        "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
      }
    },
    "analytics": {
      "scan_progress": 17,
      "state": "unknown",
```

```
      "unsupported_reason": {
        "code": "111411207",
        "message": "File system analytics cannot be enabled on volumes
that contain LUNs."
      }
    },
    "comment": "string",
    "language": "ar",
    "name": "vol_cs_dept",
    "nas": {
      "cifs": {
        "shares": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "acls": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "permission": "no_access",
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
```

```
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "vs1_root",
        "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
      },
      "path": "/user/my_volume",
      "security_style": "mixed",
      "unix_permissions": 493
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
```

```
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 0,
      "used": 0
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
  }
}
```


Response

```
Status: 202, Accepted
```

Name	Type	Description
job	job_link	

Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Response

```
Status: 201, Created
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
53411842	Consistency group does not exist.
53411843	A consistency group with specified UUID was not found.
53411844	Specified consistency group was not found in the specified SVM.
53411845	The specified UUID and name refer to different consistency groups.
53411846	Either name or UUID must be provided.
53411853	Fields provided in the request conflict with each other.

Error Code	Description
53411856	Field provided is only supported when provisioning new objects.
53411857	LUNs that are not members of the application are not supported by this API. LUNs can be added to an application by adding the volume containing the LUNs to the application.
53411860	An object with the same identifier in the same scope exists.
53411861	Volume specified does not exist in provided volume array.
53411862	Modifying existing igroups is not supported using this API.
53411864	Request content insufficient to add an existing volume to an application.
53411865	Volumes contained in one consistency group can not be added to a different consistency group.
53411866	LUNs are not supported on FlexGroups volumes.
53411867	LUN name is too long after appending a unique suffix.
53411869	Volume name is too long after appending a unique suffix.
53411870	When using the "round_robin" layout, the volume count must not be greater than the LUN count.
53411959	Volumes with Snapshot copy locking enabled cannot be added to a consistency group.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

self_link

Name	Type	Description
self	href	

application

Name	Type	Description
component_type	string	Nested consistency group tag.
type	string	Top level consistency group tag.

guarantee

Name	Type	Description
type	string	The type of space guarantee of this volume in the aggregate.

parent_consistency_group

Consistency group that is to be cloned.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

parent_snapshot

Name	Type	Description
name	string	Name of an existing Snapshot copy of a parent consistency group.

volume

Volume name suffix/prefix for the cloned volumes.

Name	Type	Description
prefix	string	Volume name prefix for cloned volumes.
suffix	string	Volume name suffix for cloned volumes.

clone

Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
guarantee	guarantee	
parent_consistency_group	parent_consistency_group	Consistency group that is to be cloned.
parent_snapshot	parent_snapshot	
split_initiated	boolean	Splits volumes after cloning. Default is false.
volume	volume	Volume name suffix/prefix for the cloned volumes.

source

The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN.

Valid in POST to create a new LUN as a clone of the source.

Valid in PATCH to overwrite an existing LUN's data as a clone of another.

Name	Type	Description
name	string	The fully qualified path name of the clone source LUN composed of a "/vol" prefix, the volume name, the (optional) qtree name, and base name of the LUN. Valid in POST and PATCH.

Name	Type	Description
uuid	string	The unique identifier of the clone source LUN. Valid in POST and PATCH.

clone

This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`.

When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`.

Persistent reservations for the patched LUN are also preserved.

Name	Type	Description
source	source	The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN. Valid in POST to create a new LUN as a clone of the source. Valid in PATCH to overwrite an existing LUN's data as a clone of another.

igroups

Name	Type	Description
_links	self_link	
name	string	The name of the initiator group.
uuid	string	The unique identifier of the initiator group.

initiators

The initiators that are members of the initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator.
name	string	Name of initiator that is a member of the initiator group.

igroup

The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator. Valid in POST and PATCH.
igroups	array[igroups]	Separate igroup definitions to include in this igroup.
initiators	array[initiators]	The initiators that are members of the group.
name	string	The name of the initiator group. Required in POST; optional in PATCH.
os_type	string	The host operating system of the initiator group. All initiators in the group should be hosts of the same operating system. Required in POST; optional in PATCH.
protocol	string	The protocols supported by the initiator group. This restricts the type of initiators that can be added to the initiator group. Optional in POST; if not supplied, this defaults to *mixed*. The protocol of an initiator group cannot be changed after creation of the group.
uuid	string	The unique identifier of the initiator group.

lun_maps

A LUN map is an association between a LUN and an initiator group.

When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.

Name	Type	Description
igroup	igroup	The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.
logical_unit_number	integer	The logical unit number assigned to the LUN when mapped to the specified initiator group. The number is used to identify the LUN to initiators in the initiator group when communicating through the Fibre Channel Protocol or iSCSI. Optional in POST; if no value is provided, ONTAP assigns the lowest available value. • Introduced in: 9.6 • readCreate: 1 • x-nullable: true

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

policy

The QoS policy

Name	Type	Description
_links	self_link	

Name	Type	Description
max_throughput_iops	integer	Specifies the maximum throughput in IOPS, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
max_throughput_mbps	integer	Specifies the maximum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_iops	integer	Specifies the minimum throughput in IOPS, 0 means none. Setting "min_throughput" is supported on AFF platforms only, unless FabricPool tiering policies are set. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_mbps	integer	Specifies the minimum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
name	string	The QoS policy group name. This is mutually exclusive with UUID and other QoS attributes during POST and PATCH.
uuid	string	The QoS policy group UUID. This is mutually exclusive with name and other QoS attributes during POST and PATCH.

qos

Name	Type	Description
policy	policy	The QoS policy

guarantee

Properties that request and report the space guarantee for the LUN.

Name	Type	Description
requested	boolean	The requested space reservation policy for the LUN. If *true*, a space reservation is requested for the LUN; if *false*, the LUN is thin provisioned. Guaranteeing a space reservation request for a LUN requires that the volume in which the LUN resides is also space reserved and that the fractional reserve for the volume is 100%. Valid in POST and PATCH.
reserved	boolean	Reports if the LUN is space guaranteed. If *true*, a space guarantee is requested and the containing volume and aggregate support the request. If *false*, a space guarantee is not requested or a space guarantee is requested and either the containing volume or aggregate do not support the request.

space

The storage space related properties of the LUN.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the LUN.

Name	Type	Description
size	integer	The total provisioned size of the LUN. The LUN size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The actual minimum and maxiumum sizes vary depending on the ONTAP version, ONTAP platform, and the available space in the containing volume and aggregate. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the LUN. This value is the total space consumed in the volume by the LUN, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways SAN filesystems and applications utilize blocks within a LUN, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the LUN blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

luns

A LUN is the logical representation of storage in a storage area network (SAN).

In ONTAP, a LUN is located within a volume. Optionally, it can be located within a qtree in a volume.

A LUN can be created to a specified size using thin or thick provisioning. A LUN can then be renamed, resized, cloned, and moved to a different volume. LUNs support the assignment of a quality of service (QoS) policy for performance management or a QoS policy can be assigned to the volume containing the LUN. See the LUN object model to learn more about each of the properties supported by the LUN REST API.

A LUN must be mapped to an initiator group to grant access to the initiator group's initiators (client hosts). Initiators can then access the LUN and perform I/O over a Fibre Channel (FC) fabric using the Fibre Channel Protocol or a TCP/IP network using iSCSI.

Name	Type	Description
clone	clone	This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`. When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`. Persistent reservations for the patched LUN are also preserved.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the LUN was created.

Name	Type	Description
enabled	boolean	The enabled state of the LUN. LUNs can be disabled to prevent access to the LUN. Certain error conditions also cause the LUN to become disabled. If the LUN is disabled, you can consult the `state` property to determine if the LUN is administratively disabled (*offline*) or has become disabled as a result of an error. A LUN in an error condition can be brought online by setting the `enabled` property to *true* or brought administratively offline by setting the `enabled` property to *false*. Upon creation, a LUN is enabled by default. Valid in PATCH.
lun_maps	array[lun_maps]	An array of LUN maps. A LUN map is an association between a LUN and an initiator group. When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.
name	string	The fully qualified path name of the LUN composed of the "/vol" prefix, the volume name, the qtree name (optional), and the base name of the LUN. Valid in POST and PATCH.
os_type	string	The operating system type of the LUN. Required in POST when creating a LUN that is not a clone of another. Disallowed in POST when creating a LUN clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
serial_number	string	The LUN serial number. The serial number is generated by ONTAP when the LUN is created. • maxLength: 12 • minLength: 12 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
space	space	The storage space related properties of the LUN.
uuid	string	The unique identifier of the LUN. The UUID is generated by ONTAP when the LUN is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

guarantee

Properties that request and report the space guarantee for the NVMe namespace.

Name	Type	Description
requested	boolean	The requested space reservation policy for the NVMe namespace. If *true*, a space reservation is requested for the namespace; if *false*, the namespace is thin provisioned. Guaranteeing a space reservation request for a namespace requires that the volume in which the namespace resides also be space reserved and that the fractional reserve for the volume be 100%. The space reservation policy for an NVMe namespace is determined by ONTAP. • Introduced in: 9.6 • x-nullable: true
reserved	boolean	Reports if the NVMe namespace is space guaranteed. This property is *true* if a space guarantee is requested and the containing volume and aggregate support the request. This property is *false* if a space guarantee is not requested or if a space guarantee is requested and either the containing volume and aggregate do not support the request.

space

The storage space related properties of the NVMe namespace.

Name	Type	Description
block_size	integer	The size of blocks in the namespace, in bytes. Valid in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone. Valid in POST.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the NVMe namespace.
size	integer	The total provisioned size of the NVMe namespace. Valid in POST and PATCH. The NVMe namespace size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The maximum size is variable with respect to large NVMe namespace support in ONTAP. If large namespaces are supported, the maximum size is 128 TB (140737488355328 bytes) and if not supported, the maximum size is just under 16 TB (17557557870592 bytes). The minimum size supported is always 4096 bytes. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the NVMe namespace. This value is the total space consumed in the volume by the NVMe namespace, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways NVMe filesystems and applications utilize blocks within a namespace, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the namespace blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

status

Status information about the NVMe namespace.

Name	Type	Description
container_state	string	The state of the volume and aggregate that contain the NVMe namespace. Namespaces are only available when their containers are available.

Name	Type	Description
mapped	boolean	Reports if the NVMe namespace is mapped to an NVMe subsystem. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
read_only	boolean	Reports if the NVMe namespace allows only read access.
state	string	The state of the NVMe namespace. Normal states for a namespace are *online* and *offline*. Other states indicate errors.

consistency_group_nvme_host

The NVMe host provisioned to access NVMe namespaces mapped to a subsystem.

Name	Type	Description
nqn	string	The NVMe qualified name (NQN) used to identify the NVMe storage target. Not allowed in POST when the `records` property is used.

consistency_group_nvme_subsystem

An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

Name	Type	Description
comment	string	A configurable comment for the NVMe subsystem. Optional in POST and PATCH.
hosts	array[consistency_group_nvme_host]	The NVMe hosts configured for access to the NVMe subsystem. Optional in POST.

Name	Type	Description
name	string	The name of the NVMe subsystem. Once created, an NVMe subsystem cannot be renamed. Required in POST.
os_type	string	The host operating system of the NVMe subsystem's hosts. Required in POST.
uuid	string	The unique identifier of the NVMe subsystem.

subsystem_map

The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems.

There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.

Name	Type	Description
_links	self_link	
anagrpid	string	The Asymmetric Namespace Access Group ID (ANAGRPID) of the NVMe namespace. The format for an ANAGRPID is 8 hexadecimal digits (zero-filled) followed by a lower case "h". There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.

Name	Type	Description
nsid	string	The NVMe namespace identifier. This is an identifier used by an NVMe controller to provide access to the NVMe namespace. The format for an NVMe namespace identifier is 8 hexadecimal digits (zero-filled) followed by a lower case "h".
subsystem	consistency_group_nvme_subsystem	An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

namespaces

An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the storage virtual machine using the NVMe over Fabrics protocol.

In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume.

An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API.

An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol.

Name	Type	Description
auto_delete	boolean	This property marks the NVMe namespace for auto deletion when the volume containing the namespace runs out of space. This is most commonly set on namespace clones. When set to *true*, the NVMe namespace becomes eligible for automatic deletion when the volume runs out of space. Auto deletion only occurs when the volume containing the namespace is also configured for auto deletion and free space in the volume decreases below a particular threshold. This property is optional in POST and PATCH. The default value for a new NVMe namespace is *false*. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the NVMe namespace was created.
enabled	boolean	The enabled state of the NVMe namespace. Certain error conditions cause the namespace to become disabled. If the namespace is disabled, you can check the `state` property to determine what error disabled the namespace. An NVMe namespace is enabled automatically when it is created.

Name	Type	Description
name	string	The fully qualified path name of the NVMe namespace composed of a "/vol" prefix, the volume name, the (optional) qtree name and base name of the namespace. Valid in POST. NVMe namespaces do not support rename, or movement between volumes.
os_type	string	The operating system type of the NVMe namespace. Required in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
space	space	The storage space related properties of the NVMe namespace.
status	status	Status information about the NVMe namespace.
subsystem_map	subsystem_map	The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems. There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.
uuid	string	The unique identifier of the NVMe namespace.

parent_consistency_group

The parent consistency group.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

storage_service

Determines the placement of any storage object created during this operation.

Name	Type	Description
name	string	Storage service name. If not specified, the default value is the most performant for the platform.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
name	string	New name for consistency group. Required to resolve naming collisions.
storage_service	storage_service	Determines the placement of any storage object created during this operation.

snapshot

A consistency group's Snapshot copy

Name	Type	Description
name	string	The name of the consistency group's Snapshot copy to restore to.

Name	Type	Description
uuid	string	The UUID of the consistency group's Snapshot copy to restore to.

restore_to

Use to restore a consistency group to a previous Snapshot copy

Name	Type	Description
snapshot	snapshot	A consistency group's Snapshot copy

_links

Name	Type	Description
self	href	

snapshot_policy_reference

This is a reference to the Snapshot copy policy.

Name	Type	Description
_links	_links	
name	string	
uuid	string	

space

Space information for the consistency group.

Name	Type	Description
available	integer	The amount of space available in the consistency group, in bytes.
size	integer	The total provisioned size of the consistency group, in bytes.
used	integer	The amount of space consumed in the consistency group, in bytes.

svm

The Storage Virtual Machine (SVM) in which the consistency group is located.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

object_stores

Name	Type	Description
name	string	The name of the object store to use. Used for placement.

tiering

The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

unsupported_reason

Name	Type	Description
code	string	If volume activity tracking is not supported on the volume, this field provides an appropriate error code.

Name	Type	Description
message	string	If volume activity tracking is not supported on the volume, this field provides an error message detailing why this is the case.

activity_tracking

The volume activity tracking configuration for this volume.

Name	Type	Description
state	string	Activity tracking state of the volume. If this value is *on*, ONTAP collects top metrics information for the volume in real time. There is a slight impact to I/O performance in order to collect this information. If this value is *off*, no activity tracking information is collected or available to view. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["off", "on"] • Introduced in: 9.10 • x-nullable: true
supported	boolean	This field indicates whether or not volume activity tracking is supported on the volume. If volume activity tracking is not supported, the reason why is provided in the `activity_tracking.unsupported_reason` field.
unsupported_reason	unsupported_reason	

unsupported_reason

Name	Type	Description
code	string	If file system analytics is not supported on the volume, this field provides the error code explaining why.
message	string	If file system analytics is not supported on the volume, this field provides the error message explaining why.

analytics

The file system analytics configuration for this volume.

Name	Type	Description
scan_progress	integer	Percentage of files in the volume that the file system analytics initialization scan has processed. Only returned when the `state` is *initializing*.

Name	Type	Description
state	string	File system analytics state of the volume. If this value is *on*, ONTAP collects extra file system analytics information for all directories on the volume. There will be a slight impact to I/O performance to collect this information. If this value is *off*, file system analytics information is not collected and not available to be viewed. If this value is *initializing*, that means file system analytics was recently turned on, and the initialization scan to gather information for all existing files and directories is currently running. If this value is *initialization_paused*, this means that the initialization scan is currently paused. If this value is 'unknown', this means that there was an internal error when determining the file system analytics state for the volume. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["unknown", "initializing", "initialization_paused", "off", "on"] • Introduced in: 9.8 • x-nullable: true
supported	boolean	This field indicates whether or not file system analytics is supported on the volume. If file system analytics is not supported, the reason will be specified in the `analytics.unsupported_reason` field.
unsupported_reason	unsupported_reason	

acls

The permissions that users and groups have on a CIFS share.

Name	Type	Description
_links	_links	
permission	string	Specifies the access rights that a user or group has on the defined CIFS Share. The following values are allowed: • no_access - User does not have CIFS share access • read - User has only read access • change - User has change access • full_control - User has full_control access
type	string	Specifies the type of the user or group to add to the access control list of a CIFS share. The following values are allowed: • windows - Windows user or group • unix_user - UNIX user • unix_group - UNIX group
user_or_group	string	Specifies the user or group name to add to the access control list of a CIFS share.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.

Name	Type	Description
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

cifs

Name	Type	Description
shares	array[consistency_group_cifs_share]	

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

export_rules

Name	Type	Description
_links	_links	
allow_device_creation	boolean	Specifies whether or not device creation is allowed.
allow_suid	boolean	Specifies whether or not SetUID bits in SETATTR Op is to be honored.
anonymous_user	string	User ID To Which Anonymous Users Are Mapped.

Name	Type	Description
chown_mode	string	Specifies who is authorized to change the ownership mode of a file.
clients	array[export_clients]	Array of client matches
index	integer	Index of the rule within the export policy.
ntfs_unix_security	string	NTFS export UNIX security options.
protocols	array[string]	
ro_rule	array[string]	Authentication flavors that the read-only access rule governs
rw_rule	array[string]	Authentication flavors that the read/write access rule governs
superuser	array[string]	Authentication flavors that the superuser security type governs

export_policy

The policy associated with volumes to export them for protocol access.

Name	Type	Description
_links	self_link	
name	string	Name of the export policy.
rules	array[export_rules]	The set of rules that govern the export policy.
uuid	string	Identifier for the export policy.

junction_parent

Name	Type	Description
_links	self_link	

Name	Type	Description
name	string	The name of the parent volume that contains the junction inode of this volume. The junction parent volume must belong to the same SVM that owns this volume.
uuid	string	Unique identifier for the parent volume.

nas

The CIFS share policy and/or export policies for this volume.

Name	Type	Description
cifs	cifs	
export_policy	export_policy	The policy associated with volumes to export them for protocol access.
gid	integer	The UNIX group ID of the volume. Valid in POST or PATCH.
junction_parent	junction_parent	
path	string	The fully-qualified path in the owning SVM's namespace at which the volume is mounted. The path is case insensitive and must be unique within an SVM's namespace. Path must begin with '/' and must not end with '/'. Only one volume can be mounted at any given junction path. An empty path in POST creates an unmounted volume. An empty path in PATCH deactivates and unmounts the volume. Taking a volume offline or restricted state removes its junction path. This attribute is reported in GET only when the volume is mounted.

Name	Type	Description
security_style	string	Security style associated with the volume. Valid in POST or PATCH. mixed ‐ Mixed-style security ntfs ‐ NTFS/WIndows-style security unified ‐ Unified-style security, unified UNIX, NFS and CIFS permissions unix ‐ UNIX-style security.
uid	integer	The UNIX user ID of the volume. Valid in POST or PATCH.
unix_permissions	integer	UNIX permissions to be viewed as an octal number, consisting of 4 digits derived by adding up bits 4 (read), 2 (write), and 1 (execute). First digit selects the set user ID (4), set group ID (2), and sticky (1) attributes. Second digit selects permission for the owner of the file. Third selects permissions for other users in the same group while the fourth selects permissions for other users not in the group. Valid in POST or PATCH. For security style "mixed" or "unix", the default setting is 0755 in octal (493 in decimal) and for security style "ntfs", the default setting is 0000. In cases where only owner, group, and other permissions are given (as in 755, representing the second, third and fourth digit), the first digit is assumed to be zero.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

Name	Type	Description
storage_service	storage_service	Determines the placement of any storage object created during this operation.

qos

The QoS policy for this volume.

Name	Type	Description
policy	policy	The QoS policy

space

Name	Type	Description
available	integer	The available space, in bytes.
size	integer	Total provisioned size, in bytes.
used	integer	The virtual space used (includes volume reserves) before storage efficiency, in bytes.

tiering

The tiering placement and policy definitions for this volume.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

volumes

Name	Type	Description
activity_tracking	activity_tracking	The volume activity tracking configuration for this volume.
analytics	analytics	The file system analytics configuration for this volume.

Name	Type	Description
comment	string	A comment for the volume. Valid in POST or PATCH.
language	string	Language encoding setting for volume. If no language is specified, the volume inherits its SVM language encoding setting.
name	string	Volume name. The name of volume must start with an alphabetic character (a to z or A to Z) or an underscore (_). The name must be 197 or fewer characters in length for FlexGroups, and 203 or fewer characters in length for all other types of volumes. Volume names must be unique within an SVM. Required on POST.
nas	nas	The CIFS share policy and/or export policies for this volume.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	The QoS policy for this volume.
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	
tiering	tiering	The tiering placement and policy definitions for this volume.
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • readOnly: 1 • Introduced in: 9.8 • x-nullable: true

consistency_groups

Name	Type	Description
_links	self_link	
application	application	
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

Name	Type	Description
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

iops

The rate of I/O operations observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency

The round trip latency observed at the storage object, in microseconds.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput

The rate of throughput bytes per second observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

metric

Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.

Name	Type	Description
_links	_links	
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
duration	string	The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations:
iops	iops	The rate of I/O operations observed at the storage object.
latency	latency	The round trip latency observed at the storage object, in microseconds.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput	throughput	The rate of throughput bytes per second observed at the storage object.
timestamp	string	The timestamp of the performance and capacity data.

Name	Type	Description
used_space	integer	The total space used in the consistency group, in bytes.

replication_relationships

Name	Type	Description
_links	self_link	
is_source	boolean	Indicates whether or not this consistency group is the source for replication.
uuid	string	The unique identifier of the SnapMirror relationship.

iops_raw

The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency_raw

The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput_raw

Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

statistics

These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.

Name	Type	Description
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
iops_raw	iops_raw	The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.
latency_raw	latency_raw	The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput_raw	throughput_raw	Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.
timestamp	string	The timestamp of the performance data.

Name	Type	Description
used_space	integer	The total used space in the consistency group, in bytes.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.

Name	Type	Description
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true

Name	Type	Description
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.

Name	Type	Description
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

consistency_group

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	

Name	Type	Description
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	

Name	Type	Description
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

job_link

Name	Type	Description
_links	_links	
uuid	string	The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message

Name	Type	Description
target	string	The target parameter that caused the error.

Delete a consistency group

DELETE `/application/consistency-groups/{uuid}`

Introduced In: 9.10

Deletes a consistency group.

> this will not delete any associated volumes or LUNs. To remove those elements, you can use the appropriate object endpoint.

Related ONTAP commands

There are no ONTAP commands for managing consistency groups.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	The unique identifier of the consistency group to delete.
delete_data	boolean	query	False	Delete the underlying storage as well as the consistency group association. **This parameter should be used with caution.** • Default value:

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Response

```
Status: 200, Ok
```

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
53411842	Consistency group does not exist.
53411843	A consistency group with specified UUID was not found.
53411844	Specified consistency group was not found in the specified SVM.
53411845	The specified UUID and name refer to different consistency groups.
53411846	Either name or UUID must be provided.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve a consistency group

GET `/application/consistency-groups/{uuid}`

Introduced In: 9.10

Retrieves a single consistency group.

Expensive properties

There is an added computational cost to retrieving values for these properties. They are not included by default in GET results and must be explicitly requested using the `fields` query parameter. See DOC Requesting specific fields to learn more.

- `volumes`
- `luns`
- `namespaces`

Related ONTAP commands

There are no ONTAP commands for managing consistency groups.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	The unique identifier of the group to retrieve.
space.available	integer	query	False	Filter by space.available
space.used	integer	query	False	Filter by space.used
space.size	integer	query	False	Filter by space.size
replication_relationships.is_source	boolean	query	False	Filter by replication_relationships.is_source • Introduced in: 9.13
replication_relationships.uuid	string	query	False	Filter by replication_relationships.uuid • Introduced in: 9.13
metric.available_space	integer	query	False	Filter by metric.available_space • Introduced in: 9.13
metric.throughput.read	integer	query	False	Filter by metric.throughput.read • Introduced in: 9.13
metric.throughput.other	integer	query	False	Filter by metric.throughput.other • Introduced in: 9.13

Name	Type	In	Required	Description
metric.throughput.write	integer	query	False	Filter by metric.throughput.write • Introduced in: 9.13
metric.throughput.total	integer	query	False	Filter by metric.throughput.total • Introduced in: 9.13
metric.duration	string	query	False	Filter by metric.duration • Introduced in: 9.13
metric.latency.read	integer	query	False	Filter by metric.latency.read • Introduced in: 9.13
metric.latency.other	integer	query	False	Filter by metric.latency.other • Introduced in: 9.13
metric.latency.write	integer	query	False	Filter by metric.latency.write • Introduced in: 9.13
metric.latency.total	integer	query	False	Filter by metric.latency.total • Introduced in: 9.13
metric.used_space	integer	query	False	Filter by metric.used_space • Introduced in: 9.13

Name	Type	In	Required	Description
metric.size	integer	query	False	Filter by metric.size • Introduced in: 9.13
metric.timestamp	string	query	False	Filter by metric.timestamp • Introduced in: 9.13
metric.status	string	query	False	Filter by metric.status • Introduced in: 9.13
metric.iops.read	integer	query	False	Filter by metric.iops.read • Introduced in: 9.13
metric.iops.other	integer	query	False	Filter by metric.iops.other • Introduced in: 9.13
metric.iops.write	integer	query	False	Filter by metric.iops.write • Introduced in: 9.13
metric.iops.total	integer	query	False	Filter by metric.iops.total • Introduced in: 9.13
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name

Name	Type	In	Required	Description
consistency_groups.parent_consistency_group.uuid	string	query	False	Filter by consistency_groups.parent_consistency_group.uuid
consistency_groups.parent_consistency_group.name	string	query	False	Filter by consistency_groups.parent_consistency_group.name
consistency_groups.namespaces.os_type	string	query	False	Filter by consistency_groups.namespaces.os_type • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.os_type	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.os_type • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.name	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.name • Introduced in: 9.12 • maxLength: 96 • minLength: 1
consistency_groups.namespaces.subsystem_map.subsystem.comment	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.comment • Introduced in: 9.12 • maxLength: 255 • minLength: 0

Name	Type	In	Required	Description
consistency_groups.namespaces.subsystem_map.subsystem.hosts.nqn	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.hosts.nqn • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.subsystem.uuid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.subsystem.uuid • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.nsid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.nsid • Introduced in: 9.12
consistency_groups.namespaces.subsystem_map.anagrpid	string	query	False	Filter by consistency_groups.namespaces.subsystem_map.anagrpid • Introduced in: 9.12
consistency_groups.namespaces.space.used	integer	query	False	Filter by consistency_groups.namespaces.space.used • Introduced in: 9.12
consistency_groups.namespaces.space.guarantee.requested	boolean	query	False	Filter by consistency_groups.namespaces.space.guarantee.requested • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.space.guarantee.reserved	boolean	query	False	Filter by consistency_groups.namespaces.space.guarantee.reserved • Introduced in: 9.12
consistency_groups.namespaces.space.size	integer	query	False	Filter by consistency_groups.namespaces.space.size • Introduced in: 9.12 • Max value: 140737488355328 • Min value: 4096
consistency_groups.namespaces.space.block_size	integer	query	False	Filter by consistency_groups.namespaces.space.block_size • Introduced in: 9.12
consistency_groups.namespaces.enabled	boolean	query	False	Filter by consistency_groups.namespaces.enabled • Introduced in: 9.12
consistency_groups.namespaces.status.mapped	boolean	query	False	Filter by consistency_groups.namespaces.status.mapped • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.status.state	string	query	False	Filter by consistency_groups.namespaces.status.state • Introduced in: 9.12
consistency_groups.namespaces.status.read_only	boolean	query	False	Filter by consistency_groups.namespaces.status.read_only • Introduced in: 9.12
consistency_groups.namespaces.status.container_state	string	query	False	Filter by consistency_groups.namespaces.status.container_state • Introduced in: 9.12
consistency_groups.namespaces.comment	string	query	False	Filter by consistency_groups.namespaces.comment • Introduced in: 9.12 • maxLength: 254 • minLength: 0
consistency_groups.namespaces.create_time	string	query	False	Filter by consistency_groups.namespaces.create_time • Introduced in: 9.12
consistency_groups.namespaces.uuid	string	query	False	Filter by consistency_groups.namespaces.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.namespaces.name	string	query	False	Filter by consistency_groups.namespaces.name • Introduced in: 9.12
consistency_groups.namespaces.auto_delete	boolean	query	False	Filter by consistency_groups.namespaces.auto_delete • Introduced in: 9.12
consistency_groups.luns.space.size	integer	query	False	Filter by consistency_groups.luns.space.size • Max value: 140737488355328 • Min value: 4096
consistency_groups.luns.space.guarantee.requested	boolean	query	False	Filter by consistency_groups.luns.space.guarantee.requested • Introduced in: 9.11
consistency_groups.luns.space.guarantee.reserved	boolean	query	False	Filter by consistency_groups.luns.space.guarantee.reserved • Introduced in: 9.11
consistency_groups.luns.space.used	integer	query	False	Filter by consistency_groups.luns.space.used
consistency_groups.luns.lun_maps.logical_unit_number	integer	query	False	Filter by consistency_groups.luns.lun_maps.logical_unit_number

Name	Type	In	Required	Description
consistency_groups.luns.lun_maps.igroup.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.name • maxLength: 96 • minLength: 1
consistency_groups.luns.lun_maps.igroup.uuid	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.uuid
consistency_groups.luns.lun_maps.igroup.comment	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.comment • Introduced in: 9.11 • maxLength: 254 • minLength: 0
consistency_groups.luns.lun_maps.igroup.os_type	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.os_type
consistency_groups.luns.lun_maps.igroup.protocol	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.protocol
consistency_groups.luns.lun_maps.igroup.igroups.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.igroups.name • maxLength: 96 • minLength: 1
consistency_groups.luns.lun_maps.igroup.igroups.uuid	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.igroups.uuid

Name	Type	In	Required	Description
consistency_groups.luns.lun_maps.igroup.initiators.comment	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.initiators.comment • maxLength: 254 • minLength: 0
consistency_groups.luns.lun_maps.igroup.initiators.name	string	query	False	Filter by consistency_groups.luns.lun_maps.igroup.initiators.name
consistency_groups.luns.os_type	string	query	False	Filter by consistency_groups.luns.os_type
consistency_groups.luns.name	string	query	False	Filter by consistency_groups.luns.name
consistency_groups.luns.uuid	string	query	False	Filter by consistency_groups.luns.uuid
consistency_groups.luns.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.luns.qos.policy.max_throughput_iops
consistency_groups.luns.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.luns.qos.policy.min_throughput_mbps
consistency_groups.luns.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.luns.qos.policy.min_throughput_iops
consistency_groups.luns.qos.policy.name	string	query	False	Filter by consistency_groups.luns.qos.policy.name

Name	Type	In	Required	Description
consistency_groups.luns.qos.policy.uuid	string	query	False	Filter by consistency_groups.luns.qos.policy.uuid
consistency_groups.luns.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.luns.qos.policy.max_throughput_mbps
consistency_groups.luns.comment	string	query	False	Filter by consistency_groups.luns.comment • maxLength: 254 • minLength: 0
consistency_groups.luns.create_time	string	query	False	Filter by consistency_groups.luns.create_time
consistency_groups.luns.enabled	boolean	query	False	Filter by consistency_groups.luns.enabled
consistency_groups.luns.serial_number	string	query	False	Filter by consistency_groups.luns.serial_number • maxLength: 12 • minLength: 12
consistency_groups.name	string	query	False	Filter by consistency_groups.name
consistency_groups.tiering.policy	string	query	False	Filter by consistency_groups.tiering.policy
consistency_groups.uuid	string	query	False	Filter by consistency_groups.uuid
consistency_groups.snapshot_policy.uuid	string	query	False	Filter by consistency_groups.snapshot_policy.uuid

Name	Type	In	Required	Description
consistency_groups.snapshot_policy.name	string	query	False	Filter by consistency_groups.snapshot_policy.name
consistency_groups.application.component_type	string	query	False	Filter by consistency_groups.application.component_type • Introduced in: 9.12
consistency_groups.application.type	string	query	False	Filter by consistency_groups.application.type • Introduced in: 9.12
consistency_groups.qos.policy.uuid	string	query	False	Filter by consistency_groups.qos.policy.uuid
consistency_groups.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.qos.policy.max_throughput_mbps
consistency_groups.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.qos.policy.min_throughput_mbps
consistency_groups.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.qos.policy.min_throughput_iops
consistency_groups.qos.policy.name	string	query	False	Filter by consistency_groups.qos.policy.name
consistency_groups.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.qos.policy.max_throughput_iops

Name	Type	In	Required	Description
consistency_groups.svm.uuid	string	query	False	Filter by consistency_groups.svm.uuid
consistency_groups.svm.name	string	query	False	Filter by consistency_groups.svm.name
consistency_groups.space.available	integer	query	False	Filter by consistency_groups.space.available
consistency_groups.space.used	integer	query	False	Filter by consistency_groups.space.used
consistency_groups.space.size	integer	query	False	Filter by consistency_groups.space.size
consistency_groups.volumes.name	string	query	False	Filter by consistency_groups.volumes.name • maxLength: 203 • minLength: 1
consistency_groups.volumes.tiering.policy	string	query	False	Filter by consistency_groups.volumes.tiering.policy
consistency_groups.volumes.uuid	string	query	False	Filter by consistency_groups.volumes.uuid
consistency_groups.volumes.snapshot_policy.uuid	string	query	False	Filter by consistency_groups.volumes.snapshot_policy.uuid
consistency_groups.volumes.snapshot_policy.name	string	query	False	Filter by consistency_groups.volumes.snapshot_policy.name

Name	Type	In	Required	Description
consistency_groups.volumes.qos.policy.uuid	string	query	False	Filter by consistency_groups.volumes.qos.policy.uuid
consistency_groups.volumes.qos.policy.max_throughput_mbps	integer	query	False	Filter by consistency_groups.volumes.qos.policy.max_throughput_mbps
consistency_groups.volumes.qos.policy.min_throughput_mbps	integer	query	False	Filter by consistency_groups.volumes.qos.policy.min_throughput_mbps
consistency_groups.volumes.qos.policy.min_throughput_iops	integer	query	False	Filter by consistency_groups.volumes.qos.policy.min_throughput_iops
consistency_groups.volumes.qos.policy.name	string	query	False	Filter by consistency_groups.volumes.qos.policy.name
consistency_groups.volumes.qos.policy.max_throughput_iops	integer	query	False	Filter by consistency_groups.volumes.qos.policy.max_throughput_iops
consistency_groups.volumes.comment	string	query	False	Filter by consistency_groups.volumes.comment • maxLength: 1023 • minLength: 0

Name	Type	In	Required	Description
consistency_groups.volumes.nas.security_style	string	query	False	Filter by consistency_groups.volumes.nas.security_style • Introduced in: 9.12
consistency_groups.volumes.nas.path	string	query	False	Filter by consistency_groups.volumes.nas.path • Introduced in: 9.12
consistency_groups.volumes.nas.gid	integer	query	False	Filter by consistency_groups.volumes.nas.gid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.uuid	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.uuid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.rw_rule	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.rw_rule • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.clients.match	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.clients.match • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.export_policy.rules.allow_suid	boolean	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.allow_suid • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.chown_mode	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.chown_mode • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.allow_device_creation	boolean	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.allow_device_creation • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.protocols	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.protocols • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.ntfs_unix_security	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.ntfs_unix_security • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.export_policy.rules.superuser	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.superuser • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.anonymous_user	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.anonymous_user • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.index	integer	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.index • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.rules.ro_rule	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.rules.ro_rule • Introduced in: 9.12
consistency_groups.volumes.nas.export_policy.name	string	query	False	Filter by consistency_groups.volumes.nas.export_policy.name • Introduced in: 9.12
consistency_groups.volumes.nas.junction_parent.name	string	query	False	Filter by consistency_groups.volumes.nas.junction_parent.name • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.junction_parent.uuid	string	query	False	Filter by consistency_groups.volumes.nas.junction_parent.uuid • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.namespace_caching	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.namespace_caching • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.encryption	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.encryption • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.no_strict_security	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.no_strict_security • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.vscan_profile	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.vscan_profile • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.allow_unencrypted_access	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.allow_unencrypted_access • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.file_umask	integer	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.file_umask • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.offline_files	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.offline_files • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.home_directory	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.home_directory • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.comment	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.comment • Introduced in: 9.12 • maxLength: 256 • minLength: 1
consistency_groups.volumes.nas.cifs.shares.change_notify	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.change_notify • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.oplocks	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.oplocks • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.unix_symlink	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.unix_symlink • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.dir_umask	integer	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.dir_umask • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.continuously_available	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.continuously_available • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.access_based_enumeration	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.access_based_enumeration • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.acls.permission	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.permission • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.acls.user_or_group	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.user_or_group • Introduced in: 9.12

Name	Type	In	Required	Description
consistency_groups.volumes.nas.cifs.shares.acls.type	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.acls.type • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.show_snapshot	boolean	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.show_snapshot • Introduced in: 9.12
consistency_groups.volumes.nas.cifs.shares.name	string	query	False	Filter by consistency_groups.volumes.nas.cifs.shares.name • Introduced in: 9.12 • maxLength: 80 • minLength: 1
consistency_groups.volumes.nas.unix_permissions	integer	query	False	Filter by consistency_groups.volumes.nas.unix_permissions • Introduced in: 9.12
consistency_groups.volumes.nas.uid	integer	query	False	Filter by consistency_groups.volumes.nas.uid • Introduced in: 9.12
consistency_groups.volumes.analytics.scan_progress	integer	query	False	Filter by consistency_groups.volumes.analytics.scan_progress • Introduced in: 9.13

Name	Type	In	Required	Description
consistency_groups.volumes.analytics.unsupported_reason.code	string	query	False	Filter by consistency_groups.volumes.analytics.unsupported_reason.code • Introduced in: 9.13
consistency_groups.volumes.analytics.unsupported_reason.message	string	query	False	Filter by consistency_groups.volumes.analytics.unsupported_reason.message • Introduced in: 9.13
consistency_groups.volumes.analytics.supported	boolean	query	False	Filter by consistency_groups.volumes.analytics.supported • Introduced in: 9.13
consistency_groups.volumes.analytics.state	string	query	False	Filter by consistency_groups.volumes.analytics.state • Introduced in: 9.13
consistency_groups.volumes.space.available	integer	query	False	Filter by consistency_groups.volumes.space.available
consistency_groups.volumes.space.used	integer	query	False	Filter by consistency_groups.volumes.space.used
consistency_groups.volumes.space.size	integer	query	False	Filter by consistency_groups.volumes.space.size

Name	Type	In	Required	Description
consistency_groups.volumes.activity_tracking.unsupported_reason.code	string	query	False	Filter by consistency_groups.volumes.activity_tracking.unsupported_reason.code • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.unsupported_reason.message	string	query	False	Filter by consistency_groups.volumes.activity_tracking.unsupported_reason.message • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.state	string	query	False	Filter by consistency_groups.volumes.activity_tracking.state • Introduced in: 9.13
consistency_groups.volumes.activity_tracking.supported	boolean	query	False	Filter by consistency_groups.volumes.activity_tracking.supported • Introduced in: 9.13
consistency_groups.volumes.language	string	query	False	Filter by consistency_groups.volumes.language
snapshot_policy.uuid	string	query	False	Filter by snapshot_policy.uuid
snapshot_policy.name	string	query	False	Filter by snapshot_policy.name
qos.policy.uuid	string	query	False	Filter by qos.policy.uuid

Name	Type	In	Required	Description
qos.policy.max_throughput_mbps	integer	query	False	Filter by qos.policy.max_throughput_mbps
qos.policy.min_throughput_mbps	integer	query	False	Filter by qos.policy.min_throughput_mbps
qos.policy.min_throughput_iops	integer	query	False	Filter by qos.policy.min_throughput_iops
qos.policy.name	string	query	False	Filter by qos.policy.name
qos.policy.max_throughput_iops	integer	query	False	Filter by qos.policy.max_throughput_iops
replication_source	boolean	query	False	Filter by replication_source
replicated	boolean	query	False	Filter by replicated
luns.space.size	integer	query	False	Filter by luns.space.size • Max value: 140737488355328 • Min value: 4096
luns.space.guarantee.requested	boolean	query	False	Filter by luns.space.guarantee.requested • Introduced in: 9.11
luns.space.guarantee.reserved	boolean	query	False	Filter by luns.space.guarantee.reserved • Introduced in: 9.11

Name	Type	In	Required	Description
luns.space.used	integer	query	False	Filter by luns.space.used
luns.lun_maps.logical_unit_number	integer	query	False	Filter by luns.lun_maps.logical_unit_number
luns.lun_maps.igroup.name	string	query	False	Filter by luns.lun_maps.igroup.name • maxLength: 96 • minLength: 1
luns.lun_maps.igroup.uuid	string	query	False	Filter by luns.lun_maps.igroup.uuid
luns.lun_maps.igroup.comment	string	query	False	Filter by luns.lun_maps.igroup.comment • Introduced in: 9.11 • maxLength: 254 • minLength: 0
luns.lun_maps.igroup.os_type	string	query	False	Filter by luns.lun_maps.igroup.os_type
luns.lun_maps.igroup.protocol	string	query	False	Filter by luns.lun_maps.igroup.protocol
luns.lun_maps.igroup.igroups.name	string	query	False	Filter by luns.lun_maps.igroup.igroups.name • maxLength: 96 • minLength: 1
luns.lun_maps.igroup.igroups.uuid	string	query	False	Filter by luns.lun_maps.igroup.igroups.uuid

Name	Type	In	Required	Description
luns.lun_maps.igroup.initiators.comment	string	query	False	Filter by luns.lun_maps.igroup.initiators.comment • maxLength: 254 • minLength: 0
luns.lun_maps.igroup.initiators.name	string	query	False	Filter by luns.lun_maps.igroup.initiators.name
luns.os_type	string	query	False	Filter by luns.os_type
luns.name	string	query	False	Filter by luns.name
luns.uuid	string	query	False	Filter by luns.uuid
luns.qos.policy.max_throughput_iops	integer	query	False	Filter by luns.qos.policy.max_throughput_iops
luns.qos.policy.min_throughput_mbps	integer	query	False	Filter by luns.qos.policy.min_throughput_mbps
luns.qos.policy.min_throughput_iops	integer	query	False	Filter by luns.qos.policy.min_throughput_iops
luns.qos.policy.name	string	query	False	Filter by luns.qos.policy.name
luns.qos.policy.uuid	string	query	False	Filter by luns.qos.policy.uuid
luns.qos.policy.max_throughput_mbps	integer	query	False	Filter by luns.qos.policy.max_throughput_mbps
luns.comment	string	query	False	Filter by luns.comment • maxLength: 254 • minLength: 0

Name	Type	In	Required	Description
luns.create_time	string	query	False	Filter by luns.create_time
luns.enabled	boolean	query	False	Filter by luns.enabled
luns.serial_number	string	query	False	Filter by luns.serial_number • maxLength: 12 • minLength: 12
statistics.size	integer	query	False	Filter by statistics.size • Introduced in: 9.13
statistics.iops_raw.read	integer	query	False	Filter by statistics.iops_raw.read • Introduced in: 9.13
statistics.iops_raw.other	integer	query	False	Filter by statistics.iops_raw.other • Introduced in: 9.13
statistics.iops_raw.write	integer	query	False	Filter by statistics.iops_raw.write • Introduced in: 9.13
statistics.iops_raw.total	integer	query	False	Filter by statistics.iops_raw.total • Introduced in: 9.13

Name	Type	In	Required	Description
statistics.used_space	integer	query	False	Filter by statistics.used_space • Introduced in: 9.13
statistics.latency_raw.read	integer	query	False	Filter by statistics.latency_raw.read • Introduced in: 9.13
statistics.latency_raw.other	integer	query	False	Filter by statistics.latency_raw.other • Introduced in: 9.13
statistics.latency_raw.write	integer	query	False	Filter by statistics.latency_raw.write • Introduced in: 9.13
statistics.latency_raw.total	integer	query	False	Filter by statistics.latency_raw.total • Introduced in: 9.13
statistics.throughput_raw.read	integer	query	False	Filter by statistics.throughput_raw.read • Introduced in: 9.13
statistics.throughput_raw.other	integer	query	False	Filter by statistics.throughput_raw.other • Introduced in: 9.13

Name	Type	In	Required	Description
statistics.throughput_raw.write	integer	query	False	Filter by statistics.throughput_raw.write • Introduced in: 9.13
statistics.throughput_raw.total	integer	query	False	Filter by statistics.throughput_raw.total • Introduced in: 9.13
statistics.available_space	integer	query	False	Filter by statistics.available_space • Introduced in: 9.13
statistics.timestamp	string	query	False	Filter by statistics.timestamp • Introduced in: 9.13
statistics.status	string	query	False	Filter by statistics.status • Introduced in: 9.13
parent_consistency_group.uuid	string	query	False	Filter by parent_consistency_group.uuid
parent_consistency_group.name	string	query	False	Filter by parent_consistency_group.name
volumes.name	string	query	False	Filter by volumes.name • maxLength: 203 • minLength: 1

Name	Type	In	Required	Description
volumes.tiering.policy	string	query	False	Filter by volumes.tiering.policy
volumes.uuid	string	query	False	Filter by volumes.uuid
volumes.snapshot_policy.uuid	string	query	False	Filter by volumes.snapshot_policy.uuid
volumes.snapshot_policy.name	string	query	False	Filter by volumes.snapshot_policy.name
volumes.qos.policy.uuid	string	query	False	Filter by volumes.qos.policy.uuid
volumes.qos.policy.max_throughput_mbps	integer	query	False	Filter by volumes.qos.policy.max_throughput_mbps
volumes.qos.policy.min_throughput_mbps	integer	query	False	Filter by volumes.qos.policy.min_throughput_mbps
volumes.qos.policy.min_throughput_iops	integer	query	False	Filter by volumes.qos.policy.min_throughput_iops
volumes.qos.policy.name	string	query	False	Filter by volumes.qos.policy.name
volumes.qos.policy.max_throughput_iops	integer	query	False	Filter by volumes.qos.policy.max_throughput_iops

Name	Type	In	Required	Description
volumes.comment	string	query	False	Filter by volumes.comment • maxLength: 1023 • minLength: 0
volumes.nas.security_style	string	query	False	Filter by volumes.nas.security_style • Introduced in: 9.12
volumes.nas.path	string	query	False	Filter by volumes.nas.path • Introduced in: 9.12
volumes.nas.gid	integer	query	False	Filter by volumes.nas.gid • Introduced in: 9.12
volumes.nas.export_policy.uuid	string	query	False	Filter by volumes.nas.export_policy.uuid • Introduced in: 9.12
volumes.nas.export_policy.rules.rw_rule	string	query	False	Filter by volumes.nas.export_policy.rules.rw_rule • Introduced in: 9.12
volumes.nas.export_policy.rules.clients.match	string	query	False	Filter by volumes.nas.export_policy.rules.clients.match • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.export_policy.rules.allow_suid	boolean	query	False	Filter by volumes.nas.export_policy.rules.allow_suid • Introduced in: 9.12
volumes.nas.export_policy.rules.chown_mode	string	query	False	Filter by volumes.nas.export_policy.rules.chown_mode • Introduced in: 9.12
volumes.nas.export_policy.rules.allow_device_creation	boolean	query	False	Filter by volumes.nas.export_policy.rules.allow_device_creation • Introduced in: 9.12
volumes.nas.export_policy.rules.protocols	string	query	False	Filter by volumes.nas.export_policy.rules.protocols • Introduced in: 9.12
volumes.nas.export_policy.rules.ntfs_unix_security	string	query	False	Filter by volumes.nas.export_policy.rules.ntfs_unix_security • Introduced in: 9.12
volumes.nas.export_policy.rules.superuser	string	query	False	Filter by volumes.nas.export_policy.rules.superuser • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.export_policy.rules.anonymous_user	string	query	False	Filter by volumes.nas.export_policy.rules.anonymous_user • Introduced in: 9.12
volumes.nas.export_policy.rules.index	integer	query	False	Filter by volumes.nas.export_policy.rules.index • Introduced in: 9.12
volumes.nas.export_policy.rules.ro_rule	string	query	False	Filter by volumes.nas.export_policy.rules.ro_rule • Introduced in: 9.12
volumes.nas.export_policy.name	string	query	False	Filter by volumes.nas.export_policy.name • Introduced in: 9.12
volumes.nas.junction_parent.name	string	query	False	Filter by volumes.nas.junction_parent.name • Introduced in: 9.12
volumes.nas.junction_parent.uuid	string	query	False	Filter by volumes.nas.junction_parent.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.namespace_caching	boolean	query	False	Filter by volumes.nas.cifs.shares.namespace_caching • Introduced in: 9.12
volumes.nas.cifs.shares.encryption	boolean	query	False	Filter by volumes.nas.cifs.shares.encryption • Introduced in: 9.12
volumes.nas.cifs.shares.no_strict_security	boolean	query	False	Filter by volumes.nas.cifs.shares.no_strict_security • Introduced in: 9.12
volumes.nas.cifs.shares.vscan_profile	string	query	False	Filter by volumes.nas.cifs.shares.vscan_profile • Introduced in: 9.12
volumes.nas.cifs.shares.allow_unencrypted_access	boolean	query	False	Filter by volumes.nas.cifs.shares.allow_unencrypted_access • Introduced in: 9.12
volumes.nas.cifs.shares.file_umask	integer	query	False	Filter by volumes.nas.cifs.shares.file_umask • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.offline_files	string	query	False	Filter by volumes.nas.cifs.shares.offline_files • Introduced in: 9.12
volumes.nas.cifs.shares.home_directory	boolean	query	False	Filter by volumes.nas.cifs.shares.home_directory • Introduced in: 9.12
volumes.nas.cifs.shares.comment	string	query	False	Filter by volumes.nas.cifs.shares.comment • Introduced in: 9.12 • maxLength: 256 • minLength: 1
volumes.nas.cifs.shares.change_notify	boolean	query	False	Filter by volumes.nas.cifs.shares.change_notify • Introduced in: 9.12
volumes.nas.cifs.shares.oplocks	boolean	query	False	Filter by volumes.nas.cifs.shares.oplocks • Introduced in: 9.12
volumes.nas.cifs.shares.unix_symlink	string	query	False	Filter by volumes.nas.cifs.shares.unix_symlink • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.dir_umask	integer	query	False	Filter by volumes.nas.cifs.shares.dir_umask • Introduced in: 9.12
volumes.nas.cifs.shares.continuously_available	boolean	query	False	Filter by volumes.nas.cifs.shares.continuously_available • Introduced in: 9.12
volumes.nas.cifs.shares.access_based_enumeration	boolean	query	False	Filter by volumes.nas.cifs.shares.access_based_enumeration • Introduced in: 9.12
volumes.nas.cifs.shares.acls.permission	string	query	False	Filter by volumes.nas.cifs.shares.acls.permission • Introduced in: 9.12
volumes.nas.cifs.shares.acls.user_or_group	string	query	False	Filter by volumes.nas.cifs.shares.acls.user_or_group • Introduced in: 9.12
volumes.nas.cifs.shares.acls.type	string	query	False	Filter by volumes.nas.cifs.shares.acls.type • Introduced in: 9.12

Name	Type	In	Required	Description
volumes.nas.cifs.shares.show_snapshot	boolean	query	False	Filter by volumes.nas.cifs.shares.show_snapshot • Introduced in: 9.12
volumes.nas.cifs.shares.name	string	query	False	Filter by volumes.nas.cifs.shares.name • Introduced in: 9.12 • maxLength: 80 • minLength: 1
volumes.nas.unix_permissions	integer	query	False	Filter by volumes.nas.unix_permissions • Introduced in: 9.12
volumes.nas.uid	integer	query	False	Filter by volumes.nas.uid • Introduced in: 9.12
volumes.analytics.scan_progress	integer	query	False	Filter by volumes.analytics.scan_progress • Introduced in: 9.13
volumes.analytics.unsupported_reason.code	string	query	False	Filter by volumes.analytics.unsupported_reason.code • Introduced in: 9.13

Name	Type	In	Required	Description
volumes.analytics.unsupported_reason.message	string	query	False	Filter by volumes.analytics.unsupported_reason.message • Introduced in: 9.13
volumes.analytics.supported	boolean	query	False	Filter by volumes.analytics.supported • Introduced in: 9.13
volumes.analytics.state	string	query	False	Filter by volumes.analytics.state • Introduced in: 9.13
volumes.space.available	integer	query	False	Filter by volumes.space.available
volumes.space.used	integer	query	False	Filter by volumes.space.used
volumes.space.size	integer	query	False	Filter by volumes.space.size
volumes.activity_tracking.unsupported_reason.code	string	query	False	Filter by volumes.activity_tracking.unsupported_reason.code • Introduced in: 9.13
volumes.activity_tracking.unsupported_reason.message	string	query	False	Filter by volumes.activity_tracking.unsupported_reason.message • Introduced in: 9.13

Name	Type	In	Required	Description
volumes.activity_tracking.state	string	query	False	Filter by volumes.activity_tracking.state • Introduced in: 9.13
volumes.activity_tracking.supported	boolean	query	False	Filter by volumes.activity_tracking.supported • Introduced in: 9.13
volumes.language	string	query	False	Filter by volumes.language
clone.split_initiated	boolean	query	False	Filter by clone.split_initiated • Introduced in: 9.12
clone.parent_snapshot.name	string	query	False	Filter by clone.parent_snapshot.name • Introduced in: 9.12
clone.guarantee.type	string	query	False	Filter by clone.guarantee.type • Introduced in: 9.12
clone.parent_consistency_group.uuid	string	query	False	Filter by clone.parent_consistency_group.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
clone.parent_consistency_group.name	string	query	False	Filter by clone.parent_consistency_group.name • Introduced in: 9.12
clone.volume.prefix	string	query	False	Filter by clone.volume.prefix • Introduced in: 9.12
clone.volume.suffix	string	query	False	Filter by clone.volume.suffix • Introduced in: 9.12
application.type	string	query	False	Filter by application.type • Introduced in: 9.12
application.component_type	string	query	False	Filter by application.component_type • Introduced in: 9.12
tiering.policy	string	query	False	Filter by tiering.policy
name	string	query	False	Filter by name
namespaces.os_type	string	query	False	Filter by namespaces.os_type • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.subsystem_map.subsystem.os_type	string	query	False	Filter by namespaces.subsystem_map.subsystem.os_type • Introduced in: 9.12
namespaces.subsystem_map.subsystem.name	string	query	False	Filter by namespaces.subsystem_map.subsystem.name • Introduced in: 9.12 • maxLength: 96 • minLength: 1
namespaces.subsystem_map.subsystem.comment	string	query	False	Filter by namespaces.subsystem_map.subsystem.comment • Introduced in: 9.12 • maxLength: 255 • minLength: 0
namespaces.subsystem_map.subsystem.hosts.nqn	string	query	False	Filter by namespaces.subsystem_map.subsystem.hosts.nqn • Introduced in: 9.12
namespaces.subsystem_map.subsystem.uuid	string	query	False	Filter by namespaces.subsystem_map.subsystem.uuid • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.subsystem_map.nsid	string	query	False	Filter by namespaces.subsystem_map.nsid • Introduced in: 9.12
namespaces.subsystem_map.anagrpid	string	query	False	Filter by namespaces.subsystem_map.anagrpid • Introduced in: 9.12
namespaces.space.used	integer	query	False	Filter by namespaces.space.used • Introduced in: 9.12
namespaces.space.guarantee.requested	boolean	query	False	Filter by namespaces.space.guarantee.requested • Introduced in: 9.12
namespaces.space.guarantee.reserved	boolean	query	False	Filter by namespaces.space.guarantee.reserved • Introduced in: 9.12
namespaces.space.size	integer	query	False	Filter by namespaces.space.size • Introduced in: 9.12 • Max value: 140737488355328 • Min value: 4096

Name	Type	In	Required	Description
namespaces.space.block_size	integer	query	False	Filter by namespaces.space.block_size • Introduced in: 9.12
namespaces.enabled	boolean	query	False	Filter by namespaces.enabled • Introduced in: 9.12
namespaces.status.mapped	boolean	query	False	Filter by namespaces.status.mapped • Introduced in: 9.12
namespaces.status.state	string	query	False	Filter by namespaces.status.state • Introduced in: 9.12
namespaces.status.read_only	boolean	query	False	Filter by namespaces.status.read_only • Introduced in: 9.12
namespaces.status.container_state	string	query	False	Filter by namespaces.status.container_state • Introduced in: 9.12

Name	Type	In	Required	Description
namespaces.comment	string	query	False	Filter by namespaces.comment • Introduced in: 9.12 • maxLength: 254 • minLength: 0
namespaces.create_time	string	query	False	Filter by namespaces.create_time • Introduced in: 9.12
namespaces.uuid	string	query	False	Filter by namespaces.uuid • Introduced in: 9.12
namespaces.name	string	query	False	Filter by namespaces.name • Introduced in: 9.12
namespaces.auto_delete	boolean	query	False	Filter by namespaces.auto_delete • Introduced in: 9.12
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Max value: 120 • Min value: 0 • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.

Name	Type	Description
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "application": {
    "component_type": "data",
    "type": "oracle"
  },
  "clone": {
    "guarantee": {
      "type": "volume"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "volume": {
      "prefix": "string"
    }
  },
  "consistency_groups": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "application": {
      "component_type": "data",
      "type": "oracle"
    },
    "luns": {
      "clone": {
        "source": {
          "name": "/vol/volume1/lun1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
```

```
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "lun_maps": {
        "igroup": {
          "comment": "string",
          "igroups": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "igroup1",
            "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
          },
          "initiators": {
            "comment": "my comment",
            "name": "iqn.1998-01.com.corp.iscsi:name1"
          },
          "name": "igroup1",
          "os_type": "aix",
          "protocol": "fcp",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "name": "/vol/volume1/lun1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "serial_number": "string",
      "space": {
```

```
        "size": 1073741824,
        "used": 0
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "namespaces": {
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "name": "/vol/volume1/qtree1/namespace1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "space": {
        "block_size": 512,
        "size": 1073741824,
        "used": 0
      },
      "status": {
        "container_state": "online",
        "state": "online"
      },
      "subsystem_map": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "anagrpid": "00103050h",
        "nsid": "00000001h",
        "subsystem": {
          "comment": "string",
          "hosts": {
            "nqn": "nqn.1992-01.example.com:string"
          },
          "name": "subsystem1",
          "os_type": "aix",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
```

```
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 5737418,
      "size": 1073741824,
      "used": 5737418
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
```

```
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "volumes": {
      "activity_tracking": {
        "state": "off",
        "unsupported_reason": {
          "code": "124518405",
          "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
        }
      },
      "analytics": {
        "scan_progress": 17,
        "state": "unknown",
        "unsupported_reason": {
          "code": "111411207",
          "message": "File system analytics cannot be enabled on
volumes that contain LUNs."
        }
      },
      "comment": "string",
      "language": "ar",
      "name": "vol_cs_dept",
      "nas": {
        "cifs": {
          "shares": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "acls": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "permission": "no_access",
```

```
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
          },
          "name": "vs1_root",
          "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
        },
        "path": "/user/my_volume",
        "security_style": "mixed",
        "unix_permissions": 493
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "space": {
        "available": 0,
        "used": 0
      },
      "tiering": {
        "control": "allowed",
        "object_stores": {
        },
```

```
        "policy": "all"
      },
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  },
  "luns": {
    "clone": {
      "source": {
        "name": "/vol/volume1/lun1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "lun_maps": {
      "igroup": {
        "comment": "string",
        "igroups": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "igroup1",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        },
        "initiators": {
          "comment": "my comment",
          "name": "iqn.1998-01.com.corp.iscsi:name1"
        },
        "name": "igroup1",
        "os_type": "aix",
        "protocol": "fcp",
        "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "name": "/vol/volume1/lun1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
```

```
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "serial_number": "string",
    "space": {
      "size": 1073741824,
      "used": 0
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "available_space": 4096,
    "duration": "PT15S",
    "iops": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "latency": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "namespaces": {
```

```
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "name": "/vol/volume1/qtree1/namespace1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "space": {
      "block_size": 512,
      "size": 1073741824,
      "used": 0
    },
    "status": {
      "container_state": "online",
      "state": "online"
    },
    "subsystem_map": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "anagrpid": "00103050h",
      "nsid": "00000001h",
      "subsystem": {
        "comment": "string",
        "hosts": {
          "nqn": "nqn.1992-01.example.com:string"
        },
        "name": "subsystem1",
        "os_type": "aix",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "parent_consistency_group": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "my_consistency_group",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "provisioning_options": {
```

```
    "action": "create",
    "storage_service": {
      "name": "extreme"
    }
  },
  "qos": {
    "policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "max_throughput_iops": 10000,
      "max_throughput_mbps": 500,
      "min_throughput_iops": 2000,
      "min_throughput_mbps": 500,
      "name": "performance",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "replication_relationships": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "snapshot_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "space": {
    "available": 5737418,
    "size": 1073741824,
    "used": 5737418
  },
  "statistics": {
    "available_space": 4096,
    "iops_raw": {
      "read": 200,
```

```
      "total": 1000,
      "write": 100
    },
    "latency_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "tiering": {
    "control": "allowed",
    "object_stores": {
    },
    "policy": "all"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "volumes": {
    "activity_tracking": {
      "state": "off",
      "unsupported_reason": {
        "code": "124518405",
        "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
      }
    },
    "analytics": {
      "scan_progress": 17,
      "state": "unknown",
```

```
      "unsupported_reason": {
        "code": "111411207",
        "message": "File system analytics cannot be enabled on volumes
that contain LUNs."
      }
    },
    "comment": "string",
    "language": "ar",
    "name": "vol_cs_dept",
    "nas": {
      "cifs": {
        "shares": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "acls": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "permission": "no_access",
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
```

```
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "vs1_root",
        "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
      },
      "path": "/user/my_volume",
      "security_style": "mixed",
      "unix_permissions": 493
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
```

```
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 0,
      "used": 0
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
  }
}
```


Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
53411842	Consistency group does not exist.
53411843	A consistency group with specified UUID was not found.
53411844	Specified consistency group was not found in the specified SVM.

Error Code	Description
53411845	The specified UUID and name refer to different consistency groups.
53411846	Either name or UUID must be provided.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

self_link

Name	Type	Description
self	href	

application

Name	Type	Description
component_type	string	Nested consistency group tag.
type	string	Top level consistency group tag.

guarantee

Name	Type	Description
type	string	The type of space guarantee of this volume in the aggregate.

parent_consistency_group

Consistency group that is to be cloned.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

parent_snapshot

Name	Type	Description
name	string	Name of an existing Snapshot copy of a parent consistency group.

volume

Volume name suffix/prefix for the cloned volumes.

Name	Type	Description
prefix	string	Volume name prefix for cloned volumes.
suffix	string	Volume name suffix for cloned volumes.

clone

Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
guarantee	guarantee	
parent_consistency_group	parent_consistency_group	Consistency group that is to be cloned.
parent_snapshot	parent_snapshot	
split_initiated	boolean	Splits volumes after cloning. Default is false.
volume	volume	Volume name suffix/prefix for the cloned volumes.

source

The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN.

Valid in POST to create a new LUN as a clone of the source.

Valid in PATCH to overwrite an existing LUN's data as a clone of another.

Name	Type	Description
name	string	The fully qualified path name of the clone source LUN composed of a "/vol" prefix, the volume name, the (optional) qtree name, and base name of the LUN. Valid in POST and PATCH.

Name	Type	Description
uuid	string	The unique identifier of the clone source LUN. Valid in POST and PATCH.

clone

This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`.

When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`.

Persistent reservations for the patched LUN are also preserved.

Name	Type	Description
source	source	The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN. Valid in POST to create a new LUN as a clone of the source. Valid in PATCH to overwrite an existing LUN's data as a clone of another.

igroups

Name	Type	Description
_links	self_link	
name	string	The name of the initiator group.
uuid	string	The unique identifier of the initiator group.

initiators

The initiators that are members of the initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator.
name	string	Name of initiator that is a member of the initiator group.

igroup

The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator. Valid in POST and PATCH.
igroups	array[igroups]	Separate igroup definitions to include in this igroup.
initiators	array[initiators]	The initiators that are members of the group.
name	string	The name of the initiator group. Required in POST; optional in PATCH.
os_type	string	The host operating system of the initiator group. All initiators in the group should be hosts of the same operating system. Required in POST; optional in PATCH.
protocol	string	The protocols supported by the initiator group. This restricts the type of initiators that can be added to the initiator group. Optional in POST; if not supplied, this defaults to *mixed*. The protocol of an initiator group cannot be changed after creation of the group.
uuid	string	The unique identifier of the initiator group.

lun_maps

A LUN map is an association between a LUN and an initiator group.

When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.

Name	Type	Description
igroup	igroup	The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.
logical_unit_number	integer	The logical unit number assigned to the LUN when mapped to the specified initiator group. The number is used to identify the LUN to initiators in the initiator group when communicating through the Fibre Channel Protocol or iSCSI. Optional in POST; if no value is provided, ONTAP assigns the lowest available value. • Introduced in: 9.6 • readCreate: 1 • x-nullable: true

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

policy

The QoS policy

Name	Type	Description
_links	self_link	

Name	Type	Description
max_throughput_iops	integer	Specifies the maximum throughput in IOPS, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
max_throughput_mbps	integer	Specifies the maximum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_iops	integer	Specifies the minimum throughput in IOPS, 0 means none. Setting "min_throughput" is supported on AFF platforms only, unless FabricPool tiering policies are set. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_mbps	integer	Specifies the minimum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
name	string	The QoS policy group name. This is mutually exclusive with UUID and other QoS attributes during POST and PATCH.
uuid	string	The QoS policy group UUID. This is mutually exclusive with name and other QoS attributes during POST and PATCH.

qos

Name	Type	Description
policy	policy	The QoS policy

guarantee

Properties that request and report the space guarantee for the LUN.

Name	Type	Description
requested	boolean	The requested space reservation policy for the LUN. If *true*, a space reservation is requested for the LUN; if *false*, the LUN is thin provisioned. Guaranteeing a space reservation request for a LUN requires that the volume in which the LUN resides is also space reserved and that the fractional reserve for the volume is 100%. Valid in POST and PATCH.
reserved	boolean	Reports if the LUN is space guaranteed. If *true*, a space guarantee is requested and the containing volume and aggregate support the request. If *false*, a space guarantee is not requested or a space guarantee is requested and either the containing volume or aggregate do not support the request.

space

The storage space related properties of the LUN.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the LUN.

Name	Type	Description
size	integer	The total provisioned size of the LUN. The LUN size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The actual minimum and maxiumum sizes vary depending on the ONTAP version, ONTAP platform, and the available space in the containing volume and aggregate. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the LUN. This value is the total space consumed in the volume by the LUN, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways SAN filesystems and applications utilize blocks within a LUN, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the LUN blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

luns

A LUN is the logical representation of storage in a storage area network (SAN).

In ONTAP, a LUN is located within a volume. Optionally, it can be located within a qtree in a volume.

A LUN can be created to a specified size using thin or thick provisioning. A LUN can then be renamed, resized, cloned, and moved to a different volume. LUNs support the assignment of a quality of service (QoS) policy for performance management or a QoS policy can be assigned to the volume containing the LUN. See the LUN object model to learn more about each of the properties supported by the LUN REST API.

A LUN must be mapped to an initiator group to grant access to the initiator group's initiators (client hosts). Initiators can then access the LUN and perform I/O over a Fibre Channel (FC) fabric using the Fibre Channel Protocol or a TCP/IP network using iSCSI.

Name	Type	Description
clone	clone	This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`. When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`. Persistent reservations for the patched LUN are also preserved.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the LUN was created.

Name	Type	Description
enabled	boolean	The enabled state of the LUN. LUNs can be disabled to prevent access to the LUN. Certain error conditions also cause the LUN to become disabled. If the LUN is disabled, you can consult the `state` property to determine if the LUN is administratively disabled (*offline*) or has become disabled as a result of an error. A LUN in an error condition can be brought online by setting the `enabled` property to *true* or brought administratively offline by setting the `enabled` property to *false*. Upon creation, a LUN is enabled by default. Valid in PATCH.
lun_maps	array[lun_maps]	An array of LUN maps. A LUN map is an association between a LUN and an initiator group. When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.
name	string	The fully qualified path name of the LUN composed of the "/vol" prefix, the volume name, the qtree name (optional), and the base name of the LUN. Valid in POST and PATCH.
os_type	string	The operating system type of the LUN. Required in POST when creating a LUN that is not a clone of another. Disallowed in POST when creating a LUN clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
serial_number	string	The LUN serial number. The serial number is generated by ONTAP when the LUN is created. • maxLength: 12 • minLength: 12 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
space	space	The storage space related properties of the LUN.
uuid	string	The unique identifier of the LUN. The UUID is generated by ONTAP when the LUN is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

guarantee

Properties that request and report the space guarantee for the NVMe namespace.

Name	Type	Description
requested	boolean	The requested space reservation policy for the NVMe namespace. If *true*, a space reservation is requested for the namespace; if *false*, the namespace is thin provisioned. Guaranteeing a space reservation request for a namespace requires that the volume in which the namespace resides also be space reserved and that the fractional reserve for the volume be 100%. The space reservation policy for an NVMe namespace is determined by ONTAP. • Introduced in: 9.6 • x-nullable: true
reserved	boolean	Reports if the NVMe namespace is space guaranteed. This property is *true* if a space guarantee is requested and the containing volume and aggregate support the request. This property is *false* if a space guarantee is not requested or if a space guarantee is requested and either the containing volume and aggregate do not support the request.

space

The storage space related properties of the NVMe namespace.

Name	Type	Description
block_size	integer	The size of blocks in the namespace, in bytes. Valid in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone. Valid in POST.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the NVMe namespace.
size	integer	The total provisioned size of the NVMe namespace. Valid in POST and PATCH. The NVMe namespace size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The maximum size is variable with respect to large NVMe namespace support in ONTAP. If large namespaces are supported, the maximum size is 128 TB (140737488355328 bytes) and if not supported, the maximum size is just under 16 TB (17557557870592 bytes). The minimum size supported is always 4096 bytes. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the NVMe namespace. This value is the total space consumed in the volume by the NVMe namespace, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways NVMe filesystems and applications utilize blocks within a namespace, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the namespace blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

status

Status information about the NVMe namespace.

Name	Type	Description
container_state	string	The state of the volume and aggregate that contain the NVMe namespace. Namespaces are only available when their containers are available.

Name	Type	Description
mapped	boolean	Reports if the NVMe namespace is mapped to an NVMe subsystem. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
read_only	boolean	Reports if the NVMe namespace allows only read access.
state	string	The state of the NVMe namespace. Normal states for a namespace are *online* and *offline*. Other states indicate errors.

consistency_group_nvme_host

The NVMe host provisioned to access NVMe namespaces mapped to a subsystem.

Name	Type	Description
nqn	string	The NVMe qualified name (NQN) used to identify the NVMe storage target. Not allowed in POST when the `records` property is used.

consistency_group_nvme_subsystem

An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

Name	Type	Description
comment	string	A configurable comment for the NVMe subsystem. Optional in POST and PATCH.
hosts	array[consistency_group_nvme_host]	The NVMe hosts configured for access to the NVMe subsystem. Optional in POST.

Name	Type	Description
name	string	The name of the NVMe subsystem. Once created, an NVMe subsystem cannot be renamed. Required in POST.
os_type	string	The host operating system of the NVMe subsystem's hosts. Required in POST.
uuid	string	The unique identifier of the NVMe subsystem.

subsystem_map

The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems.

There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.

Name	Type	Description
_links	self_link	
anagrpid	string	The Asymmetric Namespace Access Group ID (ANAGRPID) of the NVMe namespace. The format for an ANAGRPID is 8 hexadecimal digits (zero-filled) followed by a lower case "h". There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.

Name	Type	Description
nsid	string	The NVMe namespace identifier. This is an identifier used by an NVMe controller to provide access to the NVMe namespace. The format for an NVMe namespace identifier is 8 hexadecimal digits (zero-filled) followed by a lower case "h".
subsystem	consistency_group_nvme_subsystem	An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

namespaces

An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the storage virtual machine using the NVMe over Fabrics protocol.

In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume.

An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API.

An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem’s hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol.

Name	Type	Description
auto_delete	boolean	This property marks the NVMe namespace for auto deletion when the volume containing the namespace runs out of space. This is most commonly set on namespace clones. When set to *true*, the NVMe namespace becomes eligible for automatic deletion when the volume runs out of space. Auto deletion only occurs when the volume containing the namespace is also configured for auto deletion and free space in the volume decreases below a particular threshold. This property is optional in POST and PATCH. The default value for a new NVMe namespace is *false*. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the NVMe namespace was created.
enabled	boolean	The enabled state of the NVMe namespace. Certain error conditions cause the namespace to become disabled. If the namespace is disabled, you can check the `state` property to determine what error disabled the namespace. An NVMe namespace is enabled automatically when it is created.

Name	Type	Description
name	string	The fully qualified path name of the NVMe namespace composed of a "/vol" prefix, the volume name, the (optional) qtree name and base name of the namespace. Valid in POST. NVMe namespaces do not support rename, or movement between volumes.
os_type	string	The operating system type of the NVMe namespace. Required in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
space	space	The storage space related properties of the NVMe namespace.
status	status	Status information about the NVMe namespace.
subsystem_map	subsystem_map	The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems. There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.
uuid	string	The unique identifier of the NVMe namespace.

parent_consistency_group

The parent consistency group.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

storage_service

Determines the placement of any storage object created during this operation.

Name	Type	Description
name	string	Storage service name. If not specified, the default value is the most performant for the platform.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
name	string	New name for consistency group. Required to resolve naming collisions.
storage_service	storage_service	Determines the placement of any storage object created during this operation.

snapshot

A consistency group's Snapshot copy

Name	Type	Description
name	string	The name of the consistency group's Snapshot copy to restore to.

Name	Type	Description
uuid	string	The UUID of the consistency group's Snapshot copy to restore to.

restore_to

Use to restore a consistency group to a previous Snapshot copy

Name	Type	Description
snapshot	snapshot	A consistency group's Snapshot copy

_links

Name	Type	Description
self	href	

snapshot_policy_reference

This is a reference to the Snapshot copy policy.

Name	Type	Description
_links	_links	
name	string	
uuid	string	

space

Space information for the consistency group.

Name	Type	Description
available	integer	The amount of space available in the consistency group, in bytes.
size	integer	The total provisioned size of the consistency group, in bytes.
used	integer	The amount of space consumed in the consistency group, in bytes.

svm

The Storage Virtual Machine (SVM) in which the consistency group is located.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

object_stores

Name	Type	Description
name	string	The name of the object store to use. Used for placement.

tiering

The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

unsupported_reason

Name	Type	Description
code	string	If volume activity tracking is not supported on the volume, this field provides an appropriate error code.

Name	Type	Description
message	string	If volume activity tracking is not supported on the volume, this field provides an error message detailing why this is the case.

activity_tracking

The volume activity tracking configuration for this volume.

Name	Type	Description
state	string	Activity tracking state of the volume. If this value is *on*, ONTAP collects top metrics information for the volume in real time. There is a slight impact to I/O performance in order to collect this information. If this value is *off*, no activity tracking information is collected or available to view. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["off", "on"] • Introduced in: 9.10 • x-nullable: true
supported	boolean	This field indicates whether or not volume activity tracking is supported on the volume. If volume activity tracking is not supported, the reason why is provided in the `activity_tracking.unsupported_reason` field.
unsupported_reason	unsupported_reason	

unsupported_reason

Name	Type	Description
code	string	If file system analytics is not supported on the volume, this field provides the error code explaining why.
message	string	If file system analytics is not supported on the volume, this field provides the error message explaining why.

analytics

The file system analytics configuration for this volume.

Name	Type	Description
scan_progress	integer	Percentage of files in the volume that the file system analytics initialization scan has processed. Only returned when the `state` is *initializing*.

Name	Type	Description
state	string	File system analytics state of the volume. If this value is *on*, ONTAP collects extra file system analytics information for all directories on the volume. There will be a slight impact to I/O performance to collect this information. If this value is *off*, file system analytics information is not collected and not available to be viewed. If this value is *initializing*, that means file system analytics was recently turned on, and the initialization scan to gather information for all existing files and directories is currently running. If this value is *initialization_paused*, this means that the initialization scan is currently paused. If this value is 'unknown', this means that there was an internal error when determining the file system analytics state for the volume. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["unknown", "initializing", "initialization_paused", "off", "on"] • Introduced in: 9.8 • x-nullable: true
supported	boolean	This field indicates whether or not file system analytics is supported on the volume. If file system analytics is not supported, the reason will be specified in the `analytics.unsupported_reason` field.
unsupported_reason	unsupported_reason	

acls

The permissions that users and groups have on a CIFS share.

Name	Type	Description
_links	_links	
permission	string	Specifies the access rights that a user or group has on the defined CIFS Share. The following values are allowed: • no_access - User does not have CIFS share access • read - User has only read access • change - User has change access • full_control - User has full_control access
type	string	Specifies the type of the user or group to add to the access control list of a CIFS share. The following values are allowed: • windows - Windows user or group • unix_user - UNIX user • unix_group - UNIX group
user_or_group	string	Specifies the user or group name to add to the access control list of a CIFS share.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.

Name	Type	Description
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

cifs

Name	Type	Description
shares	array[consistency_group_cifs_share]	

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

export_rules

Name	Type	Description
_links	_links	
allow_device_creation	boolean	Specifies whether or not device creation is allowed.
allow_suid	boolean	Specifies whether or not SetUID bits in SETATTR Op is to be honored.
anonymous_user	string	User ID To Which Anonymous Users Are Mapped.

Name	Type	Description
chown_mode	string	Specifies who is authorized to change the ownership mode of a file.
clients	array[export_clients]	Array of client matches
index	integer	Index of the rule within the export policy.
ntfs_unix_security	string	NTFS export UNIX security options.
protocols	array[string]	
ro_rule	array[string]	Authentication flavors that the read-only access rule governs
rw_rule	array[string]	Authentication flavors that the read/write access rule governs
superuser	array[string]	Authentication flavors that the superuser security type governs

export_policy

The policy associated with volumes to export them for protocol access.

Name	Type	Description
_links	self_link	
name	string	Name of the export policy.
rules	array[export_rules]	The set of rules that govern the export policy.
uuid	string	Identifier for the export policy.

junction_parent

Name	Type	Description
_links	self_link	

Name	Type	Description
name	string	The name of the parent volume that contains the junction inode of this volume. The junction parent volume must belong to the same SVM that owns this volume.
uuid	string	Unique identifier for the parent volume.

nas

The CIFS share policy and/or export policies for this volume.

Name	Type	Description
cifs	cifs	
export_policy	export_policy	The policy associated with volumes to export them for protocol access.
gid	integer	The UNIX group ID of the volume. Valid in POST or PATCH.
junction_parent	junction_parent	
path	string	The fully-qualified path in the owning SVM's namespace at which the volume is mounted. The path is case insensitive and must be unique within an SVM's namespace. Path must begin with '/' and must not end with '/'. Only one volume can be mounted at any given junction path. An empty path in POST creates an unmounted volume. An empty path in PATCH deactivates and unmounts the volume. Taking a volume offline or restricted state removes its junction path. This attribute is reported in GET only when the volume is mounted.

Name	Type	Description
security_style	string	Security style associated with the volume. Valid in POST or PATCH. mixed ‐ Mixed-style security ntfs ‐ NTFS/WIndows-style security unified ‐ Unified-style security, unified UNIX, NFS and CIFS permissions unix ‐ UNIX-style security.
uid	integer	The UNIX user ID of the volume. Valid in POST or PATCH.
unix_permissions	integer	UNIX permissions to be viewed as an octal number, consisting of 4 digits derived by adding up bits 4 (read), 2 (write), and 1 (execute). First digit selects the set user ID (4), set group ID (2), and sticky (1) attributes. Second digit selects permission for the owner of the file. Third selects permissions for other users in the same group while the fourth selects permissions for other users not in the group. Valid in POST or PATCH. For security style "mixed" or "unix", the default setting is 0755 in octal (493 in decimal) and for security style "ntfs", the default setting is 0000. In cases where only owner, group, and other permissions are given (as in 755, representing the second, third and fourth digit), the first digit is assumed to be zero.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

Name	Type	Description
storage_service	storage_service	Determines the placement of any storage object created during this operation.

qos

The QoS policy for this volume.

Name	Type	Description
policy	policy	The QoS policy

space

Name	Type	Description
available	integer	The available space, in bytes.
size	integer	Total provisioned size, in bytes.
used	integer	The virtual space used (includes volume reserves) before storage efficiency, in bytes.

tiering

The tiering placement and policy definitions for this volume.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

volumes

Name	Type	Description
activity_tracking	activity_tracking	The volume activity tracking configuration for this volume.
analytics	analytics	The file system analytics configuration for this volume.

Name	Type	Description
comment	string	A comment for the volume. Valid in POST or PATCH.
language	string	Language encoding setting for volume. If no language is specified, the volume inherits its SVM language encoding setting.
name	string	Volume name. The name of volume must start with an alphabetic character (a to z or A to Z) or an underscore (_). The name must be 197 or fewer characters in length for FlexGroups, and 203 or fewer characters in length for all other types of volumes. Volume names must be unique within an SVM. Required on POST.
nas	nas	The CIFS share policy and/or export policies for this volume.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	The QoS policy for this volume.
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	
tiering	tiering	The tiering placement and policy definitions for this volume.
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • readOnly: 1 • Introduced in: 9.8 • x-nullable: true

consistency_groups

Name	Type	Description
_links	self_link	
application	application	
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

Name	Type	Description
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

iops

The rate of I/O operations observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency

The round trip latency observed at the storage object, in microseconds.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput

The rate of throughput bytes per second observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

metric

Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.

Name	Type	Description
_links	_links	
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
duration	string	The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations:
iops	iops	The rate of I/O operations observed at the storage object.
latency	latency	The round trip latency observed at the storage object, in microseconds.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput	throughput	The rate of throughput bytes per second observed at the storage object.
timestamp	string	The timestamp of the performance and capacity data.

Name	Type	Description
used_space	integer	The total space used in the consistency group, in bytes.

replication_relationships

Name	Type	Description
_links	self_link	
is_source	boolean	Indicates whether or not this consistency group is the source for replication.
uuid	string	The unique identifier of the SnapMirror relationship.

iops_raw

The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency_raw

The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput_raw

Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

statistics

These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.

Name	Type	Description
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
iops_raw	iops_raw	The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.
latency_raw	latency_raw	The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput_raw	throughput_raw	Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.
timestamp	string	The timestamp of the performance data.

Name	Type	Description
used_space	integer	The total used space in the consistency group, in bytes.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update a consistency group

PATCH `/application/consistency-groups/{uuid}`

Introduced In: 9.10

Updates a consistency group.

> that this operation will never delete storage elements. You can specify only elements that should be added to the consistency group regardless of existing storage objects.

Related ONTAP commands

N/A. There are no ONTAP commands for managing consistency groups.

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	The unique identifier of the consistency group to modify.
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Request Body

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "application": {
    "component_type": "data",
    "type": "oracle"
  },
  "clone": {
    "guarantee": {
      "type": "volume"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "volume": {
      "prefix": "string"
    }
  },
  "consistency_groups": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "application": {
      "component_type": "data",
      "type": "oracle"
    },
    "luns": {
      "clone": {
        "source": {
          "name": "/vol/volume1/lun1",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
```

```
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "lun_maps": {
        "igroup": {
          "comment": "string",
          "igroups": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "name": "igroup1",
            "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
          },
          "initiators": {
            "comment": "my comment",
            "name": "iqn.1998-01.com.corp.iscsi:name1"
          },
          "name": "igroup1",
          "os_type": "aix",
          "protocol": "fcp",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "name": "/vol/volume1/lun1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "serial_number": "string",
      "space": {
```

```
        "size": 1073741824,
        "used": 0
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "namespaces": {
      "comment": "string",
      "create_time": "2018-06-04 19:00:00 +0000",
      "name": "/vol/volume1/qtree1/namespace1",
      "os_type": "aix",
      "provisioning_options": {
        "action": "create"
      },
      "space": {
        "block_size": 512,
        "size": 1073741824,
        "used": 0
      },
      "status": {
        "container_state": "online",
        "state": "online"
      },
      "subsystem_map": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "anagrpid": "00103050h",
        "nsid": "00000001h",
        "subsystem": {
          "comment": "string",
          "hosts": {
            "nqn": "nqn.1992-01.example.com:string"
          },
          "name": "subsystem1",
          "os_type": "aix",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "parent_consistency_group": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
```

```
        }
      },
      "name": "my_consistency_group",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 5737418,
      "size": 1073741824,
      "used": 5737418
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
```

```
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "volumes": {
      "activity_tracking": {
        "state": "off",
        "unsupported_reason": {
          "code": "124518405",
          "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
        }
      },
      "analytics": {
        "scan_progress": 17,
        "state": "unknown",
        "unsupported_reason": {
          "code": "111411207",
          "message": "File system analytics cannot be enabled on
volumes that contain LUNs."
        }
      },
      "comment": "string",
      "language": "ar",
      "name": "vol_cs_dept",
      "nas": {
        "cifs": {
          "shares": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "acls": {
              "_links": {
                "self": {
                  "href": "/api/resourcelink"
                }
              },
              "permission": "no_access",
```

```
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
```

```
          },
          "name": "vs1_root",
          "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
        },
        "path": "/user/my_volume",
        "security_style": "mixed",
        "unix_permissions": 493
      },
      "provisioning_options": {
        "action": "create",
        "storage_service": {
          "name": "extreme"
        }
      },
      "qos": {
        "policy": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "max_throughput_iops": 10000,
          "max_throughput_mbps": 500,
          "min_throughput_iops": 2000,
          "min_throughput_mbps": 500,
          "name": "performance",
          "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
        }
      },
      "snapshot_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "default",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      },
      "space": {
        "available": 0,
        "used": 0
      },
      "tiering": {
        "control": "allowed",
        "object_stores": {
        },
```

```
        "policy": "all"
      },
      "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
    }
  },
  "luns": {
    "clone": {
      "source": {
        "name": "/vol/volume1/lun1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "lun_maps": {
      "igroup": {
        "comment": "string",
        "igroups": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "name": "igroup1",
          "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
        },
        "initiators": {
          "comment": "my comment",
          "name": "iqn.1998-01.com.corp.iscsi:name1"
        },
        "name": "igroup1",
        "os_type": "aix",
        "protocol": "fcp",
        "uuid": "4ea7a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "name": "/vol/volume1/lun1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
```

```
          }
        },
        "max_throughput_iops": 10000,
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "serial_number": "string",
    "space": {
      "size": 1073741824,
      "used": 0
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "metric": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "available_space": 4096,
    "duration": "PT15S",
    "iops": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "latency": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "namespaces": {
```

```
    "comment": "string",
    "create_time": "2018-06-04 19:00:00 +0000",
    "name": "/vol/volume1/qtree1/namespace1",
    "os_type": "aix",
    "provisioning_options": {
      "action": "create"
    },
    "space": {
      "block_size": 512,
      "size": 1073741824,
      "used": 0
    },
    "status": {
      "container_state": "online",
      "state": "online"
    },
    "subsystem_map": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "anagrpid": "00103050h",
      "nsid": "00000001h",
      "subsystem": {
        "comment": "string",
        "hosts": {
          "nqn": "nqn.1992-01.example.com:string"
        },
        "name": "subsystem1",
        "os_type": "aix",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "parent_consistency_group": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "my_consistency_group",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "provisioning_options": {
```

```
    "action": "create",
    "storage_service": {
      "name": "extreme"
    }
  },
  "qos": {
    "policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "max_throughput_iops": 10000,
      "max_throughput_mbps": 500,
      "min_throughput_iops": 2000,
      "min_throughput_mbps": 500,
      "name": "performance",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "replication_relationships": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "snapshot_policy": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "space": {
    "available": 5737418,
    "size": 1073741824,
    "used": 5737418
  },
  "statistics": {
    "available_space": 4096,
    "iops_raw": {
      "read": 200,
```

```
      "total": 1000,
      "write": 100
    },
    "latency_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "size": 4096,
    "status": "ok",
    "throughput_raw": {
      "read": 200,
      "total": 1000,
      "write": 100
    },
    "timestamp": "2017-01-25 11:20:13 +0000",
    "used_space": 4096
  },
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "tiering": {
    "control": "allowed",
    "object_stores": {
    },
    "policy": "all"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "volumes": {
    "activity_tracking": {
      "state": "off",
      "unsupported_reason": {
        "code": "124518405",
        "message": "Volume activity tracking cannot be enabled on
volumes that contain LUNs."
      }
    },
    "analytics": {
      "scan_progress": 17,
      "state": "unknown",
```

```
      "unsupported_reason": {
        "code": "111411207",
        "message": "File system analytics cannot be enabled on volumes
that contain LUNs."
      }
    },
    "comment": "string",
    "language": "ar",
    "name": "vol_cs_dept",
    "nas": {
      "cifs": {
        "shares": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
            }
          },
          "acls": {
            "_links": {
              "self": {
                "href": "/api/resourcelink"
              }
            },
            "permission": "no_access",
            "type": "windows",
            "user_or_group": "ENGDOMAIN\\ad_user"
          },
          "comment": "HR Department Share",
          "dir_umask": 18,
          "file_umask": 18,
          "name": "HR_SHARE",
          "offline_files": "none",
          "unix_symlink": "local",
          "vscan_profile": "no_scan"
        }
      },
      "export_policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "rules": {
          "_links": {
            "self": {
              "href": "/api/resourcelink"
```

```
            }
          },
          "chown_mode": "restricted",
          "clients": {
            "match": "0.0.0.0/0"
          },
          "index": 0,
          "ntfs_unix_security": "fail",
          "protocols": {
          },
          "ro_rule": {
          },
          "rw_rule": {
          },
          "superuser": {
          }
        },
        "uuid": "string"
      },
      "junction_parent": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "vs1_root",
        "uuid": "75c9cfb0-3eb4-11eb-9fb4-005056bb088a"
      },
      "path": "/user/my_volume",
      "security_style": "mixed",
      "unix_permissions": 493
    },
    "provisioning_options": {
      "action": "create",
      "storage_service": {
        "name": "extreme"
      }
    },
    "qos": {
      "policy": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "max_throughput_iops": 10000,
```

```
        "max_throughput_mbps": 500,
        "min_throughput_iops": 2000,
        "min_throughput_mbps": 500,
        "name": "performance",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "snapshot_policy": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "space": {
      "available": 0,
      "used": 0
    },
    "tiering": {
      "control": "allowed",
      "object_stores": {
      },
      "policy": "all"
    },
    "uuid": "028baa66-41bd-11e9-81d5-00a0986138f7"
  }
}
```


Response

```
Status: 200, Ok
```

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
53411842	Consistency group does not exist.
53411843	A consistency group with specified UUID was not found.
53411844	Specified consistency group was not found in the specified SVM.
53411845	The specified UUID and name refer to different consistency groups.
53411846	Either name or UUID must be provided.
53411852	A consistency group with the same identifier in the same scope exists.
53411853	Fields provided in the request conflict with each other.
53411856	Field provided is only supported when provisioning new objects.
53411857	LUNs that are not members of the application are not supported by this API. LUNs can be added to an application by adding the volume containing the LUNs to the application.
53411860	An object with the same identifier in the same scope exists.
53411861	Volume specified does not exist in provided volume array.
53411862	Modifying existing igroups is not supported using this API.
53411864	Request content insufficient to add an existing volume to an application.
53411865	Volumes contained in one consistency group cannot be added to a different consistency group.
53411866	LUNs are not supported on FlexGroup volumes.
53411867	LUN name is too long after appending a unique suffix.
53411869	Volume name is too long after appending a unique suffix.
53411870	When using the "round_robin" layout, the volume count must not be greater than the LUN count.
53411959	Volumes with Snapshot copy locking enabled cannot be added to a consistency group.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

self_link

Name	Type	Description
self	href	

application

Name	Type	Description
component_type	string	Nested consistency group tag.
type	string	Top level consistency group tag.

guarantee

Name	Type	Description
type	string	The type of space guarantee of this volume in the aggregate.

parent_consistency_group

Consistency group that is to be cloned.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

parent_snapshot

Name	Type	Description
name	string	Name of an existing Snapshot copy of a parent consistency group.

volume

Volume name suffix/prefix for the cloned volumes.

Name	Type	Description
prefix	string	Volume name prefix for cloned volumes.
suffix	string	Volume name suffix for cloned volumes.

clone

Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.

Name	Type	Description
guarantee	guarantee	
parent_consistency_group	parent_consistency_group	Consistency group that is to be cloned.
parent_snapshot	parent_snapshot	
split_initiated	boolean	Splits volumes after cloning. Default is false.
volume	volume	Volume name suffix/prefix for the cloned volumes.

source

The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN.

Valid in POST to create a new LUN as a clone of the source.

Valid in PATCH to overwrite an existing LUN's data as a clone of another.

Name	Type	Description
name	string	The fully qualified path name of the clone source LUN composed of a "/vol" prefix, the volume name, the (optional) qtree name, and base name of the LUN. Valid in POST and PATCH.

Name	Type	Description
uuid	string	The unique identifier of the clone source LUN. Valid in POST and PATCH.

clone

This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`.

When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`.

Persistent reservations for the patched LUN are also preserved.

Name	Type	Description
source	source	The source LUN for a LUN clone operation. This can be specified using property `clone.source.uuid` or `clone.source.name`. If both properties are supplied, they must refer to the same LUN. Valid in POST to create a new LUN as a clone of the source. Valid in PATCH to overwrite an existing LUN's data as a clone of another.

igroups

Name	Type	Description
_links	self_link	
name	string	The name of the initiator group.
uuid	string	The unique identifier of the initiator group.

initiators

The initiators that are members of the initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator.
name	string	Name of initiator that is a member of the initiator group.

igroup

The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.

Name	Type	Description
comment	string	A comment available for use by the administrator. Valid in POST and PATCH.
igroups	array[igroups]	Separate igroup definitions to include in this igroup.
initiators	array[initiators]	The initiators that are members of the group.
name	string	The name of the initiator group. Required in POST; optional in PATCH.
os_type	string	The host operating system of the initiator group. All initiators in the group should be hosts of the same operating system. Required in POST; optional in PATCH.
protocol	string	The protocols supported by the initiator group. This restricts the type of initiators that can be added to the initiator group. Optional in POST; if not supplied, this defaults to *mixed*. The protocol of an initiator group cannot be changed after creation of the group.
uuid	string	The unique identifier of the initiator group.

lun_maps

A LUN map is an association between a LUN and an initiator group.

When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.

Name	Type	Description
igroup	igroup	The initiator group that directly owns the initiator, which is where modification of the initiator is supported. This property will only be populated when the initiator is a member of a nested initiator group.
logical_unit_number	integer	The logical unit number assigned to the LUN when mapped to the specified initiator group. The number is used to identify the LUN to initiators in the initiator group when communicating through the Fibre Channel Protocol or iSCSI. Optional in POST; if no value is provided, ONTAP assigns the lowest available value. • Introduced in: 9.6 • readCreate: 1 • x-nullable: true

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

policy

The QoS policy

Name	Type	Description
_links	self_link	

Name	Type	Description
max_throughput_iops	integer	Specifies the maximum throughput in IOPS, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
max_throughput_mbps	integer	Specifies the maximum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_iops	integer	Specifies the minimum throughput in IOPS, 0 means none. Setting "min_throughput" is supported on AFF platforms only, unless FabricPool tiering policies are set. This is mutually exclusive with name and UUID during POST and PATCH.
min_throughput_mbps	integer	Specifies the minimum throughput in Megabytes per sec, 0 means none. This is mutually exclusive with name and UUID during POST and PATCH.
name	string	The QoS policy group name. This is mutually exclusive with UUID and other QoS attributes during POST and PATCH.
uuid	string	The QoS policy group UUID. This is mutually exclusive with name and other QoS attributes during POST and PATCH.

qos

Name	Type	Description
policy	policy	The QoS policy

guarantee

Properties that request and report the space guarantee for the LUN.

Name	Type	Description
requested	boolean	The requested space reservation policy for the LUN. If *true*, a space reservation is requested for the LUN; if *false*, the LUN is thin provisioned. Guaranteeing a space reservation request for a LUN requires that the volume in which the LUN resides is also space reserved and that the fractional reserve for the volume is 100%. Valid in POST and PATCH.
reserved	boolean	Reports if the LUN is space guaranteed. If *true*, a space guarantee is requested and the containing volume and aggregate support the request. If *false*, a space guarantee is not requested or a space guarantee is requested and either the containing volume or aggregate do not support the request.

space

The storage space related properties of the LUN.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the LUN.

Name	Type	Description
size	integer	The total provisioned size of the LUN. The LUN size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The actual minimum and maxiumum sizes vary depending on the ONTAP version, ONTAP platform, and the available space in the containing volume and aggregate. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the LUN. This value is the total space consumed in the volume by the LUN, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways SAN filesystems and applications utilize blocks within a LUN, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the LUN blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

luns

A LUN is the logical representation of storage in a storage area network (SAN).

In ONTAP, a LUN is located within a volume. Optionally, it can be located within a qtree in a volume.

A LUN can be created to a specified size using thin or thick provisioning. A LUN can then be renamed, resized, cloned, and moved to a different volume. LUNs support the assignment of a quality of service (QoS) policy for performance management or a QoS policy can be assigned to the volume containing the LUN. See the LUN object model to learn more about each of the properties supported by the LUN REST API.

A LUN must be mapped to an initiator group to grant access to the initiator group's initiators (client hosts). Initiators can then access the LUN and perform I/O over a Fibre Channel (FC) fabric using the Fibre Channel Protocol or a TCP/IP network using iSCSI.

Name	Type	Description
clone	clone	This sub-object is used in POST to create a new LUN as a clone of an existing LUN, or PATCH to overwrite an existing LUN as a clone of another. Setting a property in this sub-object indicates that a LUN clone is desired. Consider the following other properties when cloning a LUN: `auto_delete`, `qos_policy`, `space.guarantee.requested` and `space.scsi_thin_provisioning_support_enabled`. When used in a PATCH, the patched LUN's data is over-written as a clone of the source and the following properties are preserved from the patched LUN unless otherwise specified as part of the PATCH: `class`, `auto_delete`, `lun_maps`, `serial_number`, `status.state`, and `uuid`. Persistent reservations for the patched LUN are also preserved.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the LUN was created.

Name	Type	Description
enabled	boolean	The enabled state of the LUN. LUNs can be disabled to prevent access to the LUN. Certain error conditions also cause the LUN to become disabled. If the LUN is disabled, you can consult the `state` property to determine if the LUN is administratively disabled (*offline*) or has become disabled as a result of an error. A LUN in an error condition can be brought online by setting the `enabled` property to *true* or brought administratively offline by setting the `enabled` property to *false*. Upon creation, a LUN is enabled by default. Valid in PATCH.
lun_maps	array[lun_maps]	An array of LUN maps. A LUN map is an association between a LUN and an initiator group. When a LUN is mapped to an initiator group, the initiator group's initiators are granted access to the LUN. The relationship between a LUN and an initiator group is many LUNs to many initiator groups.
name	string	The fully qualified path name of the LUN composed of the "/vol" prefix, the volume name, the qtree name (optional), and the base name of the LUN. Valid in POST and PATCH.
os_type	string	The operating system type of the LUN. Required in POST when creating a LUN that is not a clone of another. Disallowed in POST when creating a LUN clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
serial_number	string	The LUN serial number. The serial number is generated by ONTAP when the LUN is created. • maxLength: 12 • minLength: 12 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
space	space	The storage space related properties of the LUN.
uuid	string	The unique identifier of the LUN. The UUID is generated by ONTAP when the LUN is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

guarantee

Properties that request and report the space guarantee for the NVMe namespace.

Name	Type	Description
requested	boolean	The requested space reservation policy for the NVMe namespace. If *true*, a space reservation is requested for the namespace; if *false*, the namespace is thin provisioned. Guaranteeing a space reservation request for a namespace requires that the volume in which the namespace resides also be space reserved and that the fractional reserve for the volume be 100%. The space reservation policy for an NVMe namespace is determined by ONTAP. • Introduced in: 9.6 • x-nullable: true
reserved	boolean	Reports if the NVMe namespace is space guaranteed. This property is *true* if a space guarantee is requested and the containing volume and aggregate support the request. This property is *false* if a space guarantee is not requested or if a space guarantee is requested and either the containing volume and aggregate do not support the request.

space

The storage space related properties of the NVMe namespace.

Name	Type	Description
block_size	integer	The size of blocks in the namespace, in bytes. Valid in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone. Valid in POST.

Name	Type	Description
guarantee	guarantee	Properties that request and report the space guarantee for the NVMe namespace.
size	integer	The total provisioned size of the NVMe namespace. Valid in POST and PATCH. The NVMe namespace size can be increased but not reduced using the REST interface. The maximum and minimum sizes listed here are the absolute maximum and absolute minimum sizes, in bytes. The maximum size is variable with respect to large NVMe namespace support in ONTAP. If large namespaces are supported, the maximum size is 128 TB (140737488355328 bytes) and if not supported, the maximum size is just under 16 TB (17557557870592 bytes). The minimum size supported is always 4096 bytes. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • example: 1073741824 • format: int64 • Max value: 140737488355328 • Min value: 4096 • Introduced in: 9.6 • x-nullable: true

Name	Type	Description
used	integer	The amount of space consumed by the main data stream of the NVMe namespace. This value is the total space consumed in the volume by the NVMe namespace, including filesystem overhead, but excluding prefix and suffix streams. Due to internal filesystem overhead and the many ways NVMe filesystems and applications utilize blocks within a namespace, this value does not necessarily reflect actual consumption/availability from the perspective of the filesystem or application. Without specific knowledge of how the namespace blocks are utilized outside of ONTAP, this property should not be used as an indicator for an out-of-space condition. For more information, see *Size properties* in the *docs* section of the ONTAP REST API documentation. • format: int64 • readOnly: 1 • Introduced in: 9.6 • x-nullable: true

status

Status information about the NVMe namespace.

Name	Type	Description
container_state	string	The state of the volume and aggregate that contain the NVMe namespace. Namespaces are only available when their containers are available.

Name	Type	Description
mapped	boolean	Reports if the NVMe namespace is mapped to an NVMe subsystem. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
read_only	boolean	Reports if the NVMe namespace allows only read access.
state	string	The state of the NVMe namespace. Normal states for a namespace are *online* and *offline*. Other states indicate errors.

consistency_group_nvme_host

The NVMe host provisioned to access NVMe namespaces mapped to a subsystem.

Name	Type	Description
nqn	string	The NVMe qualified name (NQN) used to identify the NVMe storage target. Not allowed in POST when the `records` property is used.

consistency_group_nvme_subsystem

An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

Name	Type	Description
comment	string	A configurable comment for the NVMe subsystem. Optional in POST and PATCH.
hosts	array[consistency_group_nvme_host]	The NVMe hosts configured for access to the NVMe subsystem. Optional in POST.

Name	Type	Description
name	string	The name of the NVMe subsystem. Once created, an NVMe subsystem cannot be renamed. Required in POST.
os_type	string	The host operating system of the NVMe subsystem's hosts. Required in POST.
uuid	string	The unique identifier of the NVMe subsystem.

subsystem_map

The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems.

There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.

Name	Type	Description
_links	self_link	
anagrpid	string	The Asymmetric Namespace Access Group ID (ANAGRPID) of the NVMe namespace. The format for an ANAGRPID is 8 hexadecimal digits (zero-filled) followed by a lower case "h". There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.

Name	Type	Description
nsid	string	The NVMe namespace identifier. This is an identifier used by an NVMe controller to provide access to the NVMe namespace. The format for an NVMe namespace identifier is 8 hexadecimal digits (zero-filled) followed by a lower case "h".
subsystem	consistency_group_nvme_subsystem	An NVMe subsystem maintains configuration state and namespace access control for a set of NVMe-connected hosts.

namespaces

An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the storage virtual machine using the NVMe over Fabrics protocol.

In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume.

An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API.

An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem’s hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol.

Name	Type	Description
auto_delete	boolean	This property marks the NVMe namespace for auto deletion when the volume containing the namespace runs out of space. This is most commonly set on namespace clones. When set to *true*, the NVMe namespace becomes eligible for automatic deletion when the volume runs out of space. Auto deletion only occurs when the volume containing the namespace is also configured for auto deletion and free space in the volume decreases below a particular threshold. This property is optional in POST and PATCH. The default value for a new NVMe namespace is *false*. There is an added computational cost to retrieving this property's value. It is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter. See Requesting specific fields to learn more.
comment	string	A configurable comment available for use by the administrator. Valid in POST and PATCH.
create_time	string	The time the NVMe namespace was created.
enabled	boolean	The enabled state of the NVMe namespace. Certain error conditions cause the namespace to become disabled. If the namespace is disabled, you can check the `state` property to determine what error disabled the namespace. An NVMe namespace is enabled automatically when it is created.

Name	Type	Description
name	string	The fully qualified path name of the NVMe namespace composed of a "/vol" prefix, the volume name, the (optional) qtree name and base name of the namespace. Valid in POST. NVMe namespaces do not support rename, or movement between volumes.
os_type	string	The operating system type of the NVMe namespace. Required in POST when creating an NVMe namespace that is not a clone of another. Disallowed in POST when creating a namespace clone.
provisioning_options	provisioning_options	Options that are applied to the operation.
space	space	The storage space related properties of the NVMe namespace.
status	status	Status information about the NVMe namespace.
subsystem_map	subsystem_map	The NVMe subsystem with which the NVMe namespace is associated. A namespace can be mapped to zero (0) or one (1) subsystems. There is an added computational cost to retrieving property values for `subsystem_map`. They are not populated for either a collection GET or an instance GET unless explicitly requested using the `fields` query parameter.
uuid	string	The unique identifier of the NVMe namespace.

parent_consistency_group

The parent consistency group.

Name	Type	Description
_links	self_link	
name	string	The name of the consistency group.
uuid	string	The unique identifier of the consistency group.

storage_service

Determines the placement of any storage object created during this operation.

Name	Type	Description
name	string	Storage service name. If not specified, the default value is the most performant for the platform.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
name	string	New name for consistency group. Required to resolve naming collisions.
storage_service	storage_service	Determines the placement of any storage object created during this operation.

snapshot

A consistency group's Snapshot copy

Name	Type	Description
name	string	The name of the consistency group's Snapshot copy to restore to.

Name	Type	Description
uuid	string	The UUID of the consistency group's Snapshot copy to restore to.

restore_to

Use to restore a consistency group to a previous Snapshot copy

Name	Type	Description
snapshot	snapshot	A consistency group's Snapshot copy

_links

Name	Type	Description
self	href	

snapshot_policy_reference

This is a reference to the Snapshot copy policy.

Name	Type	Description
_links	_links	
name	string	
uuid	string	

space

Space information for the consistency group.

Name	Type	Description
available	integer	The amount of space available in the consistency group, in bytes.
size	integer	The total provisioned size of the consistency group, in bytes.
used	integer	The amount of space consumed in the consistency group, in bytes.

svm

The Storage Virtual Machine (SVM) in which the consistency group is located.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

object_stores

Name	Type	Description
name	string	The name of the object store to use. Used for placement.

tiering

The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

unsupported_reason

Name	Type	Description
code	string	If volume activity tracking is not supported on the volume, this field provides an appropriate error code.

Name	Type	Description
message	string	If volume activity tracking is not supported on the volume, this field provides an error message detailing why this is the case.

activity_tracking

The volume activity tracking configuration for this volume.

Name	Type	Description
state	string	Activity tracking state of the volume. If this value is *on*, ONTAP collects top metrics information for the volume in real time. There is a slight impact to I/O performance in order to collect this information. If this value is *off*, no activity tracking information is collected or available to view. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["off", "on"] • Introduced in: 9.10 • x-nullable: true
supported	boolean	This field indicates whether or not volume activity tracking is supported on the volume. If volume activity tracking is not supported, the reason why is provided in the `activity_tracking.unsupported_reason` field.
unsupported_reason	unsupported_reason	

unsupported_reason

Name	Type	Description
code	string	If file system analytics is not supported on the volume, this field provides the error code explaining why.
message	string	If file system analytics is not supported on the volume, this field provides the error message explaining why.

analytics

The file system analytics configuration for this volume.

Name	Type	Description
scan_progress	integer	Percentage of files in the volume that the file system analytics initialization scan has processed. Only returned when the `state` is *initializing*.

Name	Type	Description
state	string	File system analytics state of the volume. If this value is *on*, ONTAP collects extra file system analytics information for all directories on the volume. There will be a slight impact to I/O performance to collect this information. If this value is *off*, file system analytics information is not collected and not available to be viewed. If this value is *initializing*, that means file system analytics was recently turned on, and the initialization scan to gather information for all existing files and directories is currently running. If this value is *initialization_paused*, this means that the initialization scan is currently paused. If this value is 'unknown', this means that there was an internal error when determining the file system analytics state for the volume. The default value is *on* for all volumes that support file system analytics. If the volume will contain LUNs or NVMe namespaces, the default value is *off*. • enum: ["unknown", "initializing", "initialization_paused", "off", "on"] • Introduced in: 9.8 • x-nullable: true
supported	boolean	This field indicates whether or not file system analytics is supported on the volume. If file system analytics is not supported, the reason will be specified in the `analytics.unsupported_reason` field.
unsupported_reason	unsupported_reason	

acls

The permissions that users and groups have on a CIFS share.

Name	Type	Description
_links	_links	
permission	string	Specifies the access rights that a user or group has on the defined CIFS Share. The following values are allowed: • no_access - User does not have CIFS share access • read - User has only read access • change - User has change access • full_control - User has full_control access
type	string	Specifies the type of the user or group to add to the access control list of a CIFS share. The following values are allowed: • windows - Windows user or group • unix_user - UNIX user • unix_group - UNIX group
user_or_group	string	Specifies the user or group name to add to the access control list of a CIFS share.

consistency_group_cifs_share

CIFS share is a named access point in a volume. Before users and applications can access data on the CIFS server over SMB, a CIFS share must be created with sufficient share permission. CIFS shares are tied to the CIFS server on the SVM. When a CIFS share is created, ONTAP creates a default ACL for the share with Full Control permissions for Everyone.

Name	Type	Description
_links	_links	
access_based_enumeration	boolean	Specifies whether all folders inside this share are visible to a user based on that individual user's access right; prevents the display of folders or other shared resources that the user does not have access to.

Name	Type	Description
acls	array[acls]	
allow_unencrypted_access	boolean	Specifies whether or not the SMB2 clients are allowed to access the encrypted share.
change_notify	boolean	Specifies whether CIFS clients can request for change notifications for directories on this share.
comment	string	Specify the CIFS share descriptions.
continuously_available	boolean	Specifies whether or not the clients connecting to this share can open files in a persistent manner. Files opened in this way are protected from disruptive events, such as, failover and giveback.
dir_umask	integer	Directory mode creation mask to be viewed as an octal number.
encryption	boolean	Specifies whether SMB encryption must be used when accessing this share. Clients that do not support encryption are not able to access this share.
file_umask	integer	File mode creation mask to be viewed as an octal number.

Name	Type	Description
home_directory	boolean	Specifies whether or not the share is a home directory share, where the share and path names are dynamic. ONTAP home directory functionality automatically offer each user a dynamic share to their home directory without creating an individual SMB share for each user. The ONTAP CIFS home directory feature enable us to configure a share that maps to different directories based on the user that connects to it. Instead of creating a separate shares for each user, a single share with a home directory parameters can be created. In a home directory share, ONTAP dynamically generates the share-name and share-path by substituting %w, %u, and %d variables with the corresponding Windows user name, UNIX user name, and domain name, respectively. • Default value: 1 • Introduced in: 9.12 • readCreate: 1 • x-nullable: true
name	string	Specifies the name of the CIFS share that you want to create. If this is a home directory share then the share name includes the pattern as %w (Windows user name), %u (UNIX user name) and %d (Windows domain name) variables in any combination with this parameter to generate shares dynamically.
namespace_caching	boolean	Specifies whether or not the SMB clients connecting to this share can cache the directory enumeration results returned by the CIFS servers.

Name	Type	Description
no_strict_security	boolean	Specifies whether or not CIFS clients can follow Unix symlinks outside the share boundaries.
offline_files	string	Offline Files The supported values are: • none - Clients are not permitted to cache files for offline access. • manual - Clients may cache files that are explicitly selected by the user for offline access. • documents - Clients may automatically cache files that are used by the user for offline access. • programs - Clients may automatically cache files that are used by the user for offline access and may use those files in an offline mode even if the share is available.
oplocks	boolean	Specifies whether opportunistic locks are enabled on this share. "Oplocks" allow clients to lock files and cache content locally, which can increase performance for file operations.
show_snapshot	boolean	Specifies whether or not the Snapshot copies can be viewed and traversed by clients.
unix_symlink	string	Controls the access of UNIX symbolic links to CIFS clients. The supported values are: • local - Enables only local symbolic links which is within the same CIFS share. • widelink - Enables both local symlinks and widelinks. • disable - Disables local symlinks and widelinks.

Name	Type	Description
vscan_profile	string	Vscan File-Operations Profile The supported values are: • no_scan - Virus scans are never triggered for accesses to this share. • standard - Virus scans can be triggered by open, close, and rename operations. • strict - Virus scans can be triggered by open, read, close, and rename operations. • writes_only - Virus scans can be triggered only when a file that has been modified is closed.

cifs

Name	Type	Description
shares	array[consistency_group_cifs_share]	

export_clients

Name	Type	Description
match	string	Client Match Hostname, IP Address, Netgroup, or Domain. You can specify the match as a string value in any of the following formats: • As a hostname; for instance, host1 • As an IPv4 address; for instance, 10.1.12.24 • As an IPv6 address; for instance, fd20:8b1e:b255:4071::100:1 • As an IPv4 address with a subnet mask expressed as a number of bits; for instance, 10.1.12.0/24 • As an IPv6 address with a subnet mask expressed as a number of bits; for instance, fd20:8b1e:b255:4071::/64 • As an IPv4 address with a network mask; for instance, 10.1.16.0/255.255.255.0 • As a netgroup, with the netgroup name preceded by the @ character; for instance, @eng • As a domain name preceded by the . character; for instance, .example.com

export_rules

Name	Type	Description
_links	_links	
allow_device_creation	boolean	Specifies whether or not device creation is allowed.
allow_suid	boolean	Specifies whether or not SetUID bits in SETATTR Op is to be honored.
anonymous_user	string	User ID To Which Anonymous Users Are Mapped.

Name	Type	Description
chown_mode	string	Specifies who is authorized to change the ownership mode of a file.
clients	array[export_clients]	Array of client matches
index	integer	Index of the rule within the export policy.
ntfs_unix_security	string	NTFS export UNIX security options.
protocols	array[string]	
ro_rule	array[string]	Authentication flavors that the read-only access rule governs
rw_rule	array[string]	Authentication flavors that the read/write access rule governs
superuser	array[string]	Authentication flavors that the superuser security type governs

export_policy

The policy associated with volumes to export them for protocol access.

Name	Type	Description
_links	self_link	
name	string	Name of the export policy.
rules	array[export_rules]	The set of rules that govern the export policy.
uuid	string	Identifier for the export policy.

junction_parent

Name	Type	Description
_links	self_link	

Name	Type	Description
name	string	The name of the parent volume that contains the junction inode of this volume. The junction parent volume must belong to the same SVM that owns this volume.
uuid	string	Unique identifier for the parent volume.

nas

The CIFS share policy and/or export policies for this volume.

Name	Type	Description
cifs	cifs	
export_policy	export_policy	The policy associated with volumes to export them for protocol access.
gid	integer	The UNIX group ID of the volume. Valid in POST or PATCH.
junction_parent	junction_parent	
path	string	The fully-qualified path in the owning SVM's namespace at which the volume is mounted. The path is case insensitive and must be unique within an SVM's namespace. Path must begin with '/' and must not end with '/'. Only one volume can be mounted at any given junction path. An empty path in POST creates an unmounted volume. An empty path in PATCH deactivates and unmounts the volume. Taking a volume offline or restricted state removes its junction path. This attribute is reported in GET only when the volume is mounted.

Name	Type	Description
security_style	string	Security style associated with the volume. Valid in POST or PATCH. mixed ‐ Mixed-style security ntfs ‐ NTFS/WIndows-style security unified ‐ Unified-style security, unified UNIX, NFS and CIFS permissions unix ‐ UNIX-style security.
uid	integer	The UNIX user ID of the volume. Valid in POST or PATCH.
unix_permissions	integer	UNIX permissions to be viewed as an octal number, consisting of 4 digits derived by adding up bits 4 (read), 2 (write), and 1 (execute). First digit selects the set user ID (4), set group ID (2), and sticky (1) attributes. Second digit selects permission for the owner of the file. Third selects permissions for other users in the same group while the fourth selects permissions for other users not in the group. Valid in POST or PATCH. For security style "mixed" or "unix", the default setting is 0755 in octal (493 in decimal) and for security style "ntfs", the default setting is 0000. In cases where only owner, group, and other permissions are given (as in 755, representing the second, third and fourth digit), the first digit is assumed to be zero.

provisioning_options

Options that are applied to the operation.

Name	Type	Description
action	string	Operation to perform
count	integer	Number of elements to perform the operation on.

Name	Type	Description
storage_service	storage_service	Determines the placement of any storage object created during this operation.

qos

The QoS policy for this volume.

Name	Type	Description
policy	policy	The QoS policy

space

Name	Type	Description
available	integer	The available space, in bytes.
size	integer	Total provisioned size, in bytes.
used	integer	The virtual space used (includes volume reserves) before storage efficiency, in bytes.

tiering

The tiering placement and policy definitions for this volume.

Name	Type	Description
control	string	Storage tiering placement rules for the object.
object_stores	array[object_stores]	Object stores to use. Used for placement.

Name	Type	Description
policy	string	Policy that determines whether the user data blocks of a volume in a FabricPool will be tiered to the cloud store when they become cold. FabricPool combines flash (performance tier) with a cloud store into a single aggregate. Temperature of a volume block increases if it is accessed frequently and decreases when it is not. Valid in POST or PATCH. all ‐ Allows tiering of both Snapshot copies and active file system user data to the cloud store as soon as possible by ignoring the temperature on the volume blocks. auto ‐ Allows tiering of both snapshot and active file system user data to the cloud store none ‐ Volume blocks are not be tiered to the cloud store. snapshot_only ‐ Allows tiering of only the volume Snapshot copies not associated with the active file system. The default tiering policy is "snapshot-only" for a FlexVol volume and "none" for a FlexGroup volume. The default minimum cooling period for the "snapshot-only" tiering policy is 2 days and for the "auto" tiering policy it is 31 days.

volumes

Name	Type	Description
activity_tracking	activity_tracking	The volume activity tracking configuration for this volume.
analytics	analytics	The file system analytics configuration for this volume.

Name	Type	Description
comment	string	A comment for the volume. Valid in POST or PATCH.
language	string	Language encoding setting for volume. If no language is specified, the volume inherits its SVM language encoding setting.
name	string	Volume name. The name of volume must start with an alphabetic character (a to z or A to Z) or an underscore (_). The name must be 197 or fewer characters in length for FlexGroups, and 203 or fewer characters in length for all other types of volumes. Volume names must be unique within an SVM. Required on POST.
nas	nas	The CIFS share policy and/or export policies for this volume.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	The QoS policy for this volume.
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	
tiering	tiering	The tiering placement and policy definitions for this volume.
uuid	string	Unique identifier for the volume. This corresponds to the instance-uuid that is exposed in the CLI and ONTAPI. It does not change due to a volume move. • example: 028baa66-41bd-11e9-81d5-00a0986138f7 • readOnly: 1 • Introduced in: 9.8 • x-nullable: true

consistency_groups

Name	Type	Description
_links	self_link	
application	application	
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true

Name	Type	Description
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

iops

The rate of I/O operations observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency

The round trip latency observed at the storage object, in microseconds.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput

The rate of throughput bytes per second observed at the storage object.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

metric

Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.

Name	Type	Description
_links	_links	
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
duration	string	The duration over which this sample is calculated. The time durations are represented in the ISO-8601 standard format. Samples can be calculated over the following durations:
iops	iops	The rate of I/O operations observed at the storage object.
latency	latency	The round trip latency observed at the storage object, in microseconds.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput	throughput	The rate of throughput bytes per second observed at the storage object.
timestamp	string	The timestamp of the performance and capacity data.

Name	Type	Description
used_space	integer	The total space used in the consistency group, in bytes.

replication_relationships

Name	Type	Description
_links	self_link	
is_source	boolean	Indicates whether or not this consistency group is the source for replication.
uuid	string	The unique identifier of the SnapMirror relationship.

iops_raw

The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

latency_raw

The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

throughput_raw

Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.

Name	Type	Description
other	integer	Performance metric for other I/O operations. Other I/O operations can be metadata operations, such as directory lookups and so on.
read	integer	Performance metric for read I/O operations.
total	integer	Performance metric aggregated over all types of I/O operations.
write	integer	Peformance metric for write I/O operations.

statistics

These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.

Name	Type	Description
available_space	integer	The total space available in the consistency group, in bytes.

Name	Type	Description
iops_raw	iops_raw	The number of I/O operations observed at the storage object. This can be used along with delta time to calculate the rate of I/O operations per unit of time.
latency_raw	latency_raw	The raw latency observed at the storage object, in microseconds. This can be divided by the raw IOPS value to calculate the average latency per I/O operation.
size	integer	The total size of the consistency group, in bytes.
status	string	Errors associated with the sample. For example, if the aggregation of data over multiple nodes fails, then any partial errors might return "ok" on success or "error" on an internal uncategorized failure. Whenever a sample collection is missed but done at a later time, it is back filled to the previous 15 second timestamp and tagged with "backfilled_data". "Inconsistent_delta_time" is encountered when the time between two collections is not the same for all nodes. Therefore, the aggregated value might be over or under inflated. "Negative_delta" is returned when an expected monotonically increasing value has decreased in value. "Inconsistent_old_data" is returned when one or more nodes do not have the latest data.
throughput_raw	throughput_raw	Throughput bytes observed at the storage object. This can be used along with delta time to calculate the rate of throughput bytes per unit of time.
timestamp	string	The timestamp of the performance data.

Name	Type	Description
used_space	integer	The total used space in the consistency group, in bytes.

consistency_group

Name	Type	Description
_links	self_link	
application	application	
clone	clone	Creates a clone of an existing consistency group from the current contents or an existing Snapshot copy.
consistency_groups	array[consistency_groups]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A consistency group can only be associated with one direct parent consistency group.
luns	array[luns]	The LUNs array can be used to create or modify LUNs in a consistency group on a new or existing volume that is a member of the consistency group. LUNs are considered members of a consistency group if they are located on a volume that is a member of the consistency group.
metric	metric	Performance and capacity numbers, such as, IOPS, latency, throughput, used space, and available space.
name	string	Name of the consistency group. The consistency group name must be unique within an SVM. If not provided and the consistency group contains only one volume, the name will be generated based on the volume name. If the consistency group contains more than one volume, the name is required.

Name	Type	Description
namespaces	array[namespaces]	An NVMe namespace is a collection of addressable logical blocks presented to hosts connected to the SVM using the NVMe over Fabrics protocol. In ONTAP, an NVMe namespace is located within a volume. Optionally, it can be located within a qtree in a volume. An NVMe namespace is created to a specified size using thin or thick provisioning as determined by the volume on which it is created. NVMe namespaces support being cloned. An NVMe namespace cannot be renamed, resized, or moved to a different volume. NVMe namespaces do not support the assignment of a QoS policy for performance management, but a QoS policy can be assigned to the volume containing the namespace. See the NVMe namespace object model to learn more about each of the properties supported by the NVMe namespace REST API. An NVMe namespace must be mapped to an NVMe subsystem to grant access to the subsystem's hosts. Hosts can then access the NVMe namespace and perform I/O using the NVMe over Fabrics protocol. • maxItems: 16 • minItems: 0 • uniqueItems: 1 • Introduced in: 9.10
parent_consistency_group	parent_consistency_group	The parent consistency group.
provisioning_options	provisioning_options	Options that are applied to the operation.
qos	qos	

Name	Type	Description
replicated	boolean	Indicates whether or not replication has been enabled on this consistency group.
replication_relationships	array[replication_relationships]	Indicates the SnapMirror relationship of this consistency group.
replication_source	boolean	Since support for this field is to be removed in the next release, use replication_relationships.is_source instead.
restore_to	restore_to	Use to restore a consistency group to a previous Snapshot copy
snapshot_policy	snapshot_policy_reference	This is a reference to the Snapshot copy policy.
space	space	Space information for the consistency group.
statistics	statistics	These are raw performance and space numbers, such as, IOPS, latency, throughput, used space, and available space. These numbers are aggregated across all nodes in the cluster and increase with the uptime of the cluster.
svm	svm	The Storage Virtual Machine (SVM) in which the consistency group is located.
tiering	tiering	The tiering placement and policy definitions for volumes in this consistency group.

Name	Type	Description
uuid	string	The unique identifier of the consistency group. The UUID is generated by ONTAP when the consistency group is created. • example: 1cd8a442-86d1-11e0-ae1c-123478563412 • readOnly: 1 • Introduced in: 9.10 • x-nullable: true
volumes	array[volumes]	A consistency group is a mutually exclusive aggregation of volumes or other consistency groups. A volume can only be associated with one direct parent consistency group. The volumes array can be used to create new volumes in the consistency group, add existing volumes to the consistency group, or modify existing volumes that are already members of the consistency group. The total number of volumes across all child consistency groups contained in a consistency group is constrained by the same limit.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code

Name	Type	Description
message	string	Error message
target	string	The target parameter that caused the error.

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